

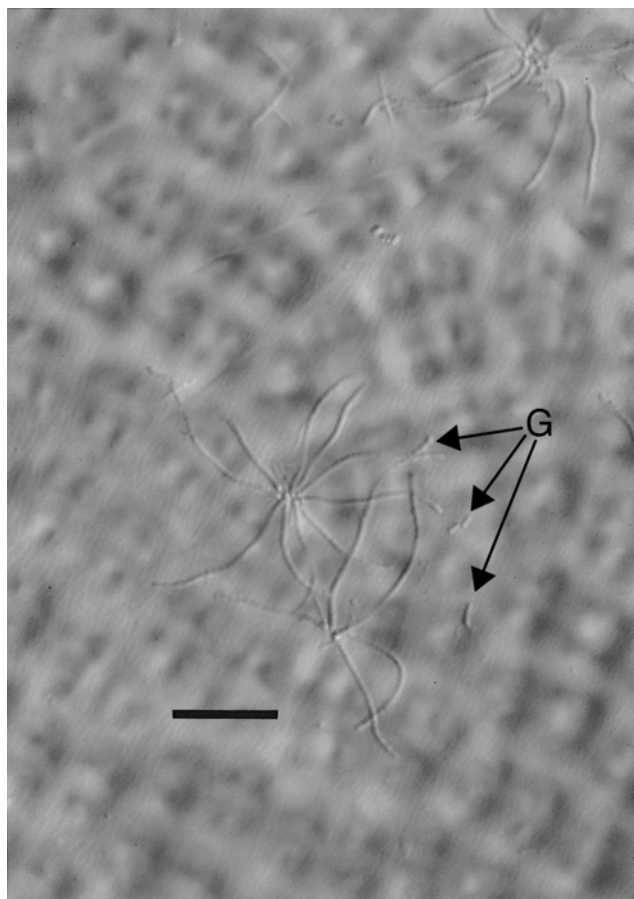


FIGURE BXII.γ.70. Rosette of *Leucothrix* on the surface of the red alga *Porphyra*. Gonidia (G) are visible. Bar = 20 μm (Nomarsky interference optics).

for growth of 25–30°C, an optimum pH for growth of 7.6, and an optimum salinity of 1.5%. Bland and Brock (1973) have shown that *Leucothrix* is able to withstand several hours of drying without any apparent reduction in viability, which is consistent with the fact that the organism extensively colonizes *Bangia fuscopurpurea*, a red algal species found high in the intertidal zone and frequently subject to drying during low tides. Radiorespirometric studies have shown that *Leucothrix* oxidizes carbohydrate substrates primarily via the Entner–Doudoroff pathway, with contributions from the pentose phosphate and the Embden–Meyerhoff pathways and the TCA cycle (Raj, 1977). Operation of the TCA cycle is consistent with the presence of isocitrate and malate dehydrogenases (Kelly and Brock, 1969a). The organism is catalase and oxidase positive (Poffe et al., 1979; Williams and Unz, 1985a). The electron transport pathway has been studied by Biggins and Dietrich (1968) using spectrophotometric methods. The terminal electron transport system is a particulate cytochrome chain containing cytochrome b_{562} , cytochrome c_{552} , and cytochrome b_{558} . Cytochrome b_{558} has been proposed to be functionally equivalent to cytochrome a , and it operates as the terminal oxidase. Cytochrome b_{558} has high CO-binding properties. The activity of the oxidase system is markedly affected by the degree of aeration during growth. Although several strains of *Leucothrix* have been tested for ability to oxidize hydrogen sulfide, these tests have all been negative (Harold and Stanier, 1955;

Williams and Unz, 1985a). However, K. Eimhjellen of the Technical University in Trondheim has found sulfide-oxidizing ability and deposition of elemental sulfur in a pure marine culture of *Leucothrix* isolated from Pacific Grove, California (Brock, 1992). He has also been able to demonstrate oxidation of thiosulfate to either tetrathionate or sulfate and followed this latter oxidation manometrically. Eimhjellen's strain is obviously different from those studied by Harold and Stanier (1955). In light of the uncertainty surrounding the ability of *Leucothrix* to oxidize sulfur compounds, more work on this aspect of *Leucothrix* metabolism is in order.

Genetics Very limited genetic information is available for *Leucothrix*. Buoyant density determinations of DNA base composition have been determined by Brock and Mandel (1966) and by Kelly and Brock (1969a) for 46 strains isolated from various locations around the world. The mol% G + C content for all strains is in the range of 46.9–51.0, indicating great homogeneity at the DNA level. However, enzyme mobility data for 35 of the strains tested by Kelly and Brock (1969a) has revealed more diversity. Electrophoresis of malate dehydrogenase (MDH) and isocitrate dehydrogenase results in 17 different banding patterns, with MDH showing the most variability. Strains from the same geographical area vary as much as those from different geographical locations.

Antibiotic sensitivity Limited antibiotic sensitivity data exist for *Leucothrix*. According to Raj (1977), growth of *Leucothrix* can be inhibited by penicillin (0.1 mg/l), streptomycin (5.0 mg/l), or chloromycetin (0.7–0.9 mg/l). An isolate tested by Williams and Unz (1985a), *Leucothrix* sp. N11, was found to be completely inhibited by <0.125 mg/l of streptomycin, gentamicin, tetracycline, ampicillin, and penicillin G. The same strain is resistant to sulfanilamide and lincomycin, at >64 mg/l for both antibiotics, and is somewhat sensitive to chloramphenicol (0.5 mg/l) and bacitracin (4.0 mg/l).

Pathogenicity *Leucothrix* is an opportunistic pathogen of marine organisms reared in aquaculture or aquaria. Johnson et al. (1971) showed that *Leucothrix* was able to grow extensively on benthic crustacea, invertebrates, and fish eggs (cod and winter flounder) in laboratory spawning tanks and aquaria where they were infected with *Leucothrix* introduced by the seawater system. Since that report, *Leucothrix* has been found to cause infestations of crustaceans and fish eggs under cultivation in high-density aquaculture situations (Shelton et al., 1975; Couch, 1978; Fisher, 1978; Solangi et al., 1979; Harper and Talbot, 1984; Dale and Blom, 1987; Hansen and Olafsen, 1989; Carr, 1996). Harper and Talbot (1984) were unable to show differences in mortalities of wild and laboratory spawned lobster eggs (*Homarus gammarus* and *H. americanus*) in cases where *Leucothrix* was one of the epiphytes observed. In contrast, Lightner et al. (1975) theorized that penaeid shrimp heavily infested with *Leucothrix* died as a result of oxygen deprivation. In a similar vein, Couch (1978) reported that mortality of penaeid shrimp was proportionate to the extent of growth of *Leucothrix* on their gills. He suggested that the mortality may have been due to the massive growth of the bacterium over the gill cuticle, blocking normal gas diffusion across the gill surfaces. Solangi et al. (1979) noted that *Leucothrix* attached to all available surfaces on cultured brine shrimp, *Artemia salina*, and caused mortality. They found that exposure of shrimp to terramycin (100 ppm) for two days eliminated *Leucothrix*, while treatment with various chemicals proved toxic to

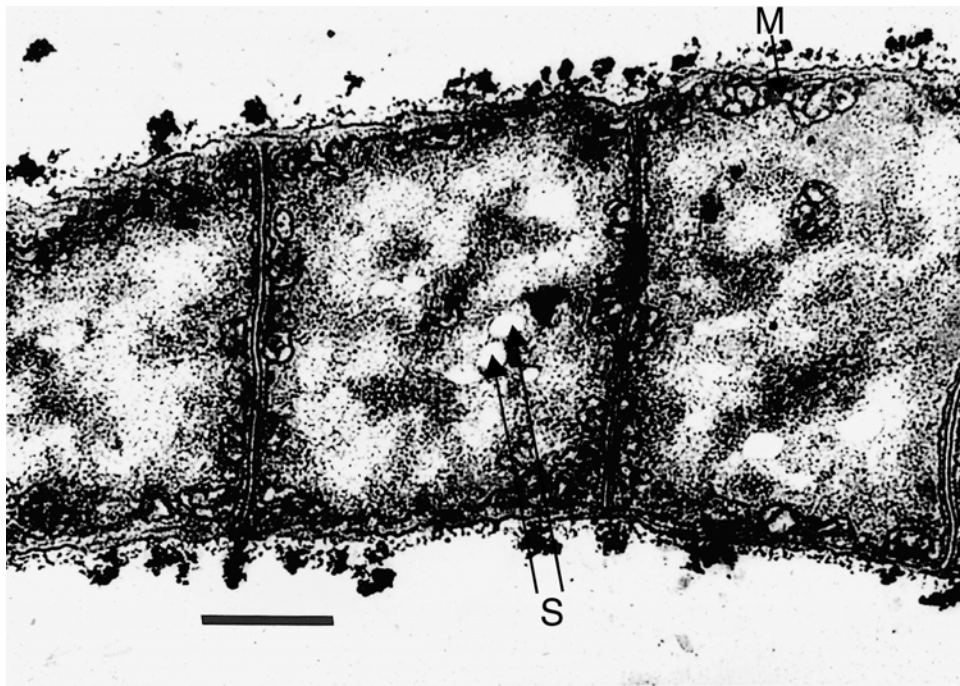


FIGURE BXII.γ.71. Electron micrograph of a thin section of a portion of a filament of *L. mucor*. Membranous vesicles (*M*) and storage vesicles (*S*) are visible. Bar = 0.5 μm. (Reproduced with permission from T.D. Brock and S.F. Conti, Archives of Microbiology 66: 76–90, 1969, ©Springer-Verlag.)

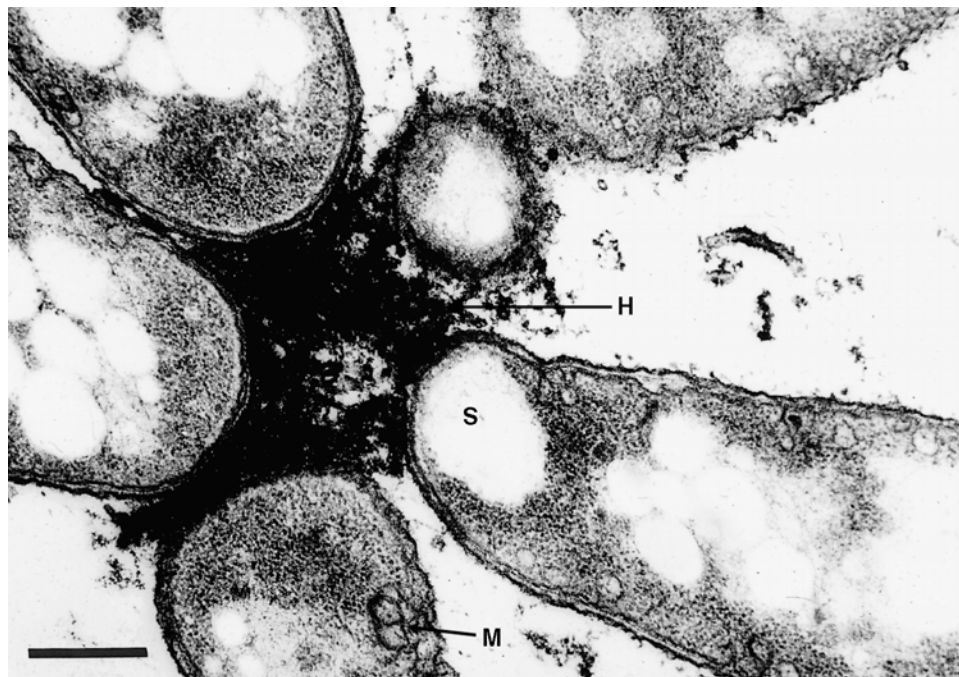


FIGURE BXII.γ.72. Electron micrograph of an *L. mucor* rosette, with the electron-dense holdfast material (*H*) shown. Membranous (*M*) and storage (*S*) vesicles are obvious in both filament cells and gonidia. Bar = 1 μm. (Reproduced with permission from T.D. Brock and S.F. Conti, Archives of Microbiology 66: 76–90, 1969, ©Springer-Verlag.)



FIGURE BXII.γ.73. *Leucothrix* filaments attached to the cuticle of *B. fuscopurpurea*. Bar = 10 μm (Nomarsky interference optics).

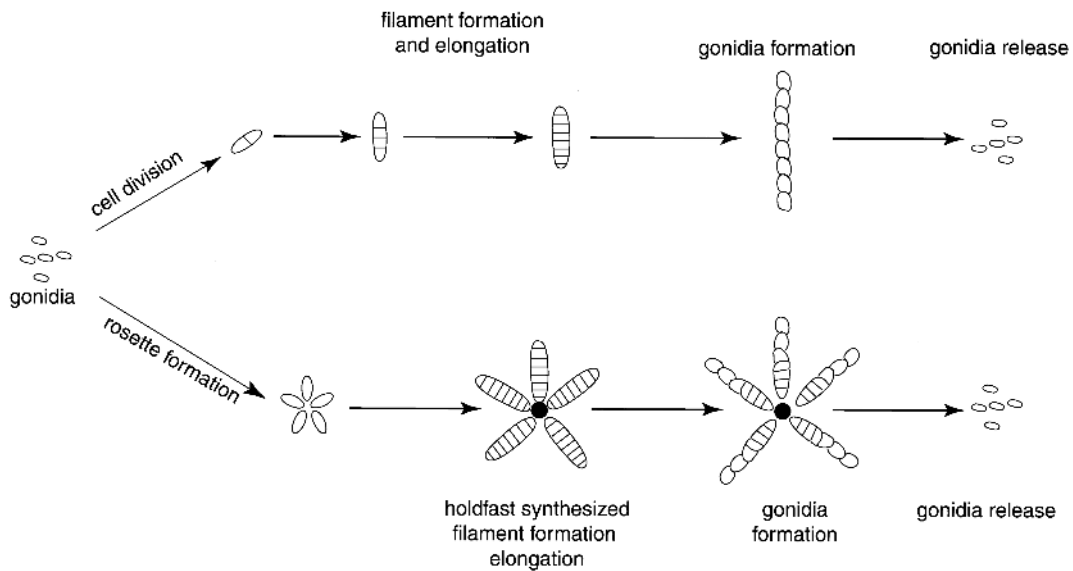


FIGURE BXII.γ.74. Life cycle of *Leucothrix*. Gonidia may attach to a substrate or to each other (rosette formation) with subsequent cell division. Ultimately, filament cells become gonidia that are released to repeat the cycle.

the shrimp. Colonization of uninfected organisms or eggs is apparently rapid. Cod and halibut eggs in aquaculture were colonized within 2 h of fertilization by a variety of bacteria including *Leucothrix* according to Hansen and Olafsen (1989). The authors further report heavier *Leucothrix* growth on cod eggs, giving them a "hairy" appearance that could be mistaken for fungal growth. Antibiotic treatment may be used to prevent such overgrowth. In addition to the antibiotics mentioned above, *Leucothrix* infec-

tions have been treated with penicillin, streptomycin, and neomycin (Johnson et al., 1971; Raj, 1977; Fisher, 1978). Also, Sadusky and Bullis (1994) report that dilute bleach solutions are effective in controlling *Leucothrix* infections in the lobster.

Ecology *Leucothrix* is found widely in the littoral zone in marine environments worldwide, growing primarily as an epiphyte of micro- and macroscopic algae or animals. The dominant

form of *Leucothrix* in the marine environment is the filament, suggesting an advantage to this morphology that might involve entrapment of available nutrients from seawater, better exposure to available oxygen, and access to any materials released by the host. The dispersive, or colonizing, form of *Leucothrix* is the gonidium. *Leucothrix* has been reported to colonize almost every type of marine algae, including species of diatoms and red, green, and brown algae (Brock, 1966; Kelly and Brock, 1969a; Johnson et al., 1971; Bland and Brock, 1973). Some algae are routinely more heavily colonized than others. Bland and Brock (1973) have reported that the red alga *Bangia fuscopurpurea*, a filamentous species living in the high intertidal region, has *Leucothrix* populations 10–30 times larger than those on other algal species. The cuticle of this alga contains mannose (Frei and Preston, 1964), a sugar readily utilized by *Leucothrix* (Bland and Brock, 1973). Johnson et al. (1971) have also reported a preference for red algal species. Brock (1966) has shown that *Leucothrix* can grow on nutrients produced or liberated from algae. However, algae also produce inhibitory chemicals, e.g., sulfuric acid, acrylic acid, and tannins (Bland and Brock, 1973). Furthermore, extracts of 5 of 39 species of corals tested were shown to inhibit a *Leucothrix* culture (Jensen et al., 1996). Although *Leucothrix* is also able to colonize artificial substrates (plastic strips) in the marine environment, Bland and Brock (1973) have shown that growth is much poorer on these strips than on seaweeds, a result that further supports the notion a living host provides a nutritional advantage for *Leucothrix*.

Johnson et al. (1971) have observed *Leucothrix* on many kinds of free-living benthic and pelagic crustaceans, including crabs, lobster, prawns, shrimp, and copepods. Filaments of the organism attach to antennae, pleopods, uropods, and egg masses. *Leucothrix* grows on cultured fish eggs (see section on Pathogenicity), but no data about its occurrence on wild fish has been found. *Leucothrix* may serve as a source of food for some crustaceans. The grass shrimp *Palaemonetes pugio* has been observed to graze on heavy growths of *Leucothrix* on its pleopods while grooming (Johnson et al., 1971). Bauer (1987) has experimentally determined that the growth of *Leucothrix* on the antennae and gills of the stomatopod *Gonodactylus oerstedii* is far heavier on organisms unable to groom than those that were. He has also observed that *Leucothrix* does not colonize the eyes of the organism or most other parts of the body, indicating that some mechanism restricts attachment or growth. McAllen and Hannah (1999) have observed that *Leucothrix* is a major biofoulant of the marine copepod *Tigriopus brevicornis*, a species that has no grooming structure. McAllen and Scott (2000) have additionally reported that the biofouling interferes with swimming behavior.

There are controversial reports about the presence of *Leucothrix* in sewage treatment processes. *Leucothrix*-like organisms have reportedly been associated with activated sludge systems, where they have been implicated in the phenomenon of bulking, and they have been observed in self-purification and other biological waste treatment processes (Eikelboom, 1975, 1977; Cyrus and Sladka, 1970). In most such cases, brackish water was involved (Eikelboom and van Buijsen, 1981), although one strictly freshwater strain, said to be *Leucothrix*, has been isolated from petrochemical wastewater undergoing activated sludge treatment (Poffe et al., 1979). Even this strain grows well at seawater salinity and thus may be considered simply more euryhaline than are other isolates. However, more recent data (Wagner et al., 1994a; Nielsen et al., 1998) from a *Leucothrix mucor* rRNA-targeted oli-

gonucleotide probe did not identify *Leucothrix* in activated sludge samples where organisms morphologically similar to *Leucothrix* were seen. All in all, investigators claiming to have isolated *Leucothrix* from freshwater environments should be encouraged to confirm the culture identity with molecular genetic tools, such as probes.

ENRICHMENT AND ISOLATION PROCEDURES

Although any marine-type seawater salts base may be used in the formation of a culture medium, pH control and the avoidance of metal precipitation are easier in a Provasoli-type culture medium (Provasoli, 1964). The salts formulation (Brock, 1966) in *Leucothrix* medium¹ has proven to be quite effective. The low phosphate concentration in this salts base is critical, as media of more typical phosphate concentrations are frequently inhibitory. Most strains have no vitamin requirements and are able to use glutamate as their sole source of carbon, nitrogen, and energy. If a richer culture medium is desired, 0.1% tryptone plus 0.1% yeast extract can be used. Other media have been described by Raj (1977) and Williams and Unz (1985a, 1989).

In the initial isolation step, it is best to keep the concentration of organic materials low to avoid problems with overgrowth by unicellular bacteria. Although Harold and Stanier (1955) describe a procedure for enrichment of *Leucothrix* from rotting seaweeds, direct isolation from fresh seaweeds is preferable. Rapidly growing, highly motile, unicellular bacteria frequently present problems when isolation of *Leucothrix* from rotted materials is attempted. An effective way of obtaining cultures of *Leucothrix* is to place relatively clean seaweeds taken directly from the sea onto an agar medium that will support the growth of *Leucothrix*. The plates are incubated overnight at 20–25°C and examined within 12–18 h for the presence of *Leucothrix* colonies. This examination is best done by using $\times 125$ magnification ($\times 12.5$ eyepiece and $\times 10$ objective) with a long working-distance condenser. It is important to use short incubation periods in order to find *Leucothrix* colonies before they become overgrown by motile contaminants. A *Leucothrix* colony may be recognized by its coiled rope or thumbprint morphology. Colonies are immediately picked by touching them with a sterile insect pin and transferring them in patches to fresh agar plates of the same composition. One advantage of isolating *Leucothrix* in this manner is that the precise habitat of the organism can be recognized; such information is of considerable value in studies on the molecular evolution of the organism (Kelly and Brock, 1969a). Transfer of agar cultures to liquid medium frequently presents problems. It has been observed that if a small inoculum is used in a large volume of liquid medium, growth often does not occur, whereas if the inoculum is placed in a small volume of medium, i.e., 1–2 ml, heavy growth occurs overnight. Once satisfactory growth has been obtained in an initial small-volume liquid culture, it is possible to make transfers to large volumes of liquid medium. In liquid medium, growth is best when flasks are shaken gently, as on a wrist-action shaker or slowly on a rotary shaker. When a rotary shaker is used, growth rate is increased if the flasks contain small internal baffles, made by pushing in the sides of the flasks while heating with an oxygen flame.

1. *Leucothrix* medium consists of (g/l deionized water, pH 7.6): Basal salts: NaCl, 11.75; MgCl₂·6H₂O, 5.35; Na₂SO₄, 2.0; CaCl₂·2H₂O, 0.75; KCl, 0.35; Tris(hydroxymethyl)aminomethane, 0.5; Na₂HPO₄, 0.05; Organic ingredients: monosodium glutamate, 1.0 or tryptone + yeast extract, 1.0 + 1.0.

Identification While an experienced researcher can identify *Leucothrix* simply based on morphological examination with the light microscope, a novice might initially want to work with a known isolate. Other than its distinct morphology and usual occurrence in the marine habitat, there are no conventional diagnostic tests specific for *Leucothrix*. In the extensive isolations of Kelly and Brock (1969a), all of the isolates recognized morphologically had virtually the same physiological properties. The morphological properties described at the beginning of this article are generally found in any *Leucothrix* strain that is isolated. The availability of 16S and 23S rRNA (or other molecular) probes make confirmation possible when identification is critical, and these should especially be used for putative freshwater isolates.

MAINTENANCE PROCEDURES

Once rapidly growing liquid cultures have been obtained (see above), there are no special maintenance problems. Cultures in the early stages of growth can be stored in the refrigerator for several months, after which they may consist mostly of gonidia. Cultures can also be lyophilized, with the best suspension medium being skim milk. When a lyophilized culture is rehydrated, it is best if the dry plug is placed in 1–2 ml of liquid medium and 0.1 ml of this suspension placed in another tube of 1–2 ml of liquid medium. Both tubes are then incubated for 1 or 2 days. If there is no visible growth in the dilution, 0.1–0.2 ml of the tube containing the original rehydrated culture should be transferred to another tube containing 1–2 ml of medium, followed by another incubation period. Only after good growth has been obtained in a small volume of medium should large-volume cultures be prepared. Contamination is easy to recognize, since *Leucothrix* cultures never show uniform turbidity. Microscopic examination of a culture should permit contamination to be readily determined. Streaking onto a seawater agar medium containing tryptone and yeast extract is also a satisfactory way of recognizing many contaminants. If contaminants do become established, they will generally grow better than the *Leucothrix* and as a result will quickly take over the culture.

DIFFERENTIATION OF THE GENUS *LEUCOTHRIX* FROM OTHER GENERA

Molecular techniques using both 16S rRNA (Woese et al., 1985) and 23S rRNA (Ludwig et al., 1995) place *Leucothrix* in a group by itself in the *Gammaproteobacteria*. Woese et al. (1985) consider *Leucothrix* to be a peripheral member of the *Gammaproteobacteria* core and specifically related to the enterics, vibrios, and oceanospirilla. Ludwig et al. (1995) define the position of *Leucothrix* as "somewhat unstable . . . This organism may represent the currently deepest branch of the gamma-subclass of the proteobac-

teria or a branch at or even below the level of gamma- and beta-subclass relatedness." In short, it is clear what *Leucothrix* is not, but not what it is. *Thiothrix*, the genus with which *Leucothrix* was once paired, is also classed within the *Gammaproteobacteria* along with other sulfur-oxidizing, gliding bacteria (*Thioploca*, *Beggiatoa*, and *Achromatium*). A head-to-head use of 16S rRNA probes for *Leucothrix* and *Thiothrix* in activated sludge has shown no relatedness, as judged by the lack of cross-reaction (Nielsen et al., 1998). The gliding characteristic is now recognized as not being unique to any group of bacteria; gliders are located widely across the spectrum of the proteobacteria. Relatedness of *Leucothrix* to the cyanobacteria has also been ruled out; Reichenbach et al. (1986), using 16S rRNA techniques, have demonstrated that *Leucothrix* is not a colorless cyanobacterium.

TAXONOMIC COMMENTS

To date, the only recognized species of *Leucothrix* is *Leucothrix mucor*, with nine marine isolates available in the American Type Culture Collection. While the genus has many phenotypic features in common with other organisms, its 16S and 23S rRNA profiles put it in a class by itself. The only characteristic in dispute is the ability of *Leucothrix* strains to oxidize sulfide. Brock (1992) cites work by Eimhjellen that demonstrated the ability of an isolate to oxidize sulfide, sulfur, and thiosulfate. The sulfide oxidation resulted in the internal deposition of sulfur granules, causing *Leucothrix* to resemble *Thiothrix*. Unfortunately, this culture is not available. One of the two organisms used by Woese et al. (1985) for the *Leucothrix* group in his classic 16S rRNA study is identified as a "*Thiomicrospira*-like isolate," a label that suggests sulfide-oxidizing ability. Overall, the *Leucothrix* genus would benefit from new experimental attention to better characterize this possible ability.

FURTHER READING

- Brock, T.D. 1992. The genus *Leucothrix*. In Balows, Trüper, Dworkin, Harder and Schleifer (Editors), *The Prokaryotes-A Handbook on the Biology of Bacteria: Ecophysiology, Isolation, Identification, Applications*, 2nd Ed., Springer-Verlag, New York. pp. 3247–3255.
- Ludwig, W., R. Rossello-Mora, R. Aznar, S. Klugbauer, S. Spring, K. Reetz, C. Beimfohr, E. Brockmann, G. Kirchhof, S. Dorn, M. Bachleitner, N. Klugbauer, N. Springer, D. Lane, R. Nietupsky, M. Weiznegger and K.-H. Schleifer. 1995. Comparative sequence analysis of 23S rRNA from *Proteobacteria*. *Syst. Appl. Microbiol.* 18: 164–188.
- Raj, H.D. 1977. *Leucothrix*. *Crit. Rev. Microbiol.* 5: 270–304.
- Woese, C.R., W.G. Weisburg, C.M. Hahn, B.J. Paster, L.B. Zablen, B.J. Lewis, T.J. Macke, W. Ludwig and E. Stackebrandt. 1985. The phylogeny of purple bacteria: the gamma subdivision. *Syst. Appl. Microbiol.* 6: 25–33.

List of species of the genus *Leucothrix*

1. *Leucothrix mucor* Oersted 1844, 44^{AL}

mu'cor. L. n. *mucor* mold; M.L. n. *mucor* a genus of molds.

The characteristics are the same as those described for the genus.

Occurs in marine habitats.

The mol% G + C of the DNA is: 46.9–51 (T_m).

Type strain: ATCC 25107, DSM 2157.

GenBank accession number (16S rRNA): X87277.

Genus V. Thiobacterium (ex Janke 1924) la Rivière and Kuenen 1989b, 496^{VP} (Effective publication: la Rivière and Kuenen 1989a, 1838)

J. GIJS KUENEN

Thi'o.bac.te'ri.um. Gr. n. *thios* sulfur; Gr. dim. n. *bakterion* a small rod; M.L. neut. n. *Thiobacterium* small sulfur rod.

Rod-shaped cells, each containing one or more sulfur globules. Cells embedded in **gelatinous masses**, which are spherical when free-floating or are dendroid when attached to a solid substrate. **Nonmotile**. No resting stages known. Has not been grown in pure culture.

Type species: Thiobacterium bovista (ex Janke 1924) la Rivière and Kuenen 1989b, 496 (Effective publication: la Rivière and Kuenen 1989a, 1838.)

FURTHER DESCRIPTIVE INFORMATION

Rod-shaped cells $0.4\text{--}1.5 \times 2.5\text{--}9 \mu\text{m}$ with up to nine sulfur inclusions. When these are present, cell masses are white in reflected light and black or bluish in transmitted light. Gram negative (Scheminzky et al., 1972). In the spherical colonies, the cells are embedded in the gelatinous walls of a bladder-like structure filled with water and up to 4 mm in diameter. Such colonies occur near the water surface and have the appearance of groups of puff balls of different sizes. Dendroid colonies show extensive branching and may reach 2–3 mm.

The only species in this genus, *T. bovista*, has been found in marine, brackish, and freshwater environments containing hy-

drogen sulfide and at temperatures of up to 45°C in thermal springs.

ENRICHMENT AND ISOLATION PROCEDURES

Molisch (1912) reported enrichment of the spherical colony type at the surface of Winogradsky columns prepared with decaying algae, mud, and seawater from the Gulf of Trieste. The dendroid form could not be enriched for but could, after being taken from nature, be kept alive for 3 months in the laboratory at 20°C in jars with water from the original habitat (Lackey and Lackey, 1961). Further details are given by la Rivière and Schmidt (1981). For micrographs, see articles by Lackey and Lackey (1961) and Scheminzky et al. (1972). No pure cultures are known.

TAXONOMIC COMMENTS

Although this genus and its type species do not appear in the Approved Lists of Bacterial Names, maintaining their descriptions in the present edition of the *Manual* and reviving the name appears justified because reports of *Thiobacterium* have occasionally been published (Scheminzky et al., 1972; Caldwell and Caldwell, 1974; Naganuma and Seki, 1991; Hedoin et al., 1996).

List of species of the genus *Thiobacterium*

1. ***Thiobacterium bovista*** (ex Janke 1924) la Rivière and Kuenen 1989b, 496^{VP} (Effective publication: la Rivière and Kuenen 1989a, 1838.)
bo.vis'ta. M.L. fem. n. *bovista* puff ball.

For a description, see that of the genus.
The mol% G + C of the DNA is: unknown.
Type strain: no culture available.

Genus VI. Thiomargarita Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999b, 1325^{VP} (Effective publication: Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999a, 493)

HEIDE N. SCHULZ AND BO BARKER JØRGENSEN

Thi'o.mar.ga.ri'ta. Gr. n. *thion* (theion) sulfur; L. n. *margarita* pearl; M.L. fem. n. *Thiomargarita* sulfur pearl.

Very large **spherical cells held together** like a string of pearls by a **common gelatinous mucus**. The cells are not joined to each other, which is the main difference to the closely related filamentous genera, *Thioploca* and *Beggiatoa*. **Numerous sulfur inclusions** occur in a thin outer layer of cytoplasm. The cells appear hollow due to a **large central vacuole in which nitrate is stored**. Most strings of cells are linear and typically contain 20–60 cells, but some are branched or coiled. Both single cells and strings of more than 100 cells occur. **Nonmotile**. Gram negative. **Oxygen tolerant**. *Thiomargarita* appear to have an autotrophic or mixotrophic, sulfide-oxidizing, nitrate-reducing metabolism, compa-

able to the large marine forms of *Thioploca* and *Beggiatoa*, which are its closest relatives according to partial 16S rDNA sequence analysis.

Type species: Thiomargarita namibiensis Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999b, 1325 (Effective publication: Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999a, 493.)

FURTHER DESCRIPTIVE INFORMATION

The cell diameters of the only described species, *Thiomargarita namibiensis*, vary between 50 and 750 μm , with the majority of

cells being between 100 and 300 μm in diameter. Although most cells are spherical, the shape of the cell is variable. Occasionally, more irregular or barrel-shaped cells (Fig. BXII.75C) occur. The mucus sheath surrounds the single cells completely, which is apparent when cells disintegrate and leave empty mucus pockets (Fig. BXII.75B, left side). The strings of cells seem to grow by lateral division (Fig. BXII.75B, cells in division). Bacteria, including filamentous forms resembling *Thiothrix* and *Desulfonema* (Fig. BXII.75A), are frequently found living attached to the outside of the mucus.

16S rRNA phylogeny and systematics Partial 16S rDNA sequence analysis shows that *Thiomargarita* belongs to the *Gammaproteobacteria* and is the closest relative to the vacuolated marine *Thioploca* and *Beggiatoa* species. Details of the phylogeny are discussed in the chapter on the genus *Thioploca*.

Morphology and cell structure Electron micrographs have shown that, similar to the nitrate-storing *Thioploca* and *Beggiatoa* species, the cytoplasm of *Thiomargarita* is restricted to a thin outer

layer of 1–2 μm thickness. About 98% of the cell is comprised of a large liquid vacuole. Apart from the sulfur inclusions along the periphery of the cells (Fig. BXII.75A, black arrow), numerous smaller inclusions which have not been identified can be observed (Fig. BXII.75A, white arrow).

Metabolism *Thiomargarita* stores nitrate at 0.1–0.8 M concentration and elemental sulfur at a concentration equivalent to 0.4–1.7 M. Growth of *Thiomargarita* can be induced by adding nitrate and sulfide to the natural sediment. Therefore, an autotrophic or mixotrophic metabolism is assumed, by which sulfide is oxidized with nitrate, comparable to the metabolism of the large marine *Thioploca* and *Beggiatoa* species. In contrast to these, however, *Thiomargarita* tolerates prolonged exposure to oxygen up to air saturation, as well as to high sulfide concentrations (~ 0.8 mM) (Schulz and de Beer, 2002). *Thiomargarita* has a remarkable capability to survive starvation. Sediment samples maintained at 5°C without addition of sulfide or nitrate contained intact cells after more than two years. Without addition of nitrate or sulfide, no dividing cells were observed and cell numbers decreased.

Ecology The only known species, *Thiomargarita namibiensis*, was found in shelf sediments off the Namibian coast. The upwelling area off Namibia is characterized by high plankton productivity, which leads to sediments rich in organic material and with high sulfate reduction rates. In contrast to the upwelling area off the Pacific coast of South America, where the related *Thioploca* species thrive, the upwelling intensity off Namibia fluctuates more. *Thiomargarita* occurs in high density (about 50 g/m²) in the area of Walvis Bay. The population seems to be associated with an unusual type of sediment, a fluid, sulfidic diatom ooze. The highest density of *Thiomargarita* was observed directly at the sediment surface, declining exponentially with depth. As *Thiomargarita* cells are not motile, but are highly resistant towards nitrate starvation and high oxygen concentrations, it is assumed that they take up nitrate only occasionally, when nitrate-rich bottom water is introduced into the sediment, possibly co-occurring with elevated oxygen concentrations. Between such events, *Thiomargarita* is able to survive high concentrations of sulfide accumulating in the sediment. Due to its large volume (50–750 μm in diameter), which is mainly due to a central vacuole with nitrate concentrations up to 800 mM, *Thiomargarita* has a large storage capacity for its electron acceptor, nitrate. Sulfide, produced by sulfate-reducing bacteria, is permanently available as electron donor in the organic-rich sediment. Thus, *Thiomargarita* can continuously oxidize sulfide, although the electron acceptor, nitrate, is only intermittently available.

MAINTENANCE PROCEDURES

Thiomargarita may be grown in a 2–3 cm thick layer of natural sediment on agar containing 0.5 mM sulfide and covered by oxic or anoxic seawater containing 1 mM nitrate. A frequent resuspension of the sediment enhances growth, whereas addition of organic substrates such as acetate or glucose has no clear effect.

DIFFERENTIATION OF THE GENUS *THIOMARGARITA* FROM OTHER GENERA

In contrast to the closely related genera *Thioploca* and *Beggiatoa*, *Thiomargarita* forms filaments, but does not form trichomes. The extremely large, nonmotile cells are surrounded by a mucus that keeps them attached to each other in a string. The ability to survive under atmospheric oxygen concentration is unique among the vacuolated, nitrate-storing sulfur *Bacteria*.

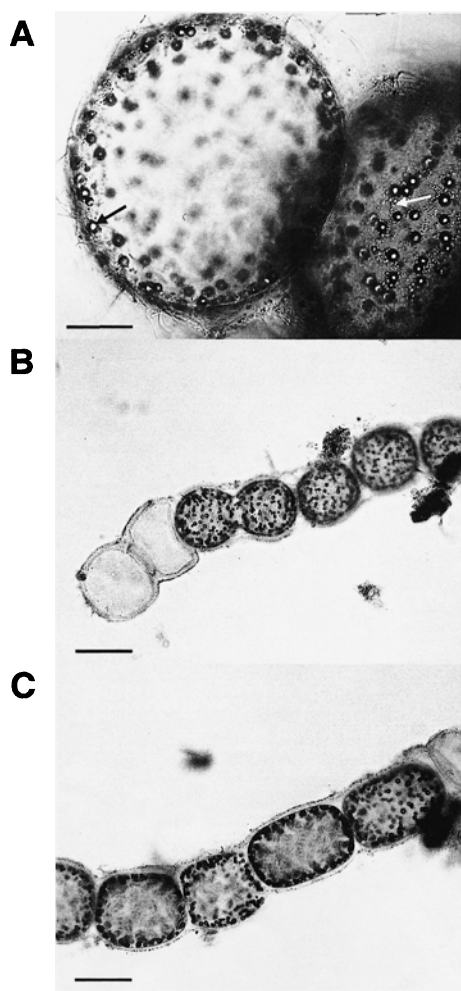


FIGURE BXII.75. *Thiomargarita namibiensis* as it appears under light microscopy. (A) The terminal cell of a string. The black arrow points to a sulfur inclusion, the white arrow to a smaller unidentified type of inclusion. The inclusions are restricted to the periphery of the cell. Outside the cell there are various filamentous bacteria, resembling *Desulfonema*, attached to the mucus sheath. Bar = 50 μm . (B) To the left, two empty sheaths; in the middle, a dividing cell. Bar = 100 μm . (C) A chain of barrel-shaped *Thiomargarita* cells. Bar = 50 μm .

List of species of the genus *Thiomargarita*

1. ***Thiomargarita namibiensis*** Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999b, 1325^{VP} (Effective publication: Schulz, Brinkhoff, Ferdelman, Hernández Mariné, Teske and Jørgensen 1999a, 493.)
na.mi.bi.en'sis. M.L. gen. n. *namibiensis* of Namibia.

For a description, see that of the genus.

The mol% G + C of the DNA is: unknown.

Type strain: No pure culture has been isolated.

Genus VII. *Thioploca* Lauterborn 1907, 242^{AL}

BO BARKER JØRGENSEN, ANDREAS TESKE AND AZEEM AHMAD

Thi.o.plo'ca. Gr. neut. n. *thion*, *theion* sulfur; Gr. fem. n. *ploke* a braid, a twist; M.L. fem. n. *Thioploca* sulfur braid.

Flexible, uniseriate, colorless filaments made up of numerous cells and enclosed by a **common gelatinous sheath**. Gram negative. Cells are disk-shaped or cylindrical, generally harbor **sulfur inclusions** in their cytoplasm, and are separated by distinct cross-walls. Cells of large marine *Thioploca* species **appear hollow due to central vacuoles in which nitrate is stored**. Individual filaments are of uniform diameter, often with **tapered terminal cells**. Filaments fall into distinct classes of **different diameters, which represent different species**. Filaments show **independent gliding movement** within a sheath. They can emerge from the end or from breaks in the sheath and may be found outside sheaths. **Chemotactic**. May be attracted by nitrate but avoid high oxygen and sulfide concentrations. *Thioploca* sheaths are mostly vertically orientated to allow the filaments to move between spatially separated pools of electron donor (sulfide) and acceptor (nitrate, possibly oxygen). The sheath surrounding several filaments is the main feature distinguishing the genus *Thioploca* from the similar, closely related genus *Beggiatoa*. Not in pure culture. Large, marine *Thioploca* species have an autotrophic or mixotrophic, sulfide-oxidizing, nitrate-reducing metabolism. By 16S rRNA, *Thioploca* is a member of the *Gammaproteobacteria* and is the closest relative of the genera *Beggiatoa* and *Thiomargarita*.

Type species: *Thioploca schmidlei* Lauterborn 1907, 242.

FURTHER DESCRIPTIVE INFORMATION

Thioploca filaments fall into distinct diameter classes, which are the traditional basis for species designation (Table BXII.γ.41, Fig. BXII.γ.76). At least two different groups of *Thioploca* species are apparent, the smaller freshwater and brackish water species, of which *T. ingrica* is the best-known representative, and the large marine species, which include *T. araucae* and *T. chileae*. Diameters of accepted species range from 0.8–43 μm, but occasionally marine sheathed *Thioploca* of up to 125 μm diameter have been observed (H.N. Schulz and B.B. Jørgensen, unpublished observations). The large, marine filaments are easily visible with the unaided eye.

Thioploca typically grow in bundles surrounded by a common sheath (Figs. BXII.γ.77 and BXII.γ.78A). The number of fila-

ments in a sheath ranges from a few to tens but may also reach a hundred. When bundles are observed (and slightly squeezed) in a microscope preparation, intertwining filaments may have the appearance of a braid, giving rise to the name *Thioploca* (Fig. BXII.γ.78C). Each bundle presumably represents a clone, and neighboring bundles may, accordingly, show small but significant differences in mean filament diameter (Schulz et al., 1996). Inspection of many sheaths of the marine *Thioploca* has revealed that ~20% of the sheaths have mixed bundles of two or three of the species *T. araucae*, *T. chileae*, and "*T. marina*". *Thioploca* filaments may leave their own sheath under environmental stress, such as sulfide accumulation in the sediment. They can also survive as free-living filaments and may perhaps subsequently enter another sheath or form a common sheath with another species. Such mixed bundles also occur in the freshwater species; these gave rise to the name "*T. mixta*" (Koppe, 1924), which is no longer in use. The transparent sheath of young and growing bundles is thin and tough but it may become very wide and loose around older and senescent bundles.

16S rRNA phylogeny and systematics Based on near-complete 16S rRNA sequences, filamentous sulfur-oxidizing bacteria of the genera *Thioploca* and *Beggiatoa* form a monophyletic, highly diversified cluster within the *Gammaproteobacteria* that currently consists of three distinct groups (Teske et al., 1995, 1999; Ahmad et al., 1999): *Thioploca* and large, marine *Beggiatoa*; heterotrophic, freshwater *Beggiatoa*; and autotrophic, marine *Beggiatoa* (Tree 1 of Fig. BXII.γ.79). The clade of marine, vacuolated, nitrate-accumulating *Thioploca* spp. (*T. araucae* and *T. chileae*), large, marine *Beggiatoa* spp., and *Thioploca ingrica* has 82–100% bootstrap support for near-complete sequences. For partial sequences (Tree 2 of Fig. BXII.γ.79), this clade includes the species *T. araucae* and *T. chileae* as well as the unicellular *Thiomargarita namibiensis* (Schulz et al., 1999a; Schulz and Jørgensen (2001)). The heterotrophic, freshwater *Beggiatoa* and the autotrophic, marine *Beggiatoa* clades do not contain *Thioploca* species (Tree 1 of Fig. BXII.γ.79). The heterotrophic, freshwater *Beggiatoa* clade contains a tight cluster of the *Beggiatoa alba* type strain B18LD (Mezzino et al., 1984) and closely related strains, always supported by

TABLE BXII.γ.41. Differential characteristics of the species of the genus *Thioploca*

Characteristic	1. <i>T. schmidlei</i>	2. <i>T. araucae</i>	3. <i>T. chileae</i>	4. <i>T. ingrica</i>	5. " <i>T. marina</i> "	6. " <i>T. minima</i> "
Size (μm)	5–9	30–43	12–20	2.0–4.5	2.5–5.0	0.8–1.5
Freshwater	+	–	–	+	–	+
Marine	–	+	+	–	+	–

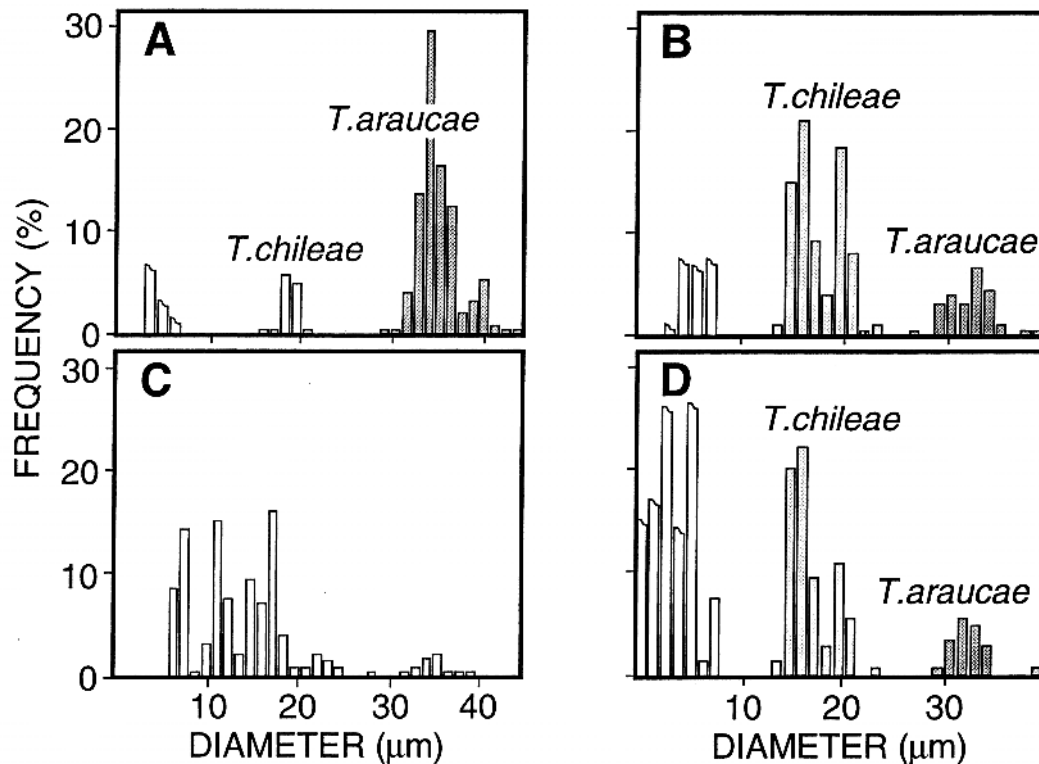


FIGURE BXII.γ.76. The presence of a common sheath and the diameters of trichomes are used to define the genus and species of *Thioploca*. The graphs illustrate the problems of this classification with regard to two marine populations from the surface layer of shelf sediments off the coast of Chile. Several hundred trichomes were randomly picked for each diagram, and the relative frequency distribution of their diameters is shown at $\sim 1 \mu\text{m}$ resolution. Frames (A) and (C) are from a station at 40 m water depth and frames (B) and (D) from a station at 80 m water depth. Frames (A) and (B) represent trichomes picked from bundles surrounded by a common sheath. Frames (C) and (D) represent trichomes picked from the abundant free trichomes without visible sheaths. The two species, *T. chileae* and *T. araucae*, are clearly distinguished, as are the filaments of a narrower species which partially corresponds to *T. marina* (the narrow species occurred in larger numbers than indicated). At the shallow station (A + C), many free trichomes were of diameters not represented by the neighboring thioplocas in sheaths (they should consequently be called *Beggiatoa* spp.). At the deep station (B + D), there was a good correspondence between diameter distributions of sheathed and free trichomes, thus indicating that the latter were thioplocas outside sheaths. As indicated by the available 16S rRNA sequence database (see Fig. BXII.γ.79) the vacuolated marine thioplocas and beggiatoas are phylogenetically intertwined and, therefore, a distinction based on morphological characteristics (sheath, diameter) is probably not possible or meaningful. (Data from Schulz et al., 1996).

100% bootstrap. The sequence data demonstrate that the physiological and morphological similarities within and among these groups result from evolutionary relatedness.

Within the *Thioploca*/*Beggiatoa* clade, representatives of both genera are phylogenetically intertwined. The current distinction between the genera *Thioploca* and *Beggiatoa* does not follow phylogenetic lines but is instead determined by a morphological characteristic, the formation of a sheath around a filament bundle. This point is illustrated by a large, marine *Beggiatoa* from the Bay of Concepcion on the Chilean shelf that resembles *T. araucae* in filament diameter, 16S rRNA sequence, nitrate content, and habitat, but that occurs in individual filaments instead of sheathed filament bundles (Teske et al., 1999). The large *Beggiatoa* spp. from Monterey Canyon with 60–80 μm filament diameter (Ahmad et al., 1999) could represent a similar case, since ensheathed *Thioploca* with the same filament diameters are

also known from this habitat (K. Buck, personal communication). Representing a possible challenge to the generality of this interpretation are massive assemblages of vacuolated *Beggiatoa* collected from the Guaymas Basin vents (Nelson et al., 1989b) that show no indication of forming sheathed filament bundles (A. Teske, unpublished data; D. C. Nelson, unpublished data). Parallel observations come from extensive *Beggiatoa* mats in the Gulf of Mexico (D.C. Nelson, unpublished data).

The 16S rRNA data support the classification of *T. araucae* and *T. chileae*, previously defined only by filament diameter, as not mere morphotypes, but separate species. However, filament diameter classes do not always correspond to genospecies or metabolic types; they may instead cover up unexpected genetic and metabolic diversity, as illustrated by the two clades of heterotrophic, freshwater *Beggiatoa* and marine, autotrophic *Beggiatoa* (Tree 1 of Fig. BXII.γ.79). With the exception of strain MS-81-

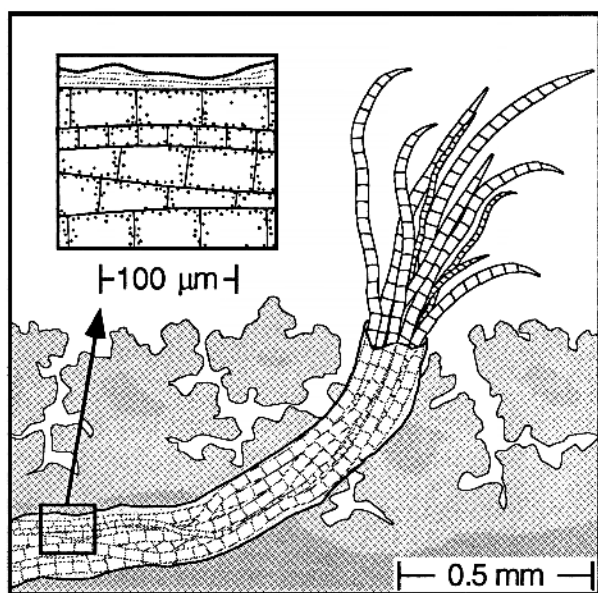


FIGURE BXII.γ.77. Schematic presentation of a bundle of marine thioplocas extending from their common sheath within the sediment and up into the overlying seawater. The insert shows the light-microscopic appearance of the trichomes, with crosswalls and fine sulfur globules along the cell periphery. (Drawing by Heide N. Schulz).

6, these *Beggiatoa* strains all share filament diameters of close to 2 μm . The genetic coherence of morphologically defined, widely distributed *Thioploca* species, for example *T. ingrica*, which is represented only by 16S rRNA sequences of specimens from a Danish brackish water habitat, is therefore not proven.

The thioplocas show some phenotypic similarities with cyanobacteria such as *Microcoleus*, both in morphology and in the formation of sheaths around bundles of filaments. The phylogenetic data clearly show, however, that the filamentous sulfide-oxidizing bacteria are unrelated to the cyanobacteria, which are not members of the *Proteobacteria*, but rather a separate monophyletic bacterial group.

Morphology and cell structure Each filament consists of a single row of cylindrical or barrel-shaped cells separated by septa. The septa may be difficult to observe microscopically in filaments rich in light-diffracting sulfur globules, but in those cases, their position is often indicated by the distribution pattern of sulfur globules (Fig. BXII.γ.78, D and F). For the smaller species with filament widths of $<10 \mu\text{m}$, the cell height is greater than the diameter, whereas the large marine species have more disk-shaped cells. Interestingly, the two diameter classes of *T. chileae* and *T. araucae* have been found to occur as two distinct populations of morphotypes: the normal type with a height-to-diameter ratio of 0.5:1.5 and a form with more rounded and flat cells having a height-to-diameter ratio of 0.3:1–0.5:1 (Schulz et al., 2000). The large, undescribed, marine thioplocas with diameters $>50 \mu\text{m}$ all belong to this short-cell type. Each trichome contains a hundred to several thousand cells. The large, marine species are generally 1–2 cm long and may reach $>5 \text{ cm}$ in length (Huetel et al., 1996).

There is a distinct difference in cell structure between the small, freshwater species and the large, marine species of *Thioploca*. Each group shows many similarities to the small and large

Beggiatoa, respectively (Maier and Murray, 1965; Strohl et al., 1982; Maier et al., 1990). In both groups, the 90–130 nm thick cell wall has a complex, four-layered structure, of which only the innermost layer and the cytoplasmic membrane continue across the septum wall separating the cells. Intracytoplasmic membranes and numerous cell inclusions form complex structures, which are apparent from TEM micrographs, and which have functions related to transport and storage. Most of these inclusions are situated in intrusions of the cytoplasmic membrane and are, therefore, extracytoplasmic (Maier et al., 1990). Most prominent are globules of S^0 , which are also visible in the light microscope. Membrane-bound vacuoles, which in *Thioploca* contain electron-dense material, are distributed throughout the cytoplasm in *T. ingrica* as well as in the 4 μm wide *Beggiatoa* (Morita and Stave, 1963; Maier and Murray, 1965).

In *T. araucae* and *T. chileae*, the cytoplasm, with its sulfur inclusions, is confined to a 0.5–1.0 μm thick layer along the periphery of each cell (Maier et al., 1990). The whole inner cell volume is filled by a single liquid vacuole surrounded by a cytoplasmic membrane without apparent connection to the cell surface (Fig. BXII.γ.78B). Such a vacuole, which comprises $>90\%$ of the cell volume, is also found in the large, marine *Beggiatoa* spp. of $>15 \mu\text{m}$ diameter (Lackey et al., 1965; Nelson et al., 1989b). It serves as a reservoir for nitrate, which the organisms use as an electron acceptor when oxidizing sulfide within the sediment (Fossing et al., 1995; McHatton et al., 1996; Zopf et al., 2001). The peripheral distribution of cytoplasm, furthermore, provides short transport distances to the external medium and, thus, counteracts the diffusion limitation of these very large procaryotes.

Metabolism Since none of the thioplocas has been isolated in pure culture, information on their physiology is based mostly on observations of whole communities and on experiments with whole bundles of filaments. However, since the organisms are large and conspicuous, it has also been possible to physically isolate individual filaments and to observe their metabolic activity, e.g., by feeding them radiolabeled substrates and subsequently performing microautoradiography.

Thioploca oxidizes H_2S to sulfate and stores S^0 as an intermediate oxidation product in the form of globules in the cells. The S^0 serves as a transportable energy source that makes the organism independent of a continuous source of H_2S . All large, marine thioplocas and *Beggiatoa* that have been examined so far contain high (100–500 mM) intracellular concentrations of nitrate, an accumulation four orders of magnitude over nitrate concentrations in the ambient sea water (Fossing et al., 1995; McHatton et al., 1996). The nitrate is stored in the large central vacuole, which functions as an anaerobic “lung” as the organisms move around in the sulfidic sediment. *Thioploca* uses nitrate as a respiratory electron acceptor for sulfide oxidation and reduces the nitrate principally to ammonium, as does *Beggiatoa* (Vargas and Strohl, 1985b; McHatton et al., 1996; Otte et al., 1999). It is unknown to what extent the smaller, freshwater or brackish water *Thioploca* species utilize nitrate as electron acceptor.

T. araucae and *T. chileae* are able to fix CO_2 and may be able to grow autotrophically but probably mostly grow mixotrophically, similar to *Beggiatoa* and *Thiothrix* (Nelson and Jannasch, 1983; Strohl and Schmidt, 1984; Otte et al., 1999). They can incorporate acetate, mixed amino acids, glycine, glucose, and thymidine in the presence of sulfide, as shown by autoradiog-

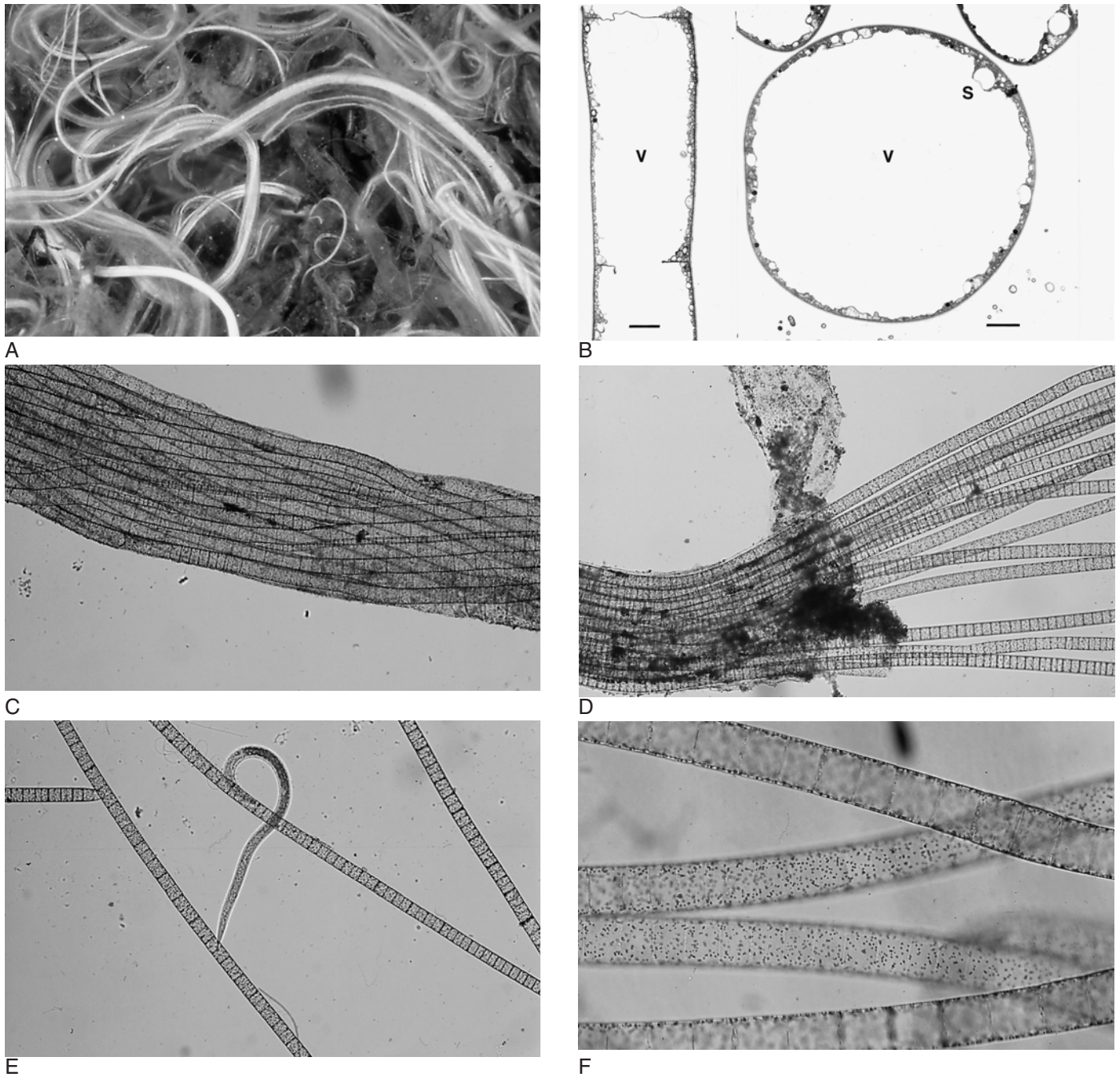


FIGURE BXII.γ.78. Marine thioplocas from the shelf off the coast of central Chile. (A) Thiplocas in gelatinous sheaths which were sieved out of the sediment. The frame is 8 mm wide. (B) Transmission electron micrographs of *Thioploca chileae* in cross-section (left) and longitudinal section (right). The cells appear empty due to the large central vacuole. Sulfur globules are stored in the thin peripheral layer of cytoplasm. The filaments are 20 μm in diameter. (C) *Thioploca* bundle densely packed in a sheath (not visible). The bundle has the appearance of a braid where the filaments cross over, hence the name *Thioploca*, meaning sulfur braid. (D) Bundle of *T. araucae* filaments extending out of their sheath. (E) Trichomes of *T. araucae* together with a marine nematode, illustrating the large size of these multicellular prokaryotes. (F) Light micrograph of *T. araucae* trichomes showing the dense sulfur globules as fine dots. In the central trichomes, the focus is on the surface, whereas in the outer trichomes, the focus is in the center, thus showing the hollow appearance with sulfur globules only along the periphery. (Part B reproduced with permission from S. Maier et al., Canadian Journal of Microbiology, 36: 438–448, 1990, ©National Research Council of Canada; A courtesy of M. Hüttel; C–F courtesy of B.B. Jørgensen.)

raphy (Maier and Gallardo, 1984a). Methylophily has also been suggested (Morita et al., 1981), but this theory appears to be incorrect since *Thioploca* cannot assimilate methane or methanol.

Furthermore, in the marine habitats of *Thioploca*, methane is produced in insufficient amounts to play a role as a carbon source (Ferdelman et al., 1997).

Ecology *Thioploca* occurs in freshwater, brackish, and marine sediments, mostly under oxygen-depleted water. The organisms seem to prefer low sulfide concentrations, although they are also found in sulfide-rich sediments along the Peruvian coast (Henrichs and Farrington, 1984). Classical localities of the freshwater species are lakes in central and northern Europe (Lauterborn, 1907; Wislouch, 1912; Koppe, 1924; Kolkwitz, 1955), but *Thioploca* is also reported from large lakes in North America, central Russia, and Japan (Maier and Preissner, 1979; Namsaraev et al., 1994; Nishino et al., 1998).

The marine species form dense and extensive communities in areas of coastal upwelling (Jørgensen and Gallardo, 1999). The largest *Thioploca* community is found associated with the Humboldt current on the Pacific continental shelf of South America (Gallardo, 1977). Biomasses here reach 120 g fresh weight/m², or up to 1 kg wet weight/m² when sheaths are included (Schulz et al., 1996). There is a seasonal variation in the population size, with the highest biomasses occurring during the summer, when the sedimentation of organic matter is highest and the overlying water is strongly depleted in oxygen (Schulz et al., 2000). The population size is strongly depressed during "El Niño," when upwelling is reduced and the overlying water remains oxygenated (V.A. Gallardo and H.N. Schulz, personal communication). Other reported marine habitats include the continental shelf underlying the Benguela current along the South Atlantic coast of Africa (Gallardo et al., 1998; H.N. Schulz, personal communication), the monsoon-driven upwelling area of the northwestern Arabian Sea (Levin et al., 1997), and hydrothermal vent sites in the eastern Mediterranean Sea (Dando and Hooper, 1996).

The upwelling habitats are characterized by very high plankton productivities, high nitrate and low oxygen concentrations in the bottom water, high sedimentation rates of organic matter, and high rates of sulfide production in the sediment (Thamdrup and Canfield, 1996; Ferdelman et al., 1997). The bacteria contribute significantly to the oxidation of sulfide in these sediments and are able to maintain very low H₂S concentrations. The winding sheaths have a mostly vertical position in the muddy sediment and may reach 2–5 cm depth for the freshwater species and 10–15 cm for the large marine species (Schulz et al., 1996; Nishino et al., 1998). The sheaths form vertical tunnels through which the filaments can migrate between the sediment-water interface and the bulk sediment (Fig. BXII.γ.80). The filaments glide with a speed of 1–3 μm/s and may thus move ~10 cm/day. Although they are not the only sulfide oxidizers in the sediment, their ability to transport intracellular nitrate reserves down into the sulfide production zone allows them to compete efficiently with other sulfide-oxidizing bacteria that require overlapping or frequently fluctuating gradients of sulfide and an appropriate electron acceptor (Jørgensen and Revsbech, 1983; Nelson et al., 1986a; Gundersen et al., 1992). Dense populations of *Thioploca* may form loose, gelatinous mats of 1–2 cm thickness at the sediment surface. Inhabited sheaths have been found to be densely covered by filamentous sulfate reducing bacteria of the genus *Desulfonema* (M. Fukui et al., 1999), thereby providing a short diffusion path between the H₂S producers and consumers.

MAINTENANCE PROCEDURES

Although thioplocas have not been isolated in pure culture, they may be maintained in undisturbed sediment in the laboratory for months or even years. Maier (1984) described the following procedures for freshwater *Thioploca*: They may be maintained in

jars overlaid with tap water at 8–20°C in the dark. At approximately yearly intervals, a few stems of extracted grass (Scotten and Stokes, 1962) may be stuck into the sediment; *Thioploca* often colonizes these stems. Alternatively, 0.2–0.3 g pulverized extracted hay is autoclaved in 60 ml tap water in 125-ml Erlenmeyer flasks and inoculated with 4–10 ml sediment. After a month of undisturbed incubation at room temperature, *Thioploca* bundles are added and incubation continues for many weeks, with intermittent inspection.

Attempts to enrich marine *Thioploca* have met little success. They may be maintained for months in undisturbed cores sampled from the natural populations and kept near the *in situ* temperature of 13°C in a basin of anoxic seawater with nitrate added (H.N. Schulz, personal communication).

DIFFERENTIATION OF THE GENUS *THIOPLOCA* FROM OTHER GENERA

Due to their conspicuous morphology and generally large size, members of the genus *Thioploca* can be easily distinguished from most other genera. However, they show a strong similarity and overlap in characteristics with *Beggiatoa* with respect to morphology, ultrastructure, physiology, ecology, and phylogeny. Tapered filament ends are characteristic of *Thioploca*, but these may also have rounded terminal cells with no tapering. *Beggiatoa*s occasionally have a tapered terminal cell, though tapering is less than that for *Thioploca*. The best criterion of differentiation is the formation of bundles surrounded by a common sheath, which has so far been attributed only to the genus *Thioploca*. Sheath-inhabiting thioplocas tend to have a more vertical orientation in sediments than do *Beggiatoa*s, which often form whitish films covering solid–water interfaces. However, the reverse orientation may also be found for both species. When triggered by environmental stress, *Thioploca* filaments often glide partly or completely out of their sheaths, and they may perhaps even establish communities without apparent sheath formation, in which case they would be recognized as *Beggiatoa* (Schulz et al., 1996). *Thioploca* can be clearly differentiated from *Thiothrix* by their gliding movement and by the lack of a terminal holdfast or rosette formation. The closely related *Thiomargarita namibiensis* has separated, nonmotile, spherical cells.

TAXONOMIC COMMENTS

Future changes in classification of *Thioploca* and *Beggiatoa* are likely. The range of strains over which the genus designation *Beggiatoa* is used is overly broad. More importantly, the differentiation between *Thioploca* and *Beggiatoa* is currently based on the formation of a common sheath surrounding filament bundles, a characteristic that might vary in response to environmental conditions. In the absence of pure cultures, it may be impossible to prove or disprove whether any natural population of vacuolated *Beggiatoa* will form sheath bundles in some specific environment. The clade (Tree 2 of Fig. BXII.γ.79) comprised of three *Thioploca* strains, two *Beggiatoa* strains, and a *Thiomargarita* strain is united by the possession of a large central vacuole. This feature currently appears to be the best morphological candidate to replace sheath formation as a marker in a revised taxonomy of the group *Beggiatoa*–*Thioploca*. This marker, in addition to being consistent with 16S rRNA phylogeny, appears to be universally connected to intracellular nitrate accumulation, presumably in the vacuole, for nitrate respiration enabling sustained anaerobic metabolism. This generalization regarding metabolic capacity remains to be confirmed for *T. ingrica*. A future revision of the

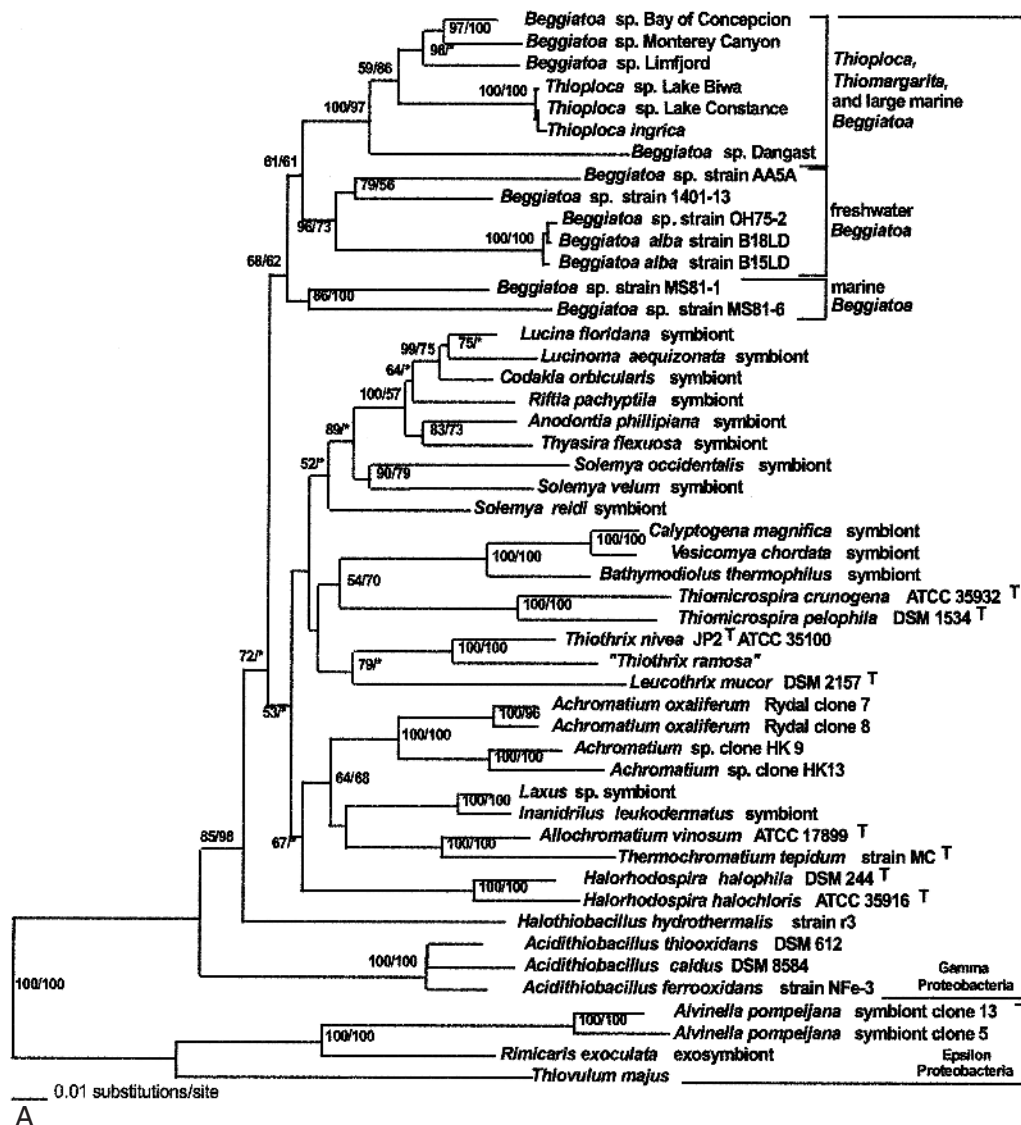


FIGURE BXII.7.79. Phylogenetic relationships of *Thioploca*, *Beggiatoa*, and *Thiomargarita*, and representative sulfur-oxidizing bacteria of the Gammaproteobacteria. The trees are based on nearly complete 16S rDNA sequences (*E. coli* positions 28–1487), (A) and on partial sequences (*E. coli* positions 341–941), including those of marine *Thioploca* species and *Thiomargarita namibiensis*, for which only partial sequences are available. (B) The trees were rooted

genus *Thioploca*, based on the vacuolated, nitrate-respiring phenotype and corresponding 16S rRNA clade, might include these gliding filaments regardless of whether they occur in sheathed bundles. For now, further investigations of natural populations of vacuolated *Beggiatoa* and *Thioploca*, as well as *Thiomargarita*, are required to check the consistency of this concept.

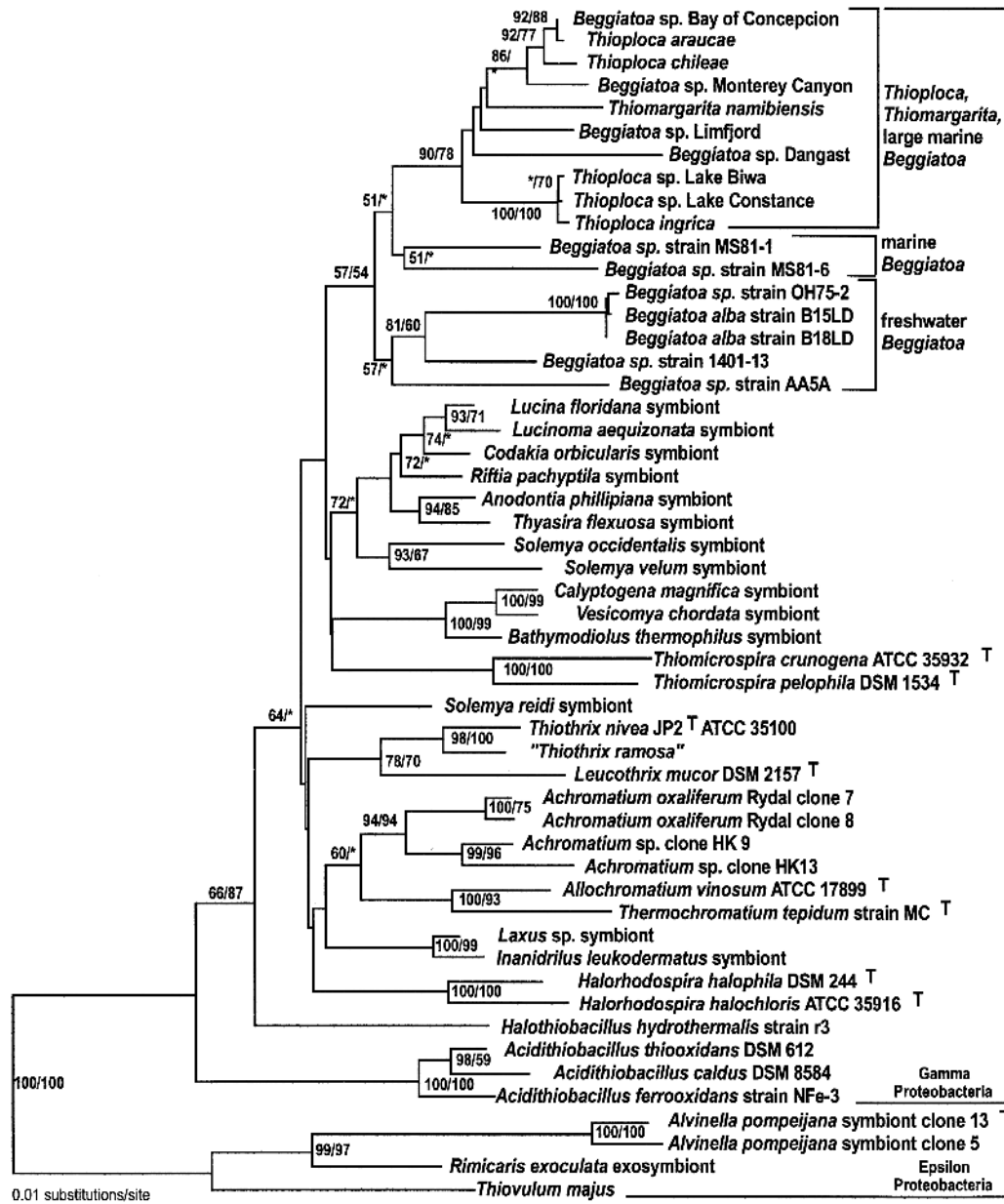
ACKNOWLEDGMENTS

Bo Barker Jørgensen was supported by the Max Planck Society and the Fonds der Chemischen Industrie. Andreas Teske was supported by a DFG postdoctoral stipend and a subsequent WHOI postdoctoral fellowship.

Azeem Ahmad, as well as a portion of the phylogenetic analyses reported here, was supported by National Science Foundation award IBN-9513962 to Douglas C. Nelson. We thank Douglas C. Nelson for important contributions to the phylogeny of *Thioploca*–*Beggiatoa* and the following colleagues for illustrative material: Markus Hüttel, Siegfried Maier, Heide N. Schulz, and Horst Völker.

FURTHER READING

Jørgensen, B.B. and V.A. Gallardo. 1999. *Thioploca* spp.: filamentous sulfur bacteria with nitrate vacuoles. FEMS Microbiol. Ecol. 28: 301–313.
Schulz, H.N. and B.B. Jørgensen. 2001. Big bacteria. Ann. Rev. Microbiol. 55: 105–137.



with *Thiovulum majus* of the *Epsilonproteobacteria* as the outgroup. Bootstrap values minimum evolution/parsimony are indicated at the nodes with more than 50% bootstrap support. Bar = 0.01 Jukes-Cantor substitutions per nucleotide.

List of species of the genus *Thioploca*

1. *Thioploca schmidlei* Lauterborn 1907, 242^{AL}
schmid' le.i. M.L. gen. n. *schmidlei* of Schmidle.

Identified from sediments of fresh and brackish water localities in Europe and from Lake Baikal, Russia. Originally found in Lake Constance, southern Germany.

The mol% G + C of the DNA is: not available.
Type strain: none isolated.

2. *Thioploca araucae* Maier and Gallardo 1984b, 417^{VP}
a.rau' cae. M.L. gen. n. *araucae* of Arauco in Central Chile.
Identified from oxygen-poor upwelling areas of the

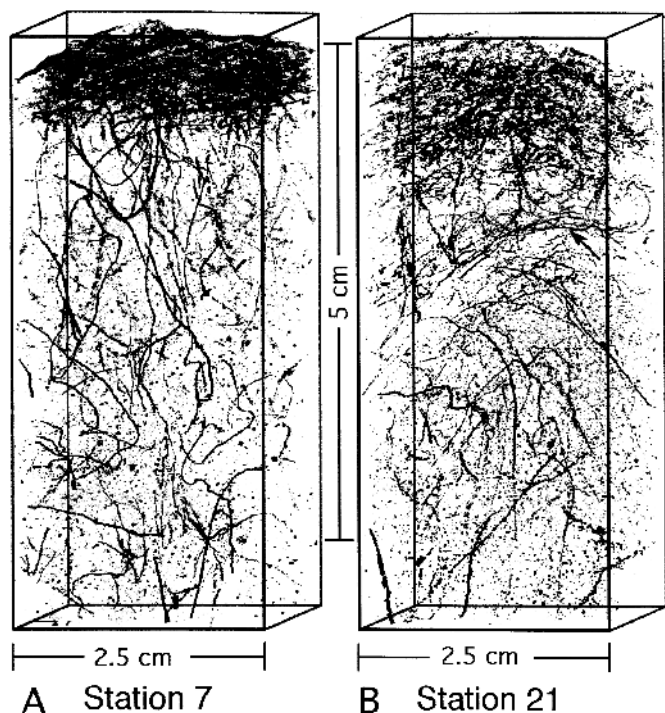


FIGURE BXII.7.80. Three-dimensional reconstruction of marine *Thioploca* sheaths (black lines) in blocks of sediment from the Chilean coast at 40 m (A) and 100 m (B) water depth. The sediment blocks of $5 \times 2.5 \times 1$ cm were rapidly frozen, cut vertically at 100 μ m increments with a cryomicrotome, and photographed in polarized light. The 3-D image was developed from 100 sequential photographs. At the sediment surface, a grayish color shows a dense mat of more horizontally oriented sheaths, plus many free filaments of either *Beggiatoa* or *Thioploca* outside their sheaths. Below the mat, dark strands of *Thioploca* sheaths extend down through the sediment with a mostly vertical orientation. (Reproduced with permission from H.N. Schulz et al., Applied and Environmental Microbiology, 62: 1855–1862, 1996, ©American Society for Microbiology.)

ocean, e.g., the Pacific continental shelf of South America, the Atlantic coast of southern Africa, and the northwest Arabian Sea.

The mol% G + C of the DNA is: not available.

Type strain: none isolated.

3. ***Thioploca chileae*** Maier and Gallardo 1984b, 417^{VP}
chi' le' ae. M.L. gen. n. *chileae* of Chile.

Identified from oxygen-poor upwelling areas of the ocean, e.g., the Pacific continental shelf of South America, the Atlantic coast of southern Africa, and the northwest Arabian Sea.

The mol% G + C of the DNA is: not available.

Type strain: none isolated.

4. ***Thioploca ingrica*** Maier 1984, 344^{VP}
in' gri. ca. M.L. adj. *ingrica* pertaining to Ingria, ancient district of Leningrad, Russia.

Identified from sediments of fresh and brackish water localities in central Europe and from Lake Erie, USA, and Lake Biwa, Japan.

The mol% G + C of the DNA is: not available.

Type strain: none isolated.

GenBank accession number (16S rRNA): L40998.

Other Organisms

1. "*Thioploca marina*"
ma. ri' na. L. adj. *marina* of marine.
Identified from oxygen-poor upwelling areas of the ocean, e.g., the Pacific continental shelf of South America (Maier and Gallardo, unpublished data).

2. "*Thioploca minima*" Koppe 1924, 630.
Identified from sediments of freshwater localities in Central Europe and Lake Erie.

Genus VIII. *Thiospira* Visloukh 1914, 48^{AL}

J. GIJS KUENEN AND GALINA A. DUBININA

Thi'o. spi' ra. Gr. n. *thios* sulfur; Gr. n. *spira* a coil; M.L. fem. *Thiospira* sulfur coil or spiral.

Spirilla, usually with pointed ends, containing **sulfur inclusions**. **Motile** by monotrichous or polytrichous **polar flagella**. No resting stages known. Isolates conforming to the descriptions of the two "species" have now been obtained (Dubinina et al., 1993).

Type species: ***Thiospira winogradskyi*** (Omeliński 1905) Visloukh 1914, 48.

FURTHER DESCRIPTIVE INFORMATION

Cells are colorless spirilla, pointed at the ends and containing sulfur globules. Polar flagella, in some cases united in a tuft, are visible in the light microscope. *Thiospira winogradskyi* is 2–2.5 μ m

wide and up to 50 μ m long. "*Thiospira bipunctata*" is 1.7–2.4 $\times$ 6.6–14 μ m.

Thiospira spp. are microaerophilic and vigorously chemotactic with respect to oxygen and possibly H₂S.

They are found in sulfurous marine and freshwater environments, including wastewater treatment systems.

The validity of the genus *Thiospira* has, in recent years, been called into question. The primary diagnostic feature of the genus was the accumulation of sulfur within spiral cells, and the separation of the two species was primarily morphological. Now that some strains have been grown in axenic cultures, it seems likely

that members of the genus should actually be included within the genus *Aquaspirillum*, especially since *Aquaspirillum* species have been shown to accumulate sulfur in their cells (Dubinina et al., 1993). However, because the taxonomic position of the genus *Aquaspirillum* is itself under investigation (see, for example, Willems et al., 1991c; Pot et al., 1992), the genus *Thiospira* has not yet been formally removed, and because many authors still report observations of *Thiospira* species (e.g., Bernard and Fenchel, 1995), the description of the genus *Thiospira* will be maintained until the situation is clarified. What follows is adapted from the chapter by la Rivière and Kuenen in the last edition of *Bergey's Manual*.

ENRICHMENT AND ISOLATION PROCEDURES

T. winogradskyi was enriched by Omelianski (1905) in Winogradsky columns kept for some months at room temperature in the dark. The cell mass was in the form of "bacterial plates", i.e., discrete dense layers of bacteria, and could be kept alive in the laboratory for 2 years. Molisch (1912) enriched "*T. bipunctata*" from Black Sea mud which, when mixed with decaying algae, was used as an H₂S-generating sediment in a seawater-filled cylinder 20–30 cm high.

Recent successful isolations (Dubinina et al., 1993) have been made from H₂S or thiosulfate-rich samples on a medium containing (g/l): (NH₄)₂SO₄, 1.0; MgSO₄·7H₂O, 1.0; casein hydrolysate, 3.0; sodium succinate, 1.0; vitamins and trace elements (Pfennig and Lippert, 1966). The pH is 7.0, and the necessary redox gradient is generated by adding a FeS suspension, which also serves as a source of sulfide.

TAXONOMIC COMMENTS

Although one of the species ("*T. bipunctata*") does not appear on the Approved Lists, it is retained in the present description of the genus in view of its marked differences from the type

species. It should be recalled that Bavendamm (1924) recognized five species.

Dubinina and Grabovich (1983) have described the isolation of heterotrophic spirilla morphologically similar to *Thiospira* and capable of forming internal sulfur granules in sulfide-containing media. Experiments indicate that the sulfur is formed through oxidation of sulfide by hydrogen peroxide.

Since other heterotrophs capable of forming sulfur inclusions in sulfide-containing media have also been found (Skerman et al., 1957), there is a distinct possibility that, on further examination, the spirilla isolated by Dubinina and Grabovich may turn out to be classifiable within the existing species of the genus *Spirillum* or, as now seems likely, *Aquaspirillum* (see above).

In spite of its present uncertain taxonomic position, the importance of the genus may prove to be considerable when further studies have been made of the population of sulfide oxidizers encountered at the deep-sea bottom around H₂S-emitting hydrothermal vents (Jannasch, 1984).

FURTHER READING

- Bavendamm, W. 1924. Die farblosen und roten Schwefelbakterien des Süß- und Salz-wassers. In Kolkwitz (Editor), Pflanzenforschung, Gustav Fischer Verlag, Jena. pp. 7–156.
- Dubinina, G.A. and M.Y. Grabovich. 1983. Isolation of pure *Thiospira* cultures and investigation of their physiology and sulfur metabolism. *Mikrobiologiya* 52: 5–12.
- Dubinina, G.A., M.Y. Grabovich, A.M. Lysenko, N.A. Chernykh and V.V. Churikova. 1993. Revision of taxonomic position of colorless sulfur spirilla of the genus *Thiospira* and description of a new species *Aquaspirillum bipunctata* comb. nov. *Microbiology* 62: 368–644.
- Jannasch, H.W. 1984. Microbial processes at deep sea hydrothermal vents. In Rona, Bostrom, Laubier and Smith (Editors), Hydrothermal Processes at Seafloor Spreading Centers, Plenum Publishing, New York. pp. 667–709.

List of species of the genus *Thiospira*

1. ***Thiospira winogradskyi*** (Omelianski 1905) Visloukh 1914, 48^{VP}

wi.no.grad'sky.i. M.L. gen. n. *winogradskyi* of Winogradsky; named for S.N. Winogradsky, a Russian microbiologist.

Found in freshwater and marine environments overlaying sulfurous muds. See Table BXII.γ.42 for differentiation of *T. winogradskyi* and "*T. bipunctata*".

The mol% G + C of the DNA is: 38 (*T_m*).

Type strain: No culture available.

2. **"*Thiospira bipunctata*"** (Molisch 1912) Visloukh 1914, 48. *bi.punc.ta'ta.* L. bis twice; L. part. adj. *pinctatus* punctate, dotted; M.L. fem. adj. *bipunctata* twice punctate.

Found in sulfurous marine and brackish waters (see Table BXII.γ.42).

TABLE BXII.γ.42. Differentiation of the species of the genus *Thiospira*

Characteristic	<i>T. winogradskyi</i>	" <i>T. bipunctata</i> "
Size	2–2.5 × up to 50 μm	1.7–2.4 × 6.6–14 μm
Number of sulfur inclusions	Numerous	Few
Volutin granules at both ends	—	+

Family II. **Piscirickettsiaceae** *fam. nov.*

JOHN L. FRYER AND CATHARINE N. LANNAN

Pis'ci.ric.kett.si.a'ce.ae. M.L. fem. n. *Piscirickettsia* type genus of the family; *-aceae* ending to denote family, M.L. fem. pl. n. *Piscirickettsiaceae* the *Piscirickettsia* family.

Gram-negative, aerobic, coccoid, rod- or spiral-shaped bacteria, occasionally pleomorphic. Isolated predominantly from the marine environment. All belong to the *Gammaproteobacteria* with 16S rRNA similarities as the major unifying factor.

Type genus: **Piscirickettsia** Fryer, Lannan, Giovannoni and Wood 1992, 123.

Further Comments “*Piscirickettsiaceae*” includes six genera, *Piscirickettsia*, *Cycloclasticus*, *Hydrogenovibrio*, *Methylophaga*, *Thioalkalimicrobium*, and *Thiomicrospira*. These groups are related phylogenetically, but share a limited number of phenotypic characters. *Piscirickettsia* is a Gram-negative, nonmotile, highly fastidious, intracellular rickettsia-like pathogen of fish. The bacterium replicates within membrane-bound cytoplasmic vacuoles in cells of infected hosts and does not grow on any known artificial medium. *Piscirickettsia* contains a single species that is known to cause

serious mortality among salmonids and other species of fish in the marine environment.

The remaining five genera are composed of rod-, comma-, or spiral-shaped bacteria. They are all motile by means of a single polar flagellum. All can be cultured on artificial media. None is pathogenic or replicates within the cells of hosts. Optimum temperatures are higher than those of *Piscirickettsia* and all were isolated from aquatic environments. *Cycloclasticus* utilizes aromatic hydrocarbons as the sole source of organic carbon; *Methylophaga* utilizes C₁ compounds; and *Hydrogenovibrio* and *Thiomicrospira* both use CO₂. Members of the genus *Thiomicrospira* are obligately autotrophic, sulfur bacteria and have been isolated from marine hydrothermal vents and mud flats. The genus *Thioalkalimicrobium* contains chemolithoautotrophs that are isolated from alkaline lakes and that oxidize sulfide and thiosulfate to sulfate.

Genus I. *Piscirickettsia* Fryer, Lannan, Giovannoni and Wood 1992, 123^{vp}

JOHN L. FRYER AND CATHARINE N. LANNAN

Pis'ci.ric.kett'si.a. L. n. *piscis* fish; generic name *Rickettsia*; N.L. fem. n. *Piscirickettsia* rickettsia-like organism affecting fish.

Pleomorphic, but predominantly coccoid organisms, ~0.5–1.5 µm in diameter. Gram negative, Gimenez negative, and stain dark blue with Giemsa’s stain. Nonmotile. Aerobic. Characteristically replicate within cytoplasmic vacuoles in susceptible fish host cells. Cultivable in fish cell cultures but not on any known host-cell-free medium. Direct host-to-host transmission occurs in the aquatic environment without the requirement of a vector. The 16S rRNA gene sequence conforms to the secondary structural models for the *Gammaproteobacteria*. No signature sequence has been identified. **The etiological agent of the disease piscirickettsiosis in fishes.**

Type species: ***Piscirickettsia salmonis*** Fryer, Lannan, Giovannoni and Wood 1992, 123.

FURTHER DESCRIPTIVE INFORMATION

At present, *Piscirickettsia salmonis* is the only described species of the genus *Piscirickettsia*. This organism has a widespread geographic distribution, and causes disease in numerous salmonid host species (Table BXII.γ.43). *Piscirickettsia salmonis*-like bacteria have also been isolated from diseased sea bass, *Atractosion nobilis*, in California, USA (Chen et al., 2000a); and in Taiwan, from grouper, *Epinephelus melanostigma* (Chen et al., 2000b), and five species of cultured tilapia (Chern and Chao, 1994).

By electron microscopy, the organisms display the typical pro-

toplasmic structure of a procaryote and the cell wall of a Gram-negative bacterium (Fig. BXII.γ.81). Although primarily coccoid, the organisms also appear as pairs of curved rods or rings (Fig. BXII.γ.82). They replicate by binary fission within membrane-bound cytoplasmic vacuoles in cells of susceptible fish hosts or fish cell lines (Fig. BXII.γ.83) where they produce a cytopathic effect (Fig. BXII.γ.84). *In vitro* replication is optimal at 15–18°C, retarded above 20°C and below 10°C, and does not occur above 25°C (Fryer et al., 1990).

Little genetic variation has been observed in *P. salmonis* isolated from differing salmonid host species or originating from diverse geographic locations. 16S rDNA sequence data are available for five *P. salmonis* isolates representing three salmonid host species and three widely separated geographic regions (North America, South America, and Europe; Fryer and Mauel, 1997). With 1527 bases compared, the five isolates form a tight monophyletic cluster. Four of the isolates show sequence similarities >99.4%. One of the five is less closely related with similarities ranging from 98.5–98.9% (Mauel et al., 1999). The genetic differences between this isolate and others in the study do not correlate with either host species or geographic region as these characters are shared among them.

Hyperimmune rabbit serum generated against purified *P. sal-*

TABLE BXII.γ.43. Geographic region and host species of *Piscirickettsia salmonis*

Geographic region	Host species	Reference
Canada, Atlantic coast	<i>Salmo salar</i>	Cusack et al. (1997)
Canada, Pacific coast	<i>Oncorhynchus gorbuscha</i> , <i>O. tshawytscha</i> , <i>S. salar</i>	Evelyn and Kent (1992)
Chile	<i>Oncorhynchus kisutch</i>	Bravo and Campos (1989)
	<i>Oncorhynchus mykiss</i> , <i>O. tshawytscha</i> , <i>S. salar</i>	Garcés et al. (1991)
	<i>Oncorhynchus masou</i>	Bravo (1994)
Ireland	<i>S. salar</i>	Rodger and Drinan (1993)
Norway	<i>S. salar</i>	Olsen et al. (1997)

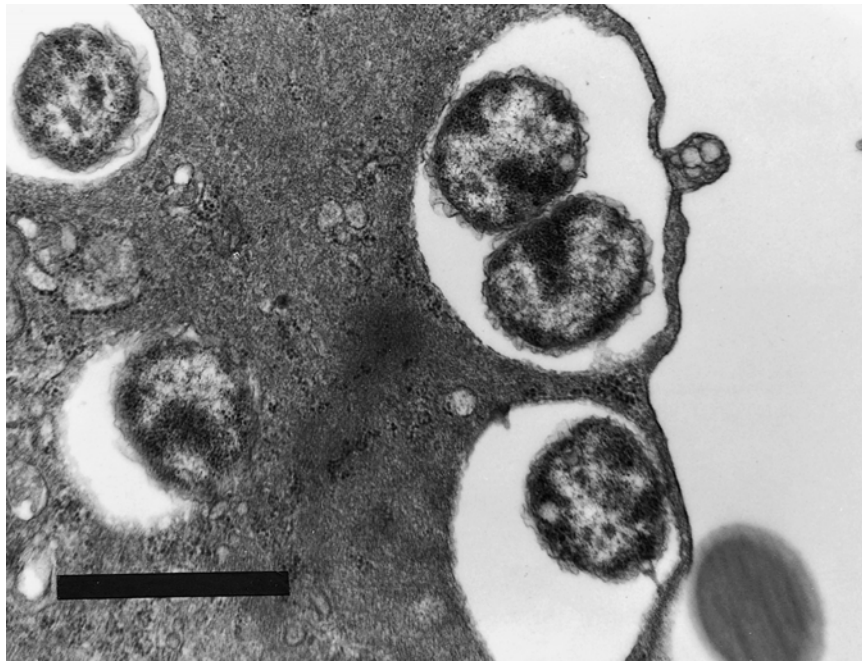


FIGURE BXII.γ.81. Ultrathin section of a cultured chinook salmon embryo (CHSE-214) cell containing *Piscirickettsia salmonis* organisms within cytoplasmic vacuoles. Bar = 1 μ m. (Reprinted with permission from J.L. Fryer et al., Fish Pathology 25: 107–114, 1990, ©The Japanese Society of Fish Pathology, Tokyo.)

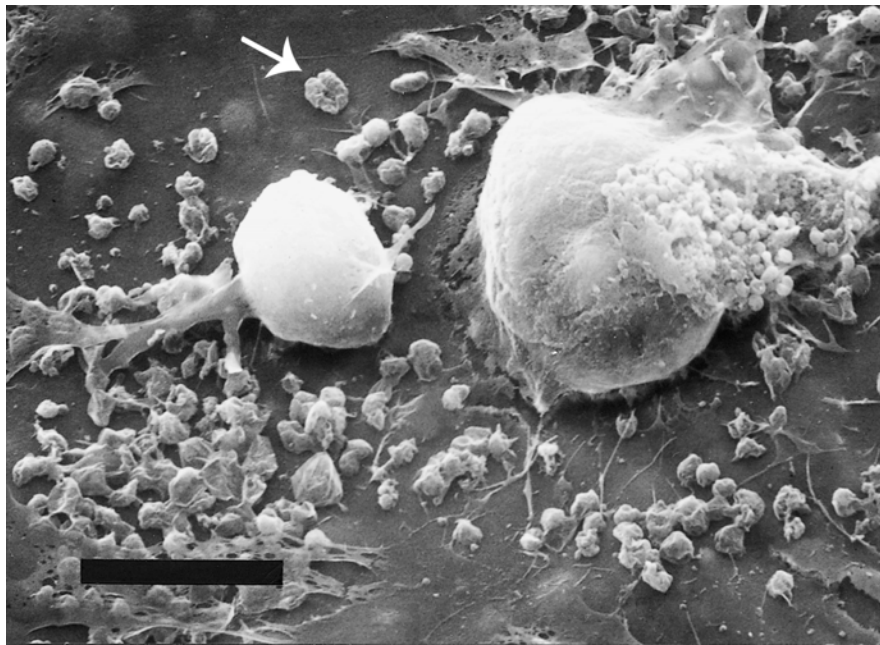


FIGURE BXII.γ.82. Scanning electron micrograph of *Piscirickettsia salmonis* organisms free or being released from cultured chinook salmon embryo (CHSE-214) cells. Note the ring form (arrow) and the varied sizes of the coccoid forms. Bar = 10 μ m. (Reprinted with permission from J.L. Fryer et al., Fish Pathology 25: 107–114, 1990, ©The Japanese Society of Fish Pathology, Tokyo.)

monis recognizes four surface-exposed protein antigens with relative molecular sizes of 65, 60, 54, and 51 kDa and two low molecular weight carbohydrate antigens, the lipopolysaccharide and a component lipo-oligosaccharide with relative molecular weights of 16 kDa and ~11 kDa, respectively (Kuzyk et al., 1996). Nevertheless, the humoral response elicited by *P. salmonis* infec-

tion in fish is apparently low. In the study by Kuzyk et al. (1996) convalescent serum from infected salmonid fish reacted with several minor immunoreactive protein antigens between 10 and 70 kDa in size as well as with a carbohydrate antigen with a relative molecular weight of ~11 kDa, but the antigens recognized by the convalescent salmonid fish serum differed greatly from those

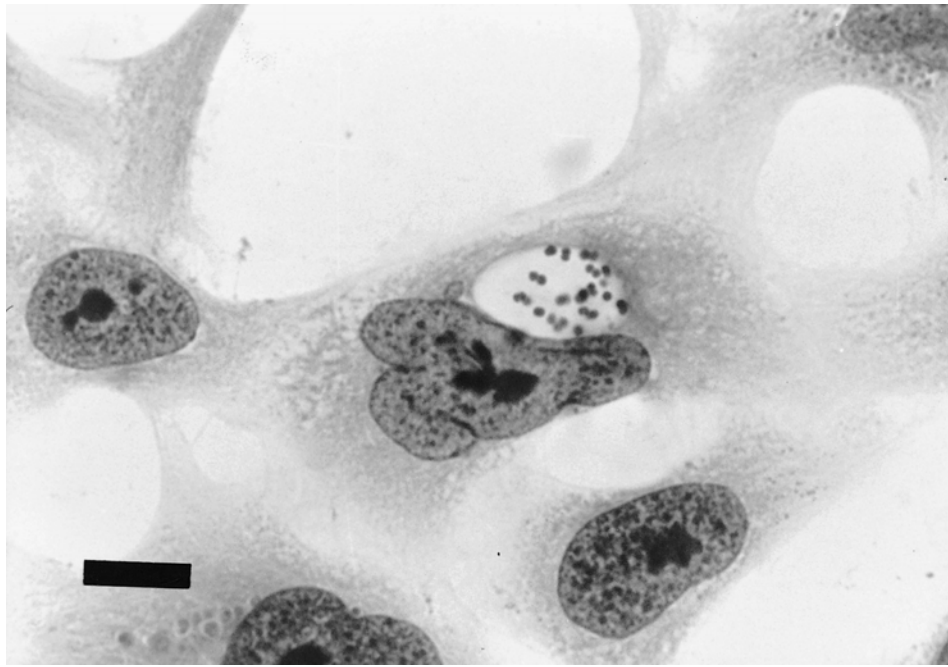


FIGURE BXII.γ.83. *Piscirickettsia salmonis* organisms within a cytoplasmic vacuole in a cultured chinook salmon embryo (CHSE-214) cell 3 d postinoculation. May-Grünwald-Giemsa stain. Bar = 10 μm. (Reprinted with permission from J.L. Fryer et al., Fish Pathology 25: 107–114, 1990, ©The Japanese Society of Fish Pathology, Tokyo.)

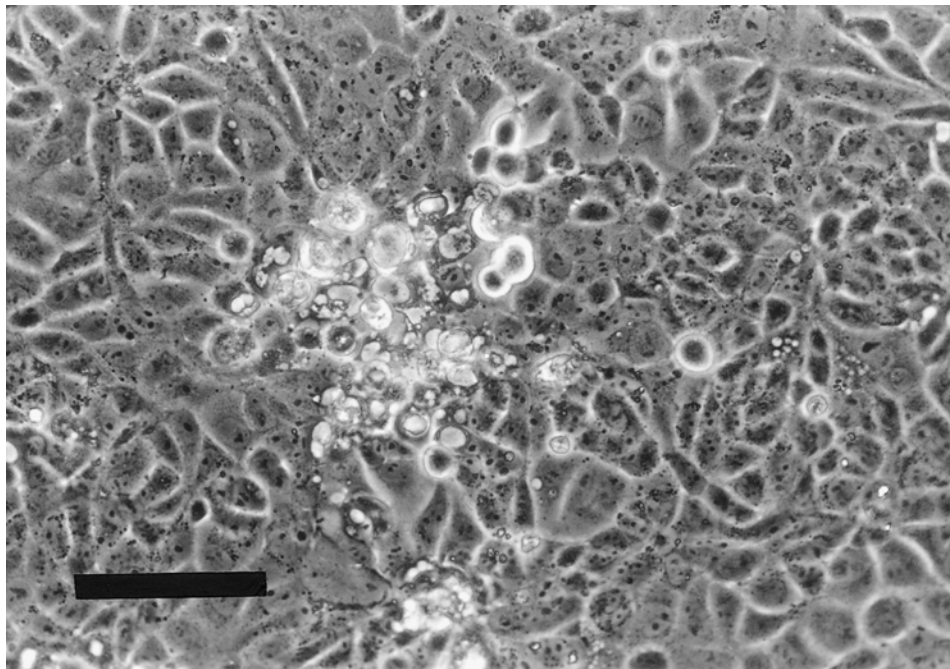


FIGURE BXII.γ.84. Cytopathic effect produced by *Piscirickettsia salmonis* in cultured chinook salmon embryo (CHSE-214) cells. Phase contrast microscopy. Bar = 100 μm. (Reprinted with permission from J.L. Fryer et al., Fish Pathology 25: 107–114, 1990, ©The Japanese Society of Fish Pathology, Tokyo.)

identified by rabbit antiserum, suggesting a modest humoral response in the fish.

Piscirickettsia salmonis replication is inhibited *in vitro* by numerous antibiotics normally used in cell culture or in human and veterinary medicine. These include gentamicin, streptomycin, erythromycin, rifampin, tetracycline, doxycycline, oxolinic

acid, and chloramphenicol (Fryer et al., 1992). Penicillin appears to have little effect. In spite of the broad antibiotic sensitivity of this organism, *in vivo* antibiotic therapy has had limited success. Intracellular concentrations of these preparations may not reach lethal levels *in vivo*, permitting replication to continue within host cells despite therapy.



FIGURE BXII.γ.85. *Oncorhynchus kisutch* infected with *Piscirickettsia salmonis*. Note the diagnostic, ring-shaped lesions on the liver. (Reprinted with permission from J.L. Fryer and M.J. Mauel, *Emerging Infectious Diseases* 3: 137–144, 1997, ©Centers for Disease Control and Prevention, Atlanta, Georgia, USA.)

Piscirickettsial infection results in a systemic, epizootic fish disease, piscirickettsiosis, which can occur in chronic or acute form. Mortality may range from low to extremely high ($\geq 90\%$ of a naturally exposed population.) Marked difference in virulence has been reported among three isolates of *P. salmonis* each obtained from a different geographic location (House et al., 1999). Frequently, skin hemorrhages and lesions ranging from small areas of raised scales to shallow ulcers up to 2 cm in diameter are the first disease signs to appear (Branson and Diaz-Munoz, 1991). Pathological changes that include necrosis and inflammation, frequently found adjacent to blood vessels, may be observed in most internal organs of infected fish, especially in kidney, liver, spleen, and intestine (Cvitanich et al., 1991). At times, epithelial hyperplasia results in lamellar fusion of the gills. Organisms are found within macrophages and in the cytoplasm of other host cells. Diagnostic, ring-shaped, cream-colored lesions are present on the livers of chronically infected fish (Fig. BXII.γ.85), but in acute cases, mortality may be the only gross sign of disease observed.

The means of transmission of *P. salmonis* in natural infections is by direct fish-to-fish passage and occurs without the need of a vector. The reservoir for the pathogen has not been determined but may include one or more species of fish or other aquatic animals. Almendras et al. (1997) demonstrated direct horizontal transmission of *P. salmonis* under experimental conditions in both fresh and salt water. Vertical transmission from parent to offspring may also occur. The organisms can be found in the gonad of infected fish, and Larenas et al. (1996) reported *P. salmonis* organisms in the fertilized ova from experimentally infected parents. Despite this, piscirickettsiosis is less frequently observed in anadromous salmonids during the freshwater stage of their life cycle than after entry into the marine environment, suggesting that vertical transmission may be an uncommon occurrence.

ENRICHMENT AND ISOLATION PROCEDURES

Piscirickettsia salmonis can be isolated from the internal organs of infected fish by inoculation of homogenized tissues onto cell cultures derived from susceptible fish hosts. The cultures are incubated at 15–18°C for up to 28 d and microscopically observed

for development of the characteristic cytopathic effect. However, isolation of these agents from clinical specimens presents a considerable challenge. Aseptic collection of diagnostic specimens in the field is difficult, enhancing the likelihood of adventitious bacterial contamination of the inoculum. In spite of this, the broad antibiotic sensitivity of *P. salmonis* dictates that the cultures be maintained in antibiotic-free medium. For this reason, preliminary diagnosis of piscirickettsiosis is normally made by examination of Gram stain, Giemsa stain, methylene blue, or acridine orange-stained kidney imprints or smears, with confirmation of identity provided by serological methods, e.g., immunofluorescence (Lannan et al., 1991) or immunohistochemistry (Alday-Sanz et al., 1994).

MAINTENANCE PROCEDURES

Piscirickettsia salmonis can be subcultured with the inoculation of spent growth medium from infected cultures onto fresh fish cell cultures in antibiotic-free medium. Infectious organisms can be recovered from spent culture medium stored for up to 3 weeks at 5°C (Lannan and Fryer, 1994). Long-term survival in culture medium requires the addition of 10% DMSO as a cryopreservative followed by storage at –70°C or in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *PISCIRICKETTSIA* FROM OTHER GENERA

Piscirickettsia salmonis might be confused with *Neorickettsia helminthoeca*, the coccoid, Gram-negative, intracellular bacterium that is the etiologic agent of salmon-poisoning disease in canids. Although *N. helminthoeca* is not a parasite of salmonid fishes, salmonids are one host in the complex life cycle of *Nanophyetus salmincola*, the trematode vector of *N. helminthoeca*. Unlike *P. salmonis*, geographic distribution of *N. helminthoeca* is restricted to a portion of the Pacific coast of the United States. The differential characteristics of these two genera are listed in Table BXII.γ.44.

TAXONOMIC COMMENTS

The genus *Piscirickettsia* currently contains one species. Additional microorganisms assigned to this genus must be intracel-

TABLE BXII.γ.44. Differential characteristics of *Piscirickettsia salmonis* and *Neorickettsia helminthoeca*

Characteristic	<i>P. salmonis</i>	<i>N. helminthoeca</i>
Replicates within fish cells	+	—
Replicates at 37°C	—	+
Widespread geographic distribution	+	—
Pathogenic for <i>Canidae</i>	—	+
Reacts with convalescent serum from canine salmon poisoning disease	—	+
<i>Alphaproteobacteria</i>	—	+
<i>Gamma</i> proteobacteria	+	—

lular, Gram-negative, procaryotic parasites of fishes and perhaps other aquatic animals. These bacteria must be specifically related to *P. salmonis*. Members of the species *P. salmonis* are those recognized by specific antiserum against the type strain, LF-89.

Based on the intracellular location of *P. salmonis* and phenotypic similarities between this organism and the genus *Ehrlichia*, it was originally proposed that *Piscirickettsia* should be placed with *Ehrlichia* in the family *Rickettsiaceae* (Fryer et al., 1992). However, microorganisms historically included in the *Rickettsiaceae* appear to be polyphyletic (Weisburg et al., 1989). The majority fall in the *Alphaproteobacteria*, but certain of these intracellular bacteria, including *P. salmonis*, *Rickettsiella grylli* (Roux et al., 1997), *Coxiella burnetii*, and the bacterium designated *Wolbachia persica* (Weisburg et al., 1989), belong to the *Gamma*proteobacteria. Therefore, these organisms are phylogenetically remote from others in the family. *Piscirickettsia* appears to group with the genera *Cycloclasticus*, *Hydrogenovibrio*, *Methylophaga*, and *Thiomicrospira*, and is somewhat more distantly related to the *Legionella*, *Francisella*, and *Coxiella*. *Piscirickettsia* is now assigned to a new family, the “*Piscirickettsiaceae*”.

Gram-negative obligate intracellular bacteria were not recognized as pathogens of fish before 1989, when *P. salmonis* was isolated from diseased salmonid fish in Chile (Fryer et al., 1990). Since that time, bacteria of this type have been observed or isolated from other species of salmonid and nonsalmonid fish at numerous locations (Brocklebank et al., 1992; Rodger and Drinnan, 1993; Chern and Chao, 1994; Olsen et al., 1997; Chen et al., 2000a, b). Many of the isolates from salmonid fish have been serologically identified as *P. salmonis*, but the taxonomic place-

ment of those found in other fishes has not been determined. Whether these organisms can be classified as *P. salmonis*, or as additional species in the genus *Piscirickettsia*, or whether they more properly belong to other genera as yet undescribed remains unresolved.

ACKNOWLEDGMENTS

We thank Sandra Bravo of Puerto Montt, Chile, who first reported the disease that was later identified as piscirickettsiosis, and L. H. Garcés, J. J. Larenas, and P. A. Smith of the Veterinary Sciences Faculty of the University of Chile, Santiago, all of whom have provided specimens, counsel, and technical help in the identification and characterization of *P. salmonis*. In addition, we thank Ena Urbach of the Department of Microbiology, Oregon State University, for aid in the phylogenetic description of this organism. This is Oregon Agricultural Experiment Station technical paper no. 11325.

FURTHER READING

Cvitanich, J.D., O.N. Garate and C.E. Smith. 1991. The isolation of a rickettsia-like organism causing disease and mortality in Chilean salmonids and its confirmation by Koch postulate. *J. Fish Dis.* 14: 121–145.
Fryer, J.L. and R.P. Hedrick. 2003. *Piscirickettsia salmonis*: a Gram-negative intracellular bacterial pathogen of fish. *J. Fish Dis.* 26: 251–262.
Fryer, J.L., C.N. Lannan, L.H. Garcés, J.J. Larenas and P.A. Smith. 1990. Isolation of a rickettsiales-like organism from diseased coho salmon (*Oncorhynchus kisutch*) in Chile. *Fish Pathology* 25: 107–114.
Fryer, J.L., C.N. Lannan, J. Giovannoni and N.D. Wood. 1992. *Piscirickettsia salmonis* gen. nov., sp. nov., the causative agent of an epizootic disease in salmonid fishes. *Int. J. Syst. Bacteriol.* 42: 120–126.

List of species of the genus *Piscirickettsia*

1. ***Piscirickettsia salmonis*** Fryer, Lannan, Giovannoni and Wood 1992, 123^{VP}
sal.mo' nis. L. gen. n. *salmonis* of salmon.

The characteristics are as described for the genus.
The mol% G + C of the DNA is: unknown.
Type strain: LF-89, ATCC(R) VR 1361.
GenBank accession number (16S rRNA): U36941.

Genus II. ***Cycloclasticus*** Dyksterhouse, Gray, Herwig, Lara and Staley 1995, 120^{VP}

ALLISON D. GEISELBRECHT

Cy.clo.clas'ti.cus. Gr. n. *kyklos* circle or rings; Gr. adj. *klastos* broken; M.L. masc. n. *cycloclasticus* ring-breaker.

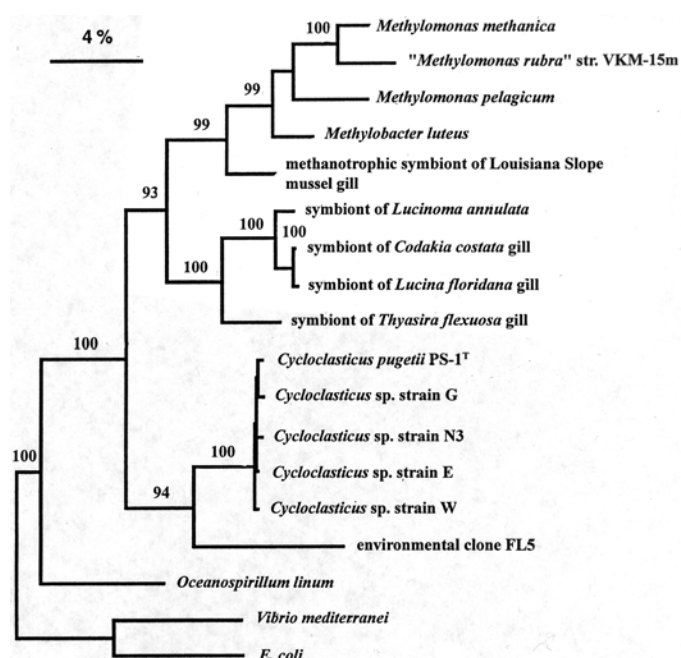
Unicellular, rod-shaped bacterium, 0.5 × 1.0–2.0 μm. **Gram negative**. Motile by means of a single polar flagellum. **Obligately aerobic**, nonfermentative, **heterotrophic**. **Grows poorly on complex bacteriological media containing no aromatic compounds**. Requires at least 1% salinity for growth. Aromatic compounds, including biphenyl, naphthalene, phenanthrene, anthracene, and toluene, used as sole or principal carbon sources for growth.

In addition, strains use selected fatty acids and amino acids, including acetate, propionate, and glutamate. **Oxidase** and **catalase** positive.

The mol% G + C of the DNA is: 37.
Type species: ***Cycloclasticus pugetii*** Dyksterhouse, Gray, Herwig, Lara and Staley 1995, 120.

TABLE BXII.γ.45. Characteristics of the genus *Cycloclasticus*^a

Characteristic	<i>Cycloclasticus pugetii</i>
Cell shape	Straight rods
Motility	+
Flagellar arrangement	Polar, single
Poly-β-hydroxybutyrate accumulation	—
Na ⁺ required for growth	+
NO ₃ ⁻ reduced to NO ₂ ⁻	+
Oxidase	+
Catalase	+
Lipase	+
Amylase	—
Gelatinase	—
Intracellular membranes	—
Utilization of:	
Methane	—
Methanol	—

^aFrom Dyksterhouse et al. (1995).**FIGURE BXII.γ.86.** Phylogenetic relationships based on 16S rDNA sequences of *Cycloclasticus* and closely related species found in the Ribosomal Database Project, release 6.0 (Maidak et al., 1999). Environmental clone FL5 is an rDNA clone amplified from free-living bacteria in the marine water column (DeLong et al., 1993). This tree was the most parsimonious tree generated with PAUP, version 3.0s (Swofford, 1993). Heuristic unweighted bootstrap values are shown near the clades; only values greater than 50% are shown. Bar = ~0.04 change per nucleotide position.

FURTHER DESCRIPTIVE INFORMATION

The genus *Cycloclasticus* comprises a collection of marine strains with one validly named species. Phylogenetically, the closest relatives of these organisms are members of the methane-oxidizing genera *Methylomonas* and *Methylobacter*, a group of uncultured sulfur-oxidizing symbionts found in marine invertebrates, and a 16S rRNA environmental clone designated FL5. However, the mol% G + C DNA base composition of *Cycloclasticus* and *Methylomonas methanica* differ substantially (37 and 52, respectively), and *Cycloclasticus* cannot utilize methane or other C₁ compounds

TABLE BXII.γ.46. Predominant whole-cell fatty acids and percent composition of *Cycloclasticus* strains^a

Fatty acid	% total fatty acids in:	
	<i>C. pugetii</i> PS-1 ^b	<i>Cycloclasticus</i> sp. N3 ^c
Unknown ^d	6	6
C _{12:0}	2.5	3
C _{12:0} 3OH	2.8	3
C _{16:0} iso	nd ^e	3
C _{16:1} ω7c	61.5	53
C _{16:0}	23	21
Total identified	95.7	89

^aOnly fatty acids representing 2.5% or more of the total fatty acid composition of each strain are listed. Cellular fatty acids were extracted, methylated, and detected with a Hewlett Packard model 5890 series II gas chromatograph as previously described (MIDI Inc., 1993).^bData from Dyksterhouse et al. (1995). Strains were grown in liquid culture on 50 ml of Bacto Marine 2216 broth containing biphenyl crystals for 3 d with agitation, centrifuged, washed twice with ONR7a. 40 mg (wet weight) of the cell pellet was used for the analysis.^cData from Geiselbrecht et al. (1998). Strains were grown on ONR7a plus Casamino acids (Difco) and salicylate (0.01% each) plates for 2 weeks at 20°C. Cells were harvested by scraping the entire plate surface.^dUnidentified peak at an equivalent carbon length of 11:798.^end, not detected.

as carbon sources. In addition, *Cycloclasticus* does not utilize thiosulfate as an energy source. Furthermore, existing strains are not phylogenetically closely related to other marine genera of aerobic, heterotrophic, motile, rod-shaped bacteria belonging to the *Gammaproteobacteria* (see Table BXII.γ.45 and Fig. BXII.γ.86) (Dyksterhouse et al., 1995).

Transmission electron microscopy of whole cells shows a single polar flagellum with several fimbriae originating from the cell surface and a cell envelope structure typical of Gram-negative bacteria, with a cell size of 0.5 × 1.0–2.0 μm. Colonies are small, smooth, round, and raised, with slow growth on solid media, and in some cases requiring up to 2 weeks for visible colony formation. Cultures have been shown to grow at temperatures ranging from 4–28°C, in the presence of salinity ranging from 1–7% (Dyksterhouse et al., 1995).

Whole cell fatty acid analyses show similar compositions between strains (Table BXII.γ.46). The predominant fatty acids in these and two additional strains tested were similar; C_{16:1} ω7c was the most prevalent fatty acid.

A characteristic of this genus is an apparent preference for aromatic compounds as carbon sources. Carbon source utilization studies indicate that only a few additional carbon sources, including selected fatty acids and amino acids, can be used as sole carbon sources (Table BXII.γ.47). The concentrations of the carbon sources are crucial in determining carbon source utilization: strains responded differently to 0.01% and 0.1% w/v amounts of carbon source (Geiselbrecht, 1998).

Polyaromatic hydrocarbon (PAH) degradation studies performed with *Cycloclasticus* strains indicate similar, broad PAH-substrate ranges including naphthalene, substituted naphthalenes, phenanthrene, fluorene, biphenyl, and acenaphthene (Table BXII.γ.48) (Geiselbrecht et al., 1998). "*Cycloclasticus oligotrophus*" RB1 is capable of using a variety of aromatic compounds, such as toluene and *o*-, *m*-, and *p*-xylenes, as sole carbon sources (Wang et al., 1996).

Wang et al. (1996) cloned and sequenced a 5.7-kb DNA fragment from "*C. oligotrophus*" RB1. This DNA region contains genes that encode proteins somewhat homologous to the large-subunit (XylC1) and small-subunit (XylC2) biphenyl dioxygenase of ter-

TABLE BXII.γ.47. Nonaromatic carbon sources utilized by selected *Cycloclasticus* strains

Carbon source	<i>C. pugetii</i> ^a	<i>Cycloclasticus</i> strain N3 ^b	<i>Cycloclasticus</i> strain G ^b
D-Glucose	—	—	nd
D-Fructose	—	—	nd
D-Mannitol	—	nd	nd
DL-Malate	—	nd	nd
α-Ketoglutarate	—	nd	nd
Salicylate	+	—	—
Benzoate	+	nd	nd
Acetate	+	+ ^c	+ ^d
Propionate	+	+ ^d	+ ^d
Glutamate	+	+ ^c	—
Succinate	nd	+ ^c	+ ^d
Pyruvate	—	—	+

^aData from Dyksterhouse et al. (1995).^bData from Geiselbrecht (1998). Strain N3 was isolated from Puget Sound sediments; strain G from Gulf of Mexico sediments.^cGrowth occurred when carbon source was provided at 0.1% but not at 0.01% w/v.^dGrowth occurred when carbon source was provided at 0.01% but not at 0.1% w/v.**TABLE BXII.γ.48.** PAHs used as sole carbon sources by *Cycloclasticus* strains

PAH	<i>C. pugetii</i> ^a	<i>Cycloclasticus</i> strain N3 ^b	<i>Cycloclasticus</i> strain W ^b
Naphthalene	+	+	+
1-Methylnaphthalene	nd	+	+
2-Methylnaphthalene	nd	+	+
Acenaphthene	nd	+ ^c	+ ^c
2,6-Dimethylnaphthalene	nd	+	+
Biphenyl	+	+ ^c	+
Phenanthrene	+	+	+
Anthracene	+	+	nd
Fluorine	nd	+	+

^aPAH provided in excess of its saturation level as the sole carbon source in ONR7a broth. Growth indicates use of PAH as sole carbon source. From Dyksterhouse et al. (1995).^bPAH was provided as sole carbon source at or below its saturation level in ONR7a broth. The disappearance of PAH was monitored via gas chromatography after 7 d.^cPAH was only partially removed.

restrial bacteria, although these sequences are quite divergent from the sequences of known dioxygenases. By utilizing degenerate PCR primers, partial ISP sequences were obtained from *C. pugetii* and *Cycloclasticus* strain W that indicated the presence of *bph*-type dioxygenase ISP sequences similar to the region cloned in “*C. oligotrophus*” RB1, as well as a sequence corresponding to an additional PAH dioxygenase type more closely related to naphthalene type dioxygenases (Geiselbrecht et al., 1998). Although this study did not examine the functionality of these sequences, these data may indicate that *Cycloclasticus* contains multiple dioxygenases.

16S rDNA sequence analysis performed on three strains of *Cycloclasticus pugetii* revealed that 16S rDNA sequences from strain PS-1^T, -2, and -3 are nearly identical. Furthermore, *Cycloclasticus* sp. N3 contains only 3 base pair differences from *C. pugetii*^T (Geiselbrecht et al., 1996). “*C. oligotrophus*” RB1 is also phylogenetically very closely related to *C. pugetii*^T. All of these isolates share a distinctive 16S rDNA sequence at approximately base pair position 825, relative to *E. coli* (Gutell, 1993) of 5'-CUAACUG-UUGGCGGGUUUCCGC-3'. No DNA-DNA reassociation data exist for strains of the genus.

Cycloclasticus strains have been isolated from coastal marine sediments and waters in Puget Sound, the Gulf of Mexico, and Resurrection Bay, Alaska (Button et al., 1993, 1998; Dyksterhouse et al., 1995; Geiselbrecht et al., 1996, 1998). In Puget Sound, *Cycloclasticus* is widespread in both contaminated and uncontaminated sediments (Geiselbrecht et al., 1996). Furthermore, they

occur in the Puget Sound water column (Geiselbrecht, 1998). “*C. oligotrophus*” RB1 was isolated from the water column in Resurrection Bay, Alaska, following a 10⁸ dilution, indicating that they can potentially be numerous in the marine water column (Button et al., 1993). In the Gulf of Mexico, they have been isolated from offshore and near-shore sites at sediment dilution numbers ranging from 2 × 10³ to 2 × 10⁵, indicating that they are somewhat numerous in these sediments (Geiselbrecht et al., 1998).

ENRICHMENT AND ISOLATION PROCEDURES

An artificial seawater mineral salts medium called ONR7a has been used for most *Cycloclasticus* isolations. This medium contains all of the major cations and anions that are present in concentrations greater than 1 mg/l in seawater. ONR7a contains (per liter of distilled or deionized water): NaCl, 22.79 g; MgCl₂·6H₂O, 11.18 g; Na₂SO₄, 3.98 g; CaCl₂·2H₂O, 1.46 g; TAPSO (3-[N-tris(hydroxymethyl)methylamino]-2-hydroxypropanesulfonic acid), 1.3 g; KCl, 0.72; NH₄Cl, 0.27 g; Na₂HPO₄·7H₂O, 89 mg; NaBr, 83 mg; NaHCO₃, 31 mg; H₃BO₃, 27 mg; SrCl₂·6H₂O, 24 mg; NaF, 2.6 mg; FeCl₂·4H₂O, 2.0 mg. To prevent precipitation of ONR7a during autoclaving, three separate solutions are prepared and then mixed together after autoclaving when the solutions are cooled to at least 50°C: one solution contains NaCl, Na₂SO₄, KCl, NaBr, Na₂HPO₄, H₃BO₃, NaF, NH₄Cl, Na₂HPO₄, and TAPSO (pH adjusted to 7.6 with NaOH, then autoclaved); the second solution contains MgCl₂, CaCl₂, and SrCl₂ (divalent

cation salts made as a 50× stock and autoclaved); and the third solution contains FeCl₂ (made as a 200× stock and then preferably filter-sterilized). For solid media, Bacto Agar (Difco) or agarose (Sigma) is used as a solidifying agent. ONR8a contains all the compounds listed above for the ONR7a plus 10 ml/l of a vitamin solution (Staley, 1968) and 10 ml/l of a trace element solution (Dyksterhouse et al., 1995). The type strain of the genus *Cycloclasticus*, *C. pugetii*, was isolated by blending Puget Sound sediments, diluting these sediments 1:5, and spreading the dilution onto ONR8a containing 0.2 g of biphenyl on the Petri dish lid. Colonies surrounded by a diffusible yellow zone indicative of PAH degradation were selected for further isolation (Dyksterhouse et al., 1995). The vitamin and trace element solutions were later found to be unnecessary for growth, and were eliminated.

Cycloclasticus has also been isolated following enrichment in dilute most probable number (MPN) tubes specific for PAH-degrading marine bacteria. Surficial (aerobic) 1-cm portions of sediments were used as the inocula for these MPN preparations. Briefly, this technique involved serially diluting wet sediments with ONR7a containing PAHs as sole carbon sources. Tubes were scored positive or negative based on PAH degradation. After incubation for approximately 5 months at 15°C, the most dilute tubes that showed color changes indicative of PAH degradation were used for isolation. Samples from positive MPN tubes were streaked onto plates containing ONR7a solidified with 0.8% agarose. Phenanthrene or naphthalene were used as the sole carbon sources; these PAHs are provided in the vapor phase by placing approximately 1 mg of PAH in the lid of each plate and storing the plates inverted and wrapped in Parafilm[®]. Colonies that were capable of growing with either of these PAHs as sole carbon sources are picked and inoculated into an ONR7a solution containing the appropriate PAH in crystal form at a concentration of approximately 1 mg/ml (Geiselbrecht et al., 1998). *Cycloclasticus* spp. have also been isolated from enrichments made by inoculating ONR7a + PAHs with fresh sediments, incubating for 2 weeks at room temperature, and plating onto ONR7a solidified with 0.8% agarose (Geiselbrecht et al., 1996).

"*C. oligotrophus*" RB1 was isolated following a 10⁸ dilution of Resurrection Bay seawater into unamended seawater prepared by filtration through fired Gelman 1/E 47-mm-diameter filters, autoclaving, refiltering through other fired Gelman filters, and aseptically siphoning the sterile filtrate into incubation chambers (Button et al., 1993).

MAINTENANCE PROCEDURES

Although initial isolations were at 15°C, subsequent studies indicated a temperature growth range of 4–28°C, with strong growth at room temperature. Routine growth occurs in either ONR7a with PAHs as a sole carbon source, or in a seawater medium amended with acetate (Button et al., 1998). Strains have also been grown on Bacto Marine 2216 Petri plates with biphenyl

in the vapor phase (Dyksterhouse et al., 1995). Cultures can be frozen and stored at –80°C using either DMSO or glycerol.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

The use of PAHs as a sole carbon source is a characteristic of this genus. PAH degradation can be confirmed by several methods, but a simple method involves growing cultures with individual PAHs as sole carbon sources and monitoring the disappearance of the PAHs with a gas chromatograph equipped with a flame ionization detector in comparison to sterile controls. Briefly, 20-ml Balch tubes fitted with Teflon-lined stoppers can be used, and PAHs added as concentrated solutions in methylene chloride. After the methylene chloride is evaporated, 5 ml of a seawater medium (such as ONR7a) is added. Tubes are vortexed and shaken to allow the PAHs to solubilize. The tubes are inoculated with an exponential-phase culture of *Cycloclasticus*, grown on either acetate or a different PAH. Control tubes receive no inoculum. The tubes are incubated in the dark with shaking (for aeration), and then the cultures are extracted with hexane. Each extract can be analyzed with a gas chromatograph equipped with a 30-m DB-5 column (diameter, 0.5 mm) or comparable column, and the recovery of parent compound recovered versus controls indicate PAH degradation (Geiselbrecht et al., 1998).

DIFFERENTIATION OF THE GENUS *CYCLOCLASTICUS* FROM OTHER GENERA

The ability to utilize PAHs is not unique to this genus; other marine PAH-degrading, Gram-negative rods have been isolated and include members of the genus *Pseudomonas* (Shiaris and Cooney, 1983; Tagger et al., 1990), *Flavobacterium* (Shiaris and Cooney, 1983; Okpokwasili et al., 1984), *Moraxella* (Tagger et al., 1990), *Marinobacter* (Gauthier et al., 1992), *Vibrio* (West et al., 1984), *Sphingomonas* (Zylstra et al., 1997), and a new genus *Nephtunomonas* (Hedlund et al., 1999a). *Cycloclasticus* can be distinguished from some of these genera by its 1% salt requirement. *Cycloclasticus* can be distinguished from all of these genera as well as other aerobic, heterotrophic, motile, rod-shaped marine bacteria by its distinct phylogenetic location.

TAXONOMIC COMMENTS

As noted, several strains of *Cycloclasticus* have been isolated. "*Cycloclasticus oligotrophus*" RB1 has been described as a distinct species based on its small size and inability to utilize polar substrates such as amino acids (Button et al., 1993, 1998). Although "*C. oligotrophus*" RB1 is phylogenetically closely related to *C. pugetii*^T (Button et al., 1998), it has not yet been validly named.

FURTHER READING

Dyksterhouse, S.E., J.P. Gray, R.P. Herwig, J.C. Lara and J.T. Staley. 1995. *Cycloclasticus pugetii* gen. nov., sp. nov., an aromatic hydrocarbon-degrading bacterium from marine sediments. *Int. J. Syst. Bacteriol.* 45: 116–123.

List of species of the genus *Cycloclasticus*

1. *Cycloclasticus pugetii* Dyksterhouse, Gray, Herwig, Lara and Staley 1995, 120^{VP}
pu.get' i.i. M.L. gen. n. *pugetii* of Puget, named in honor of Peter Puget, a British naval officer who participated in the Vancouver Expedition and for whom Puget Sound was named.

The characteristics are as described for the genus.

Isolated from surficial sediments of Sinclair Inlet, which is located on Puget Sound near the city of Bremerton, Washington.

The mol% G + C of the DNA is: 37 (HPLC).

Type strain: PS-1, ATCC 51542.

GenBank accession number (16S rRNA): U12624, L34955.

Other Organisms

A collection of 15 strains of *Cycloclasticus* was isolated from several Puget Sound locations subsequent to the isolation of *Cycloclasticus pugetii*. These were grouped on the basis of fatty acid analyses and a representative strain's 16S rDNA was sequenced, *Cycloclasticus* sp. strain N3 (GenBank U57920) (Geiselbrecht et al., 1996). This strain differs from *C. pugetii* at only 3 base pair positions in the 16S rDNA. Twenty-three strains of *Cycloclasticus* spp. were isolated from offshore and near-shore Gulf of Mexico

sediments (Geiselbrecht et al., 1998). Partial 16S rDNA sequencing revealed that all 23 were members of the *Cycloclasticus* genus, and the 16S rDNA was sequenced in entirety from three strains: *Cycloclasticus* sp. strain G (GenBank AF093002), *Cycloclasticus* sp. strain E (GenBank AF093003), and *Cycloclasticus* sp. strain W (GenBank AF093004). All three are closely related to *C. pugetii*. At this time, it is unclear whether any of these strains are potentially new species of *Cycloclasticus*.

Genus III. *Hydrogenovibrio* Nishihara, Igarashi and Kodama 1991b, 132^{VP}

HIROFUMI NISHIHARA

Hydro.ge.no.vib'ri.o. Gr. n. *hydro* water; Gr. n. *genus* offspring; M.L. masc. n. *hydrogenium* hydrogen, that which produces water; M.L. masc. n. *vibrio* that which vibrates; M.L. masc. n. *Hydrogenovibrio* hydrogen vibrio.

Cells are **comma-shaped rods** (0.2–0.5 × 1–2 µm) that occur singly (Fig. BXII.γ.87). **Gram negative**. Motile by means of a polar flagellum. No resting stages are known. **Aerobic**. **Chemolitho-autotrophic, using H₂ or reduced inorganic sulfur compounds**, such as S⁰, thiosulfate, and tetrathionate, as the electron donor **and CO₂** as the carbon source. CO₂ is fixed via the **Calvin-Benson cycle**. Chemoorganotrophic growth has not been observed. The optimum temperature for growth is ~37°C. The optimum pH for growth is ~6.5. NaCl is required for growth with an optimum concentration of 0.5 M.

Straight chain saturated C_{16:0} and C_{18:0} acids and a straight-chain unsaturated C_{16:1} acid are the major components of the cellular fatty acids. Ubiquinone-8 is the major component of the quinone system.

The mol% G + C of the DNA is: 44.1.

Type species: Hydrogenovibrio marinus Nishihara, Igarashi and Kodama 1991b, 132.

FURTHER DESCRIPTIVE INFORMATION

Reddish-brown, smooth colonies are formed on inorganic basal medium solidified with agar in an atmosphere of H₂/O₂/CO₂ (85:5:10) (1 atm at room temperature). Colonies are not formed when the partial pressure of oxygen is more than 10%. The colonies grow up to approximately 1 mm in diameter after a week of incubation at 26.5°C. No colony is formed heterotrophically on nutrient broth medium supplemented with 0.5 M NaCl.

H. marinus grows rapidly and vigorously in the presence of

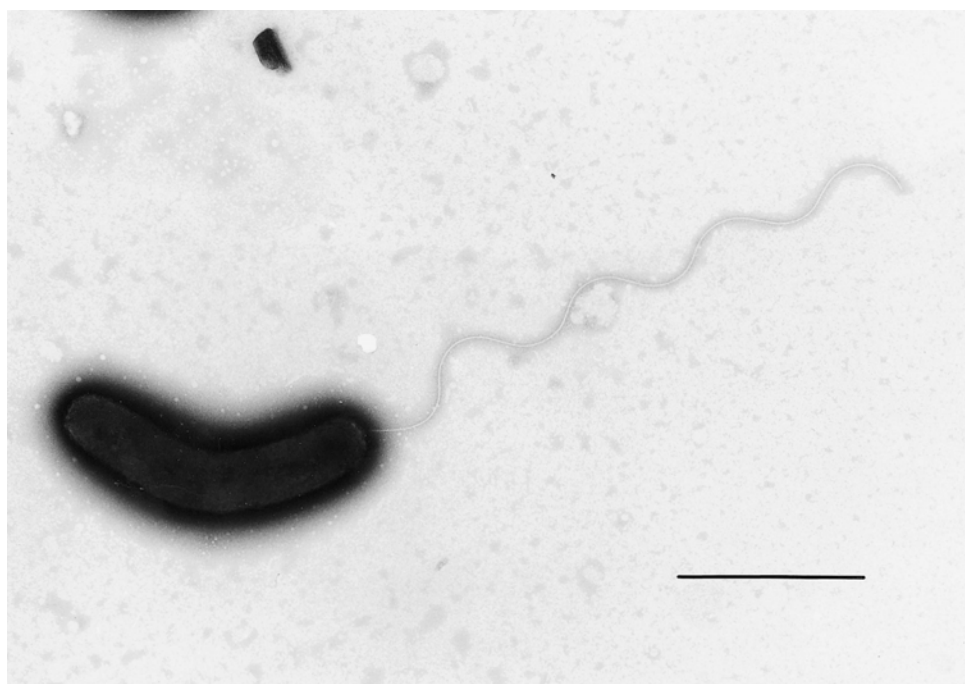


FIGURE BXII.γ.87. Electron micrograph of *H. marinus* strain MH-110 fixed with glutaraldehyde and negatively stained. Bar = 1 µm.

H₂ and CO₂ as the sole energy and carbon sources, respectively. Lag time of growth is not observed at any tension of oxygen (1–40%), nevertheless hydrogen-oxidizing bacteria are usually microaerophilic. Specific growth rate in the initial stage of growth is very high (0.60–0.67 h⁻¹ at 37, pH 6.5) irrespective of any oxygen tension. The cell density reaches more than 20 g/l dry weight after 24 h cultivation under a continuous gas flow of H₂/O₂/CO₂ in a jar fermentor (Nishihara et al., 1991a). A reduced inorganic sulfur compound such as S⁰, thiosulfate, or tetrathionate is also utilized as the sole energy source under both aerobic (shaking, pO₂ = 20%, pCO₂ = 10%) and microaerobic (standing, pO₂ = 5%, pCO₂ = 10%) conditions in flask cultures, but the growth is very poor. A decrease in pH value occurs when *H. marinus* is cultivated on these sulfur compounds. S⁰ is deposited extracellularly when cultivated with thiosulfate or tetrathionate. Ferrous, manganous, ammonium, or nitrite ions are not utilized as an energy source under both aerobic and microaerobic conditions (Nishihara et al., 1989).

Chemoorganotrophic growth is not observed on the following organic compounds (0.1%, w/v) or media: acetate, propionate, oxalate, succinate, maleate, fumarate, DL-malate, α-ketoglutarate, DL-lactate, citrate, pyruvate, formate (5–100 mM), glycollate, DL-β-hydroxybutyrate, *p*-hydroxybenzoate, gluconate, D-xylose, D-galactose, sucrose, D-glucose, D-mannose, D-fructose, maltose, L-rhamnose, trehalose, D-raffinose, soluble starch, methanol, ethanol, mannitol, glycine, L-glutamate, L-aspartate, L-serine, L-leucine, L-valine, L-lysine, L-tryptophan, L-histidine, L-alanine, L-proline, L-arginine, L-threonine, L-phenylalanine, nutrient broth medium, yeast extract-peptone-glucose medium, brain heart infusion medium, and ZoBell 2216E medium (Nishihara et al., 1989).

Ammonium ions and urea are utilized as sole nitrogen sources, but nitrate ions, nitrite ions, and gaseous nitrogen are not. Nitrite inhibits growth. No growth is observed in the absence of NaCl, and Na⁺ is not replaceable by K⁺ or Li⁺. The optimum NaCl concentration is 0.5 M. No growth factors are required (Nishihara et al., 1989).

Glycogen is accumulated when *H. marinus* is autotrophically cultivated on hydrogen and CO₂ under nitrogen or magnesium starvation or oxygen limitation. The maximum absorption of the iodine complex is observed at 453 nm (Nishihara et al., 1993).

The major hydrogenase is membrane-bound type. The natural electron carrier has not been identified. Pyridine nucleotides reducing activity is not observed in either the membrane or the soluble fraction. Type *b*, *c*, and *o* cytochromes have been found. The membrane-bound hydrogenase is characterized as highly oxygen-tolerant and thermostable. The optimum temperature for hydrogen-oxidation activity using methylviologen as an electron acceptor is 90°C (Nishihara et al., 1997).

CO₂ is fixed via the Calvin-Benson cycle. Three different molecules of ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCo) have been found in *H. marinus* cells. Two of them are form I RuBisCos (CbbLS-1 and CbbLS-2), which are composed of eight large and eight small subunits, and the other is form II

RuBisCo (CbbM), which is composed of only large subunits (Chung et al., 1993a). The specific factors (τ values) of CbbLS-1 and CbbLS-2 are higher than that of CbbM, but lower than those of other form I RuBisCos from various organisms (Hayashi et al., 1998). The gene sequences of CbbLS-1, CbbLS-2, and CbbM from *H. marinus* are deposited in the DDBJ database under accession numbers D43621, D43622, and D28135, respectively. The deduced amino acid sequences of both CbbLS-1 and CbbLS-2 exhibited particularly high similarities to those of *Chromatium vinosum* and sulfur-oxidizing bacteria (*Thiobacillus*) (Nishihara et al., 1998). Large subunits of form I RuBisCos (CbbLs) from these bacteria formed a monophyletic group in the maximum parsimony tree of *cbbL* amino acids (Delwiche and Palmer, 1996).

The 16S rRNA gene sequence of *H. marinus* is deposited in the DDBJ database under accession number D86374. According to the sequence, *H. marinus* is affiliated with purple sulfur bacteria and sulfur-oxidizing bacteria belonging to the *Gammaproteobacteria*, members of *Thiomicrospira* being the closest relatives (Nishihara et al., 1998). Sequence similarities of the 16S rRNA genes and of the *cbbL* genes among these *Gammaproteobacteria* suggest a common autotrophic ancestry.

H. marinus was isolated from seawater (surface or shallow water) collected at a coastal region in Japan.

ENRICHMENT AND ISOLATION PROCEDURES

The following inorganic medium was used for the enrichment of *H. marinus*: K₂HPO₄, 2.5 g/l; KH₂PO₄, 0.5 g/l; (NH₄)₂SO₄, 2.0 g/l; NaCl, 29.3 g/l; MgSO₄·7H₂O, 0.2 g/l; CaCl₂, 10.0 mg/l; FeSO₄·7H₂O, 10.0 mg/l; and NiSO₄·7H₂O, 0.6 mg/l; plus trace elements (pH 7.0) (Nishihara et al., 1989). The flasks were gassed with a gas mixture of H₂/O₂/CO₂ (75:15:10) (1 atm at room temperature) and shaken at 25°C. Single colony isolation was performed on plate medium solidified with 1.5% agar. The inoculated plates were placed in a desiccator and the gas phase was replaced by a gas mixture of H₂/O₂/CO₂ (85:5:10) (1 atm at room temperature).

MAINTENANCE PROCEDURES

H. marinus can be preserved by a freeze-drying method. 50 mM phosphate buffer (pH 7.0) containing 2% sodium glutamate was used as the suspending medium. The cultures can also be stored in 10–15% (v/v) glycerol at –80°C or in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *HYDROGENOVIBRIO* FROM OTHER GENERA

Hydrogenovibrio is the only taxon for mesophilic and obligately chemolithoautotrophic hydrogen bacteria. *Thiomicrospira* is the closest relative of *Hydrogenovibrio* according to the 16S rRNA gene sequence analysis. Hydrogen autotrophy is the main feature that differentiates these two genera. *H. marinus* grows best on hydrogen as an energy source. On the other hand, growth of *Thiomicrospira* on hydrogen has not been observed (Nishihara et al., 1991b) (see also the table in the description of the genus *Thiomicrospira*).

List of species of the genus *Hydrogenovibrio*

1. ***Hydrogenovibrio marinus*** Nishihara, Igarashi and Kodama 1991b, 132^{VP}
*ma.ri'*nus. L. adj. *marinus* marine, of the sea.

The description of this species is the same as that of the genus.

The mol% G + C of the DNA is: 44.1 (HPLC).

Type strain: MH-110, DSM 11271; JCM 7688.

GenBank accession number (16S rRNA): D86374.

Genus IV. Methylophaga Janvier, Frehel, Grimont and Gasser 1985, 138^{VP}

JOHN P. BOWMAN

Me.thyl.o'pha.ga. Fr. n. *methyl* methyl radical; Gr. v. *phagein* to eat; M.L. n. *Methylophaga* methyl eating.

Rod-shaped cells $0.2 \times 0.9\text{--}1.0 \mu\text{m}$, occurring singly or in pairs. **Motile by means of a single polar flagellum.** **Gram negative.** Spores and other resting stages are not formed. **Chemoheterotrophic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor.** **Restricted methylotroph,** able to utilize only a limited range of substrates, mostly C_1 substrates including methanol, methylamine, dimethylamine, and trimethylamine. The only C_{2+} compounds used for growth are D-fructose and sucrose. The **ribulose monophosphate pathway** is used for assimilation of C_1 compounds. One species can utilize dimethylsulfide and grow mixotrophically by oxidizing H_2S to thiosulfate. Catalase and oxidase positive. **Na^+ and Mg^{2+} are required for optimum growth.** Requires vitamin B_{12} for growth. Major fatty acids are $C_{16:0}$ and $C_{16:1\omega7c}$. The major co-enzyme Q is ubiquinone-8 (Q-8). Belongs to the *Gamma*proteobacteria. Isolated from seawater, brackish waters and marine benthic mats.

The mol% G + C of the DNA is: 42–45.

Type species: **Methylophaga marina** Janvier, Frehel, Grimont and Gasser 1985, 138.

FURTHER DESCRIPTIVE INFORMATION

Morphology Cells lack differentiated resting cells, prostheca, and other ultrastructural peculiarities except for having a somewhat wide periplasmic space with a width of 20–30 nm (Janvier et al., 1985). Cell walls are of the usual type for Gram-negative cells.

Cultural characteristics *Methylophaga* strains grow well on artificial seawater (ASW)-methanol medium¹ but do not grow on most standard complex marine media such as marine agar (ZoBell, 1941). The colonies of *Methylophaga* strains are circular and convex, possess entire edges and a butyrous consistency, and have a white to pale pink color.

Methylophaga strains require Na^+ and Mg^{2+} ions (20 mM and 5 mM, respectively) for optimal growth (Janvier et al., 1985), though some strains still grow weakly in their absence (Urakami and Komagata, 1987). Most *Methylophaga* strains are able to tolerate up to 9–10% NaCl. "*M. limanica*" exhibits optimum growth with 0.5–1.0 M NaCl (Doronina and Trotsenko, 1997). Strains usually possess a requirement for cyanocobalamin (vitamin B_{12}). The optimal pH for growth is equivalent to that of seawater (7.4–7.8) and the optimal growth temperature is 30–37°C. *M. thalassica* is slightly more thermotolerant than *M. marina* and can grow at 42°C (Urakami and Komagata, 1987).

Carbon and energy sources Only a few substrates are utilized for carbon and energy. These are mainly C_1 compounds including methanol and methylamine, which are utilized by all *Methylophaga* strains (Table BXII.γ.49). No strains grow on methane or methanesulfonate. Among a large number of C_{2+} compounds tested, only D-fructose supports growth, except for "*M. limanica*", which can also use sucrose (Janvier et al., 1985; de Zwart et al., 1996; Doronina et al., 1997).

M. sulfidovorans has the unusual ability to grow on dimethylsulfide (DMS) and oxidize it to thiosulfate. The doubling time is approximately 15 h compared with 2–3 h on methanol. This species can also grow mixotrophically by oxidizing H_2S to thiosulfate when in the presence of a carbon substrate such as DMS, which acts as a source of sulfide and carbon (de Zwart and Kuenen, 1997). Sulfide concentrations of 200 $\mu\text{mol/l}$ reduce DMS oxidation by about 50%. The sulfide oxidation rate of *M. sulfidovorans* is equivalent to that in other sulfide-oxidizers such as *Thiobacillus thioparus* (de Zwart et al., 1997).

C_1 metabolism *Methylophaga* strains possess a dissimilatory pathway for methanol oxidation like that found in other methylotrophic bacteria. In this pathway, methanol is oxidized to formaldehyde by methanol dehydrogenase; then the formaldehyde is either oxidized to formate (and then to CO_2) or is fixed as cellular carbon. The methanol dehydrogenase in *Methylophaga marina* has a structure similar to that found in other methylotrophic bacteria (Janvier and Gasser, 1987). Mutagenesis studies indicate that apo-methanol dehydrogenase and pyrroloquinol quinone—the cofactor typically associated with methanol dehydrogenase—are insufficient to constitute an active enzyme and that another unknown factor is required (Janvier et al., 1992). *Methylophaga* strains use the 2-keto-3-deoxy-6-phosphogluconate aldolase variant of the ribulose monophosphate (RuMP) pathway for formaldehyde fixation (Doronina et al., 1997). The assimilation of methylamine, however, uses a variety of enzymes that differ between strains. Methylamine dehydrogenase is the primary methylamine-oxidizing enzyme in *M. marina* and "*M. limanica*" (Doronina et al., 1997), whereas in *M. thalassica* the predominant enzyme is N-methylglutamate dehydrogenase (Janvier et al., 1985). This differentiation is not absolute, however, because "*M. limanica*" and some strains of *M. marina* also exhibit N-methylglutamate dehydrogenase activity (Janvier et al., 1985; Doronina et al., 1997). Methylamine dehydrogenase converts methylamine directly to formaldehyde and ammonia using cytochrome *c* as an electron acceptor (Mehta, 1977). Methylamine can also be converted to γ -glutamylmethylamide (γ -GMA) by the addition of glutamate via γ -GMA synthetase. γ -GMA is then cleaved by γ -GMA lyase to yield ammonia and N-methylglutamate, and the latter is oxidized by N-methylglutamate dehydrogenase to produce formaldehyde and glutamate (Hou, 1984; Doronina et al., 1997). An alternative pathway has been discovered for *Methylophaga* strain AA-30 in which γ -GMA is cleaved into ammonia, formaldehyde, and 2-oxoglutarate; the latter compound is then transaminated to form glutamate (Kimura et al., 1990b, 1995). As in other methylotrophs possessing the RuMP pathway, the tricarboxylic acid cycle lacks 2-oxoglutarate dehydrogenase; moreover, strains lack pyruvate kinase and glyoxylate cycle enzymes (Doronina et al., 1997).

1. Artificial seawater (ASW)-methanol medium contains (g/l of natural or artificial seawater) KH_2PO_4 , 0.14; Bis-Tris, 2.0; and ferric ammonium citrate, 0.06. The pH is adjusted to ~7.5 and 1.6% agar is added (optional). After autoclaving and cooling to ~50°C, methanol and vitamin B_{12} are added to final concentrations of 0.2% and 1 $\mu\text{g/ml}$, respectively. Artificial seawater consists of (g/l distilled water) NaCl, 24; $MgCl_2 \cdot 6H_2O$, 3; $MgSO_4 \cdot 7H_2O$, 2; KCl, 0.5; $CaCl_2 \cdot 2H_2O$, 1; Bis-Tris, 0.5; and 1 ml of trace element solution. The trace element solution (Widdel and Pfennig, 1981) consists of (per liter of distilled water) 25% HCl, 10 ml; $FeCl_2 \cdot 4H_2O$, 1.5 g; $CoCl_2 \cdot 6H_2O$, 0.19 g; $MnCl_2 \cdot 4H_2O$, 0.1 g; $ZnCl_2$, 0.07 g; H_3BO_3 , 0.062 g; $Na_2MoO_4 \cdot 2H_2O$, 0.036; $NiCl_2 \cdot 6H_2O$, 0.024 g; and $CuCl_2 \cdot 2H_2O$, 0.017 g. The ferrous chloride is dissolved in HCl and then added to the distilled water with the other reagents.

TABLE BXII.γ.49. Phenotypic characteristics of the species of the genus *Methylophaga*^a

Characteristic	1. <i>M. marina</i>	2. <i>M. sulfidovorans</i>	3. <i>M. thalassica</i>	4. " <i>M. limanica</i> "
Nitrate reduction	—	—	d (weak)	—
Mixotrophy (H ₂ S oxidation)	—	+	—	nd
Catalase, oxidase	+	+	+	+
Optimal temperature (°C)	30–37	22–24	30–37	30–37
Optimal pH	7.5	7.4–7.8	7.5	6.7–7.2
Growth at 42°C	—	—	+	—
Growth with 9% NaCl	+	—	d	+
Methylamine dehydrogenase	+	nd	—	+
N-Methylglutamate dehydrogenase	d	nd	+	+
<i>Utilizable carbon sources:</i>				
Methanol, methylamine	+	+	+	+
Dimethylamine	d	+	+	—
Trimethylamine	—	—	d	—
Dimethylsulfide	—	+	—	nd
D-Fructose	+	—	d	+
Sucrose	—	—	—	+
Mol% G + C of DNA (<i>T_m</i>)	43–44	42	44–45	43–44

^aFor symbols see standard definitions; nd, no data.

Nitrogen metabolism *Methylophaga* strains utilize a variety of nitrogen sources including ammonia, nitrate, short chain amines (Kimura et al., 1990a), amino acids, and yeast extract. In general, strains cannot reduce nitrate to nitrite, though some *Methylophaga thalassica* strains have been reported to produce weak positive results in this test (Urakami and Komagata, 1987). Ammonia is assimilated by the glutamate cycle (glutamate synthase and glutamine synthetase) (Doronina et al., 1997).

DNA composition The mol% G + C of the DNA has been analyzed for several *Methylophaga* strains. In *M. marina* the values range from 43–44. *M. thalassica* has slightly higher values of 44–45. Additional *Methylophaga*-like strains analyzed by Urakami and Komagata (1987) have values as high as 48.

Fatty acids and quinones Chemotaxonomically the strains of *Methylophaga* form a homogeneous group. The major fatty acids include C_{12:0} (4–5%), C_{14:0} (3–4%), C_{16:0} (33–35%), C_{16:1 ω7c} (37–44%), C_{18:1 ω7c} (4–11%) and C_{10:0 3OH} (3–5%). The major isoprenoid quinone is ubiquinone-8 and all strains contain significant levels of squalene (Urakami and Komagata, 1986c, 1987).

Ecology Limited information is available on the distribution and ecology of *Methylophaga*. Studies using indirect immunofluorescence have shown that *M. marina* strains can be most frequently isolated from open seawater rather than estuarine waters (Sieburth et al., 1993). The ability of *M. sulfidovorans* to utilize dimethylsulfide and use H₂S for energy suggests the species has a significant role in cycling of DMS and sulfide in marine ecosystems (de Zwart et al., 1997). DMS is the breakdown product of dimethylsulfoniopropionate (DMSP), an important osmolyte produced copiously by various phytoplankton species (Ledyard et al., 1993; Visscher and Taylor, 1994). Other bacteria that can metabolize DMS, such as *Hyphomicrobium* and *Thiobacillus* spp. (Kanagawa and Mikami, 1989; Visscher and Taylor, 1994), have been isolated from freshwater sites. "*M. limanica*" was isolated from the hypersaline waters of a liman adjoining the Black Sea (Doronina and Trotsenko, 1997).

ENRICHMENT AND ISOLATION PROCEDURES

M. marina and *M. thalassica* can be isolated following initial enrichment of marine samples in ASW-methanol broth for 3–6 d at about 30°C (Janvier et al., 1985). Enrichment cultures are then serially diluted onto ASW-methanol agar and incubated at

30°C for about 1 week. The appearance of pale pink colonies 0.5–2 mm in size are usually indicative of *Methylophaga* strains. A similar approach can be taken for "*M. limanica*" except NaCl levels used should match sample salinity levels (from 1.0–2.0 M). *M. sulfidovorans* can be isolated from benthic mats and sediment by enriching samples in ASW-dimethylsulfide (DMS)² broth at 20–25°C (de Zwart et al., 1996). A sample slurry is serially diluted in ASW-DMS broth and growth from the highest dilution is then transferred to fresh medium. Isolation must be performed in liquid medium because no growth occurs on DSM agar media. The DMS-utilizing culture is passaged until it becomes morphologically homogeneous. Alternatively, purification can be achieved on ASW-methanol agar, with checks made to ensure DMS utilization is maintained.

MAINTENANCE PROCEDURES

Methylophaga can be preserved at –80°C in ASW-methanol media supplemented with 30% DMSO. Strains can also be freeze-dried using 20% skim milk as a cryoprotectant.

DIFFERENTIATION OF THE GENUS *METHYLOPHAGA* FROM OTHER GENERA

To date, *Methylophaga* is the only known restricted methylotroph isolated from marine habitats. A requirement for Na⁺ ions differentiates this genus from other metabolically similar restricted methylotrophs belonging to the *Betaproteobacteria*, i.e., *Methylophilus*, *Methylobacillus*, and *Methylovorus*.

TAXONOMIC COMMENTS

16S rRNA sequences are available for *Methylophaga marina*, *Methylophaga thalassica*, and *Methylophaga sulfidovorans*, and these species form a tightly knit group with an overall sequence divergence of approximately 4–5%. *Methylophaga* forms a distinct lineage within the *Gamma*proteobacteria (Ando et al., 1989; Janvier and

2. ASW-dimethylsulfide (DMS) broth consists of (g/l) NaCl, 15–25; NH₄SO₄, 0.5; CaCl₂·6H₂O, 0.33; KCl, 0.2; MgSO₄·7H₂O, 1; KH₂PO₄, 0.02; Na₂CO₃, 2; FeSO₄·7H₂O, 1 mg; trace element solution (similar to that given for ASW-methanol medium except that it is supplemented with 6 mM Na₂SeO₄), 1 ml. The pH is adjusted to ~7.5 and 10- or 20-ml portions of the medium are dispensed into 60- or 100-ml serum vials, sealed by butyl-rubber stoppers, and autoclaved. After the medium is cooled, DMS and vitamin B₁₂ are added to final concentrations of 1 mM and 1 µg/ml, respectively.

Grimont, 1995; de Zwart et al., 1996). The closest relatives include the marine genus *Cycloclasticus* and various type I methanotrophs including *Methylomonas* and *Methylobacter*. *Methylophaga* are unrelated to other restricted methylotrophic bacteria, including *Methylophilus* (Jenkins et al., 1987), *Methylobacillus* (Urakami and Komagata, 1986b), and *Methylovorus* (Govorukhina and Trotsenko, 1991), which belong to the *Betaproteobacteria* and which are nonhalophilic.

"*Alteromonas thalassomethanolica*" (e.g., ATCC 33145 and NCIMB 2160) and "*Methylomonas thalassica*" (e.g., NCIMB 2162, ATCC 33146) described originally by Yamamoto et al. (1980) were renamed *Methylophaga thalassica* by Janvier et al. (1985). Further phenotypic analysis by Urakami and Komagata (1987)

indicates that *M. marina* and *M. thalassica* are very similar phenotypically and that they can only be identified definitively by a limited number of biochemical characteristics (Table BXII.γ.49), 16S rRNA analysis, and DNA–DNA hybridization. DNA–DNA hybridization studies indicate that *M. marina* strains share 81 to >100% hybridization, *M. thalassica* strains share 61–74% hybridization, and that no significant DNA hybridization occurs between the species.

The putative species "*M. limanica*" has yet to be validated. Doronina et al. (1997) reported that "*M. limanica*" strain hal-6 exhibited 30–40% DNA hybridization with *M. marina* and *M. thalassica*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *METHYLOPHAGA*

Phenotypic characteristics differentiating *Methylophaga* species are shown in Table BXII.γ.49. Electrophoresis of enzymes has also been used to distinguish *Methylophaga* species. For example, the electrophoretic mobility of glucose-6-phosphate dehydrogen-

ase is slightly greater for *M. thalassica* strains vs. *M. marina* strains. The methanol dehydrogenases of these species also differ in terms of mobility (Janvier et al., 1985).

List of species of the genus *Methylophaga*

1. ***Methylophaga marina*** Janvier, Frehel, Grimont and Gasser 1985, 138^{VP}
ma.ri'na. L. adj. *marinus* of the sea.

Short, slender, Gram-negative rods $0.2 \times 1 \mu\text{m}$. Colonies on artificial seawater-methanol agar are white to pale pink, convex, and circular with an entire edge and a creamy consistency. Na^+ ions, Mg^{2+} ions, and vitamin B_{12} are required for growth.

Optimal growth temperature, 30–37°C; range, 10–40°C. Optimal pH, ~7.5.

Other characteristics are as given in the genus description and are presented in Table BXII.γ.49.

Isolated from seawater and marine sediment.

The mol% G + C of the DNA is: 43–44 (T_m).

Type strain: 222, ATCC 35842, DSM 5989, NCIMB 2244.

GenBank accession number (16S rRNA): X87338, X95459.

2. ***Methylophaga sulfidovorans*** de Zwart, Nelisse and Kuenen 1998, 1083^{VP} (Effective publication: de Zwart, Nelisse and Kuenen 1996, 266.)

sul.fi.do'vo.rans. L. n. *sulfur* sulfur; L. prefix *sulf-* + *ide* probably originally formed from German *sulfid*; L. v. *voro* to devour; M.L. part. adj. *sulfidovorans* sulfide-devouring.

Short, slender, Gram-negative motile rods, approximately $0.9 \times 0.2 \mu\text{m}$. Na^+ ions, Mg^{2+} ions, and vitamin B_{12} are required for growth. Other characteristics are as given in the genus description and are presented in Table BXII.γ.49.

Isolated from a marine benthic mat off the coast of the Netherlands.

The mol% G + C of the DNA is: 42 (T_m).

Type strain: YK-4015, ATCC 33146, DSM 5690, IAM 12458, LMG 4055, NCIMB 2163.

GenBank accession number (16S rRNA): X87339, X95460.

3. ***Methylophaga thalassica*** Janvier, Frehel, Grimont and Gasser 1985, 138^{VP}

tha.las'si.ca. Gr. n. *thalassica* of the sea.

Short, slender, Gram-negative rods, $0.2 \times 1 \mu\text{m}$. Colonies on artificial seawater-methanol agar are white to pale pink, convex, and circular with an entire edge and a creamy consistency. Na^+ ions, Mg^{2+} ions, and vitamin B_{12} are required for growth.

Optimal temperature, 30–37°C; range, 10–42°C. Optimal pH, ~7.5.

Other characteristics are as given in the genus description and are presented in Table BXII.γ.49.

Isolated from seawater.

The mol% G + C of the DNA is: 44–45 (T_m).

Type strain: YK-4015, ATCC 33146, DSM 5690, IAM 12458, LMG 4055, NCIMB 2163.

GenBank accession number (16S rRNA): X87339, X95460.

Species Incertae Sedis

1. **"*Methylophaga limanica*"** Doronina, Krauzova and Trotsenko 1997, 438.

li.ma'ni.ca. M.L. n. *limanica* of a liman (a salty lagoon-like water body).

Short Gram-negative rods possessing a single polar flagellum. Do not form spores or prostheca.

Colonies on artificial seawater salts-methanol agar appear opaque, white, and circular, with an entire edge and a creamy consistency. No growth occurs on nutrient agar.

Requires vitamin B_{12} and Na^+ ions for growth (0.5–1.0 M is optimal). Optimal temperature and pH for growth are 30–37°C and pH 6.7–7.2, respectively.

Isolated from a Black Sea liman.

Other characteristics are as given in the genus description and are presented in Table BXII.γ.49.

The mol% G + C of the DNA is: 43.5 (T_m).

Deposited strain: hal 6.

Genus V. Thioalkalimicrobium Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 576^{VP}

THE EDITORIAL BOARD

Thi.o.al.kal.i.mi.cro'bi.um. Gr. n. *thios* sulfur; N.L. n. *alkali* Arabic *al the*; *qaliy* soda ash; Gr. adj. *micros* small; Gr. masc. n. *bios* life; M.L. n. *Thioalkalimicrobium* sulfur alkaline microbe.

Motile Gram-negative rods, curved rods, or spirilla (0.4–0.5 × 0.8–1.5 µm). 1–3 monopolar flagella. Possess **carboxysomes**. **Chemolithoautotrophs; obligate aerobes. Oxidize sulfide and thiosulfate to sulfate**; do not oxidize sulfite. Carbon assimilation via Calvin cycle. Halotolerant up to 1.2–1.5 M Na⁺; require at least 0.2–0.3 M NaCl.

The mol% G + C of the DNA is: 48–51.2.

Type species: Thioalkalimicrobium aerophilum Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578 (*Thioalkalimicrobium aerophilum* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578.)

FURTHER DESCRIPTIVE INFORMATION

These organisms were found in the water and sediments of alkaline lakes in which sulfide is produced in the underlying sediments. Growth in the laboratory was studied in carbonate/bicarbonate buffered medium containing sulfide, thiosulfate, and polysulfide. Tetrathionate and elemental sulfur were not used by most strains. The pH optimum was 9–10. Ammonia, nitrite, ni-

trate, and thiocyanate were used as nitrogen sources (Sorokin et al., 2001a).

Analysis of the 16S rDNA sequence of the type strains of *T. aerophilum* and *T. sibiricum* revealed a close relationship with the genus *Thiomicrospira*, particularly with *Thiomicrospira pelophila*. Thus, the genus *Thioalkalimicrobium* is a member of the family "Piscirickettsiaceae" in the *Gammaproteobacteria* (Sorokin et al., 2001a).

ENRICHMENT AND ISOLATION PROCEDURES

These bacteria were obtained by enrichment and isolation in a carbonate/bicarbonate buffered liquid salts medium containing thiosulfate. For some strains, the amount of NaCl in the base medium had to be increased (Sorokin et al., 2001a).

MAINTENANCE PROCEDURES

Thioalkalimicrobium spp. can be stored at 4°C for 2 months between transfers in liquid salts medium (pH 10) containing 5mM MgCl₂ to stabilize the cell walls. They can also be stored in 10% glycerol (v/v) at –80°C (Sorokin et al., 2001a).

List of species of the genus Thioalkalimicrobium

1. **Thioalkalimicrobium aerophilum** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578^{VP} (*Thioalkalimicrobium aerophilum* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578.)
ae.ro.phi' lum. Gr. masc. n. *aër* gas Gr. adj. *philus* loving; M.L. n. *aerophilum* air-loving.

Description as for the genus with the following additional characteristics. Grow preferentially in atmospheric O₂ levels. Tetrathionate oxidized by most strains. Colonies flat and spreading.

The mol% G + C of the DNA is: 48–51.2 (*T_m*).

Type strain: AL 3, DSM 13739.

GenBank accession number (16S rRNA): AF126548.

2. **Thioalkalimicrobium sibiricum** Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578^{VP} (*Thioalkalimicrobium sibiricum* (sic) Sorokin, Lysenko, Mityushina, Tourova, Jones, Rainey, Robertson and Kuenen 2001a, 578.)
si.be'ri.cum. M.L. masc. adj. *sibericus* from Siberia.

Description as for the genus with the following additional characteristics. Grow preferentially at less than atmospheric O₂ levels. Tetrathionate not oxidized by most strains. Colonies compact.

The mol% G + C of the DNA is: 48.2–49.5 (*T_m*).

Type strain: AL 7, DSM 13740.

GenBank accession number (16S rRNA): AF126549.

Genus VI. Thiomicrospira Kuenen and Veldkamp 1972, 253^{AL}

THORSTEN BRINKHOFF, JAN KUEVER, GERARD MUYZER AND HOLGER W. JANNASCH

Thi.o.mi.crospi'ra. Gr. n. *thios* sulfur; Gr. adj. *micros* small, little; Gr. n. *spira* spiral; M.L. fem. n. *Thiomicrospira* small sulfur spiral.

Small **spiral-, comma-, or rod-shaped cells** (0.2–0.5 × 0.8–3.0 µm) that occur singly. Gram negative. Motile by means of a polar flagellum or nonmotile. No resting stages are known. **Aerobic. Chemolithoautotrophic, using reduced inorganic sulfur compounds, such as S⁰, sulfide, tetrathionate, thiosulfate** (but not sulfite or thiocyanate), **and CO₂ as carbon source.** CO₂ is fixed via the **Calvin-Benson cycle** (ribulose 1,5-bisphosphate carboxylase is present). The final oxidation product is sulfate, but S⁰ may accumulate in the medium. During growth acid is produced. Ammonium salts serve as nitrogen source. Na⁺ ions are required for growth. The temperature optimum lies between 28°C and

40°C. The optimal pH varies from 6.0 to 8.0. Metabolic properties are very similar to those of the genus *Thiobacillus*.

Q-8 is the major ubiquinone. Members of a **monophyletic group** within the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 39.6–49.9.

Type species: Thiomicrospira pelophila Kuenen and Veldkamp 1972, 253.

FURTHER DESCRIPTIVE INFORMATION

All described *Thiomicrospira* spp. are obligate chemolithotrophs, and thus use CO₂ as their main carbon source. They do not have

functional citric acid or glyoxylate cycles. However, *Thiomicrospira pelophila* can assimilate small amounts of organic substrates, such as acetate, while growing autotrophically under thiosulfate limitation (Kuenen and Veldkamp, 1972). Heterotrophic growth, as reported for *Thiomicrospira thyasirae* (Wood and Kelly, 1989) could not be confirmed with a culture obtained from the Deutsche Sammlung für Mikroorganismen und Zellkulturen (DSMZ) (Kuever, unpublished results). Strains of *Thiomicrospira crunogena* can use polymetal sulfides as electron donors (Eberhard et al., 1995; Wirsén et al., 1998). Members of the genus *Thiomicrospira* require oxygen as terminal electron acceptor for respiration. Good growth occurs at low to saturated oxygen conditions.

Temperature ranges of species of this genus are from 3.5 to 40°C. At optimal pH the maximum growth rates on TP or T-ASW medium¹ at 30°C ranged from 0.3 to 0.8 h⁻¹. The optimal pH values are between 6.0 and 8.0; however, growth of particular species was reported down to values of 4.0 or up to 9.0.

On 1% (w/v) agar plates with TP medium small, smooth, whitish colonies are first obtained. Later, bigger yellowish colored colonies with a rough surface are formed due to intensive sulfur precipitation. Production and consumption of S⁰ globules is both pH and thiosulfate dependent (Javor et al., 1990). Under stress conditions, pleomorphism and spirals up to 5–15 µm length were observed for *T. thyasirae* (Wood and Kelly, 1993).

Although *Thiomicrospira* was present in one freshwater habitat (Brinkhoff and Muyzer, 1997), so far, all described species and strains were isolated from marine environments. Na⁺ ions are required, with an optimal concentration of around 470 mM. The named species were obtained from intertidal mud flats, a continental shelf, and deep-sea hydrothermal vents. Deep-sea strains are barotolerant, but not barophilic, and are known to excrete approximately 10% of their fixed carbon as low molecular weight organic compounds (Ruby and Jannasch, 1982; Jannasch et al., 1985; Wirsén et al., 1998).

Thiomicrospira appears to be ecologically very significant in hydrothermal vent systems (e.g., Muyzer et al., 1995; Brinkhoff et al., 1999c), while in an intertidal mud flat habitat the genus was found in much lower densities than other sulfur-oxidizing bacteria (Brinkhoff et al., 1998).

ENRICHMENT AND ISOLATION PROCEDURES

Procedures for the enrichment and isolation of these organisms were described several times (Ruby et al., 1981; Kuenen et al., 1992; Brinkhoff and Muyzer, 1997). *Thiomicrospira* can be obtained after inoculation of a general marine medium for chemolithoautotrophic bacteria containing thiosulfate as the sole electron donor with a water or sediment sample. Incubation is

aerobic at room temperature. After growth is obtained, indicated by a decreasing pH value, and often sulfur precipitation, an aliquot should be transferred to a 1–1.5% (w/v) agar plate, containing the same medium, to promote formation of the distinctive colony types. For some isolates enrichment by filtration of the sample before inoculation of the medium was described. A pore size of 0.22 µm was recommended for *T. pelophila* (Kuenen and Veldkamp, 1972) and one of 0.45 µm for *T. chilensis* (Brinkhoff and Muyzer, 1997).

MAINTENANCE PROCEDURES

Active cultures can be stored at 4°C for 2 weeks, but should then be transferred in fresh medium, to prevent acid formation that might easily result in irreversible inhibition of growth. For long-term storage, members of this genus are best maintained in liquid nitrogen or at –80°C in TP medium containing 20% (v/v) glycerol or 5% (v/v) DMSO (final concentration). They can be subcultured on mineral salts medium with sulfide, thiosulfate, or S⁰.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Two oligonucleotide probes specific for the genus *Thiomicrospira* were designed targeting the 16S rRNA gene (Brinkhoff and Muyzer, 1997). The probes are located at positions 128–145 (TMS128F: 5'-GAA TCT RCC CTT TAG TTG-3') and 830–849 (TMS849R: 5'-CTT TTT AAT AAG RCC AAC AG-3') according to *Escherichia coli* numbering (Brosius et al., 1981). They were checked against the small-subunit rRNA sequences stored in the Ribosomal Database Project (RDP) using the Check Probe software program (Maidak et al., 1997) and contained at least two mismatches to all non-*Thiomicrospira* sequences. The probes require a hybridization temperature of 52°C for specific dot-blot hybridization with nearly complete PCR-amplified 16S rRNA gene fragments. If both probes are used combined as PCR primers, an annealing temperature of 44°C is required for specific amplification of *Thiomicrospira* 16S rRNA genes (Brinkhoff and Muyzer, 1997).

Two slightly different versions of probe TMS849R, which differ by an A/G mismatch, were developed for use in rRNA slot-blot hybridization (TMS849A: 5'-CTT TTT AAT AAG ACC AAC A-3'; TMS849G: 5'-CTT TTT AAT AAG GCC AAC A-3') (Brinkhoff et al., 1998). Both probes contain one mismatch to several *Thiomicrospira* sequences. The combination of both probes matches all known species. Specificity was tested with rRNA isolated from *Thiomicrospira* species having either no mismatch or one. Both probes have a T_d of 41°C. Hybridization is done at 30°C, and washing at 41°C (Brinkhoff et al., 1998).

DIFFERENTIATION OF THE GENUS *THIOMICROSPIRA* FROM OTHER GENERA

Thiomicrospira species can be differentiated from the related genus *Thiobacillus* by their lower mol% G + C content and their higher sulfide tolerance (800–1000 µM) (Kuenen and Veldkamp, 1972; Ruby and Jannasch, 1982; Jannasch et al., 1985). Not all *Thiomicrospira* species have a spiral cell form; *T. chilensis* and *T. frisia* are rod-shaped. Therefore, morphology should not be used for distinguishing *Thiomicrospira* from thiobacilli. The most reliable method for a clear identification is the use of the genus-specific probes, as described above, or sequencing of the 16S rRNA gene followed by phylogenetic analysis.

1. TP (*Thiomicrospira pelophila*) medium (Kuenen and Veldkamp, 1972, modified by Brinkhoff et al., 1999b): synthetic seawater medium contained (g/l deionized water) NaCl, 25.0; (NH₄)₂SO₄, 1.0; MgSO₄·7H₂O, 1.5; CaCl₂·2H₂O, 0.42; K₂HPO₄, 0.5; Na₂S₂O₃·5H₂O, 5.0. One ml of a trace element solution containing EDTA (Widdel and Bak, 1992) was used. Vitamin B₁₂ is added after sterile filtration (final concentration 50 ng/l). Bromothymol blue (4 mg/l) is used as pH indicator. K₂HPO₄ and Na₂S₂O₃·5H₂O are autoclaved separately, each in 10% of the final volume. After autoclaving the pH is adjusted to 7.2 with sterile 0.4% (w/v) Na₂CO₃ solution.

T-ASW (thiosulfate artificial seawater) medium (Jannasch et al., 1985): synthetic seawater medium contained (g/l deionized water) NaCl, 25.0; (NH₄)₂SO₄, 1.0; MgSO₄·7H₂O, 1.5; CaCl₂·2H₂O, 0.3; KH₂PO₄, 0.42; NaHCO₃, 0.2; Trizma hydrochloride, 3.0; Na₂S₂O₃, 1.58. One ml trace element solution SL-8 (Biebl and Pfennig, 1978) was used. Phenol red (2 ml of a 0.5% [w/v] solution) is used as pH indicator. KH₂PO₄ and Na₂S₂O₃ are autoclaved separately, each in 10% of the final volume. After autoclaving the pH is adjusted to 7.5 with sterile 0.4% (w/v) Na₂CO₃ solution.

TAXONOMIC COMMENTS

Phylogenetic analysis of nearly complete 16S rRNA gene sequences showed that all *Thiomicrospira* spp., with the exception of *Thiomicrospira denitrificans* (Timmer-ten Hoor, 1975), form a monophyletic group (bootstrap value of 100%) within the *Gammaproteobacteria* (Fig. BXII.γ.88). *T. denitrificans* belongs to the *Epsilonproteobacteria* (Muyzer et al., 1995). This organism requires reclassification as a new genus. Even though *T. denitrificans*, based on phylogenetic and physiological analysis, does not belong to the genus *Thiomicrospira*, it is listed here as a species. The same description of this organism as given by Kuenen and Robertson (1989) was used. However, its characteristics are not included in the general description of the genus *Thiomicrospira* and the specific oligonucleotides do not match the 16S rRNA gene of *T. denitrificans*.

The *Thiomicrospira* group includes *Hydrogenovibrio marinus*, which differs from *Thiomicrospira* spp. only in the ability to use hydrogen as electron donor (Table BXII.γ.50) (Nishihara et al., 1991b, 1998). Since hydrogen utilization is a ubiquitous trait among many phylogenetically diverse bacteria, hydrogen utilization alone may be insufficient to separate this organism as a separate genus.

The rod-shaped *Thiomicrospira* species and strains, i.e., *T. frisia*, *T. chilensis*, *Thiomicrospira* spp. str. Art-3, and *Thiomicrospira* spp. str. Milos T-2, form a subcluster within the *Thiomicrospira* branch. This indicates a common ancestor of these organisms (Kuever, unpublished).

The reliable differentiation of *Thiomicrospira* species requires integration of physiological and genomic data. In cases of high 16S rRNA gene homology among closely related strains, DNA-

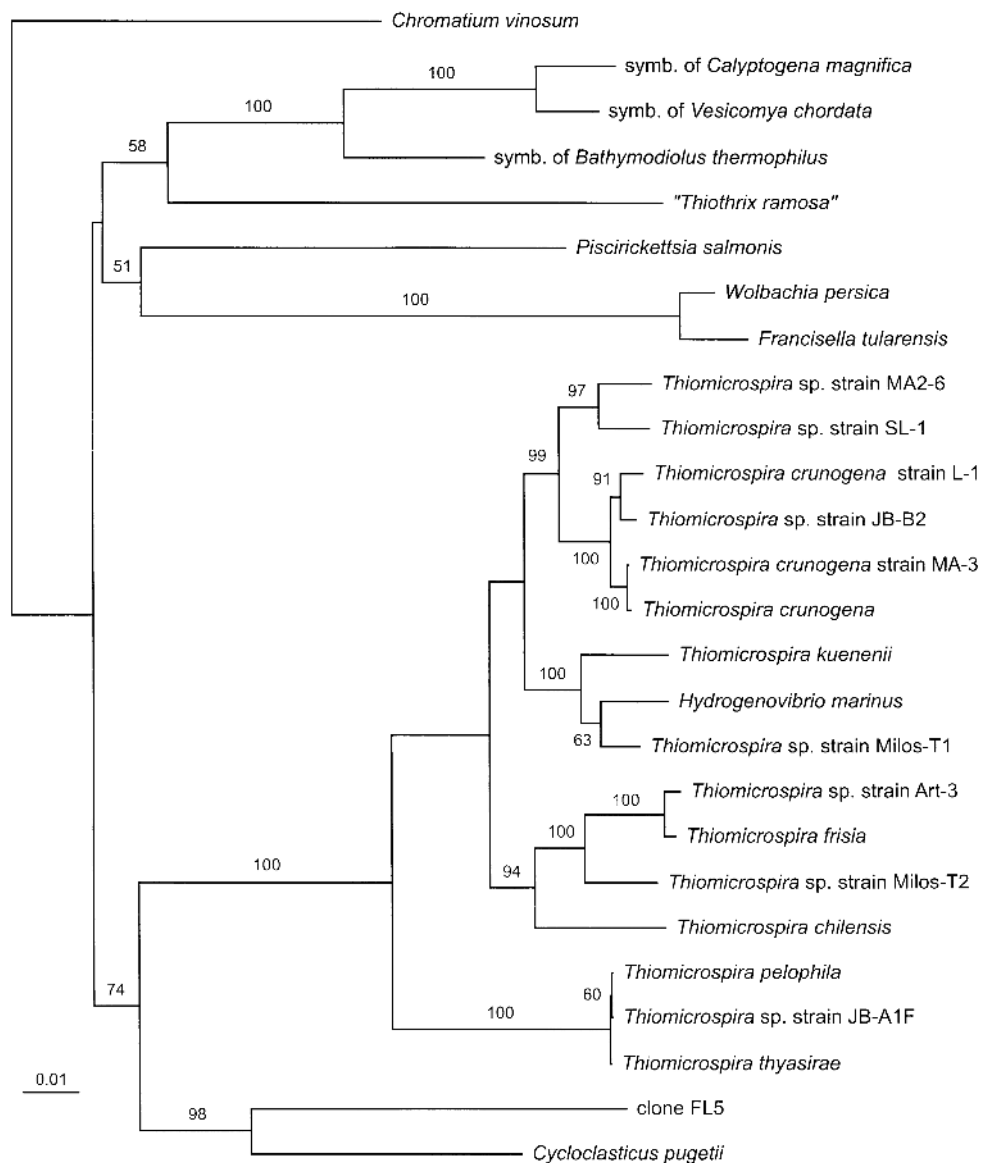


FIGURE BXII.γ.88. Neighbor joining tree based on nearly complete 16S rRNA gene sequences, showing the phylogenetic relationships between different *Thiomicrospira* spp. and closely related bacteria. The *Thiomicrospira* sequences were aligned to prealigned sequences taken from the Ribosomal Database Project (RDP; Maidak et al., 1997). *Chromatium vinosum* was taken as the outgroup. The numbers on the branches refer to bootstrap values; only values of 50% and higher are shown. Bar = 0.01 estimated nucleotide change per sequence position.

TABLE BXII.γ.50. Morphological and physiological characteristics of the species composing the genus *Thiomicrospira*, *Hydrogenovibrio marinus*, and *Thiomicrospira denitrificans*^a

Characteristic	1. <i>T. pelophila</i>	2. <i>T. chilensis</i>	3. <i>T. crunogena</i>	4. <i>T. frisia</i>	5. <i>T. kuenenii</i>	6. <i>T. thyasirae</i>	7. <i>H. marinus</i>	8. <i>T. denitrificans</i>
Class:								
<i>Gammaproteobacteria</i>	+	+	+	+	+	+	+	
<i>Epsilonproteobacteria</i>								+
Shape	Vibrio	Rod	Vibrio	Rod	Vibrio	Vibrio	Vibrio	Spiral
Width (μm)	0.2–0.3	0.3–0.5	0.4–0.5	0.3–0.5	0.3–0.4	0.3	0.2–0.5	0.3
Length (μm)	1.0–2.0	0.8–2.0	1.5–3.0	1.0–2.7	1.0–2.5	0.8–1.2	1.0–2.0	variable
Motility	+	+	+	+	+	–	+	–
Mol% G + C content	45.7 (44) ^b	49.9	44.2 (42) ^b	39.6	42.4	45.6 ^c (52) ^b	44.1	36.0
Major ubiquinone	Q-8	Q-8	Q-8	Q-8	Q-8	Q-8 ^d	Q-8	nd
Maximum growth rate with thiosulfate, h ⁻¹	0.3	0.4	0.8	0.45	0.35	0.07 ^e	0.6	0.06
Optimal pH	7.0 ^f	7.0	7.5–8.0	6.5	6.0	7.5	6.5	7.0
pH range	5.6–9.0 ^f	5.3–8.5	5.0–8.5	4.2–8.5	4.0–7.5	7.0–8.4	nd	nd
Optimal temperature (°C)	28–30	32–37	28–32	32–35	29–33.5	35–40	37	22
Temperature range (°C)	3.5–42	3.5–42	4–38.5	3.5–39	3.5–42	10–45	nd	nd
Na ⁺ requirement	+	+	+	+	+	+	+	–
Optimal Na ⁺ -concentration (mM)	470	470	nd	470	470	430	500	nd
Na ⁺ -concentration range (mM)	40–1240 ^f	100–1240	at least 45	100–1240	100–640	250–3000	nd	nd
Vitamin B ₁₂ -dependent	+	–	–	–	–	–	–	nd
Calvin Benson cycle	+	+	+	+	+	+	+	+
Formation of sulfur from thiosulfate at pH 7.0 in liquid medium	+	+	+	–	–	+	+	–
Growth on H ₂ as the sole electron donor	– ^c	–	– ^c	–	–	– ^c	+	nd
Fully aerobic growth	+	+	+	+	+	+	+	–
Denitrification	–	–	–	–	–	–	–	+
Heterotrophic growth	–	–	–	–	–	– ^c	–	nd

^aSymbols: +, 90% or more strains are positive; –, 90% or more strains are negative; nd, not determined; data from Kuenen and Veldkamp, 1972, 1973; Timmer-ten Hoor, 1975; Ruby and Jannasch, 1982; Jannasch et al., 1985; Kuenen and Robertson, 1989; Wood and Kelly, 1989, 1993; Nishihara et al., 1991b, 1998; Brinkhoff et al. 1999a, 1999b; Kuever, unpublished data.

^bThe values in brackets are from the original description, determined by thermal denaturation.

^cJ. Kuever, unpublished results.

^dIn the original description Q-10 was found (Wood and Kelly, 1989); we could only detect Q-8 and traces of Q-7 (J. Kuever, unpublished results).

^eIn the original description the maximum growth rate on acetate was 0.53 (h⁻¹) (Wood and Kelly, 1989). Growth rates on thiosulfate seemed to be similar to *T. pelophila* in the range of 0.2–0.3 (Kuever, unpublished results).

^fBrinkhoff et al., 1999a.

DNA hybridization was necessary to decide whether these strains belong to one species or not. This method revealed that *Thiomicrospira* strains, which were obtained from different hydrothermal vent systems, and which showed specific metabolic adaptations to their environments, belong to one species, *T. crunogena* (Wirsén et al., 1998). DNA–DNA hybridization also confirmed *T. pelophila*, *T. crunogena*, *T. kuenenii*, *T. frisia*, and *T. chilensis* as distinct species (Brinkhoff et al., 1999a, 1999b).

ACKNOWLEDGMENTS

The authors wish to thank Andreas Teske, Carl O. Wirsén, and Stefan Sievert for help and advice. The help of Hirofumi Nishihara and Yasuo Igarashi is greatly appreciated, because they supplied us with unpublished

results and submitted manuscripts. We are grateful to Gijs Kuenen for providing the photograph of *Thiomicrospira pelophila*.

FURTHER READING

- Brinkhoff, T. and G. Muyzer. 1997. Increased species diversity and extended habitat range of sulfur-oxidizing *Thiomicrospira* spp. Appl. Environ. Microbiol. 63: 3789–3796.
- Kuenen, J.G., L.A. Robertson and O.H. Tuovinen. 1992. The genera *Thiobacillus*, *Thiomicrospira*, and *Thiosphaera*. In Balows, Trüper, Dworkin, Harder and Schleifer (Editors), The Prokaryotes. A Handbook on the Biology of Bacteria: Ecophysiology, Isolation, Identification, Applications, 2nd Ed., Vol. III, Springer Verlag, New York. pp. 2638–2657.

List of species of the genus *Thiomicrospira*

1. ***Thiomicrospira pelophila*** Kuenen and Veldkamp 1972, 253^{AL}.
pe.lo'phi.la. Gr. n. *pelos* mud; Gr. adj. *phila* loving; N.L. adj. *pelophila* mud-loving.

Colonies on thiosulfate agar are small, white, and often surrounded by a halo of sulfur. On soft thiosulfate agar, colonies may be up to 1 cm in diameter and grow down into the agar as large disks. These disks are white from

precipitated sulfur and only sparsely populated with bacteria. In media containing more than 0.2–0.3 mM thiosulfate, sulfur is formed. The type strain requires vitamin B₁₂; others do not. Sulfide tolerance is high in this species compared with *Thiobacillus* species (Kuenen and Veldkamp, 1972). Even though CO₂, fixed via the Calvin-Benson cycle, is the primary carbon source, small amounts of organic compounds such as acetate and succinate may serve as a

secondary source under energy-limited conditions (Kuenen and Veldkamp, 1973). See Table BXII.γ.50 for further physiological and biochemical details; Fig. BXII.γ.89 illustrates morphology and flagellation.

Isolated from a marine intertidal mud flat of the Dutch Wadden Sea.

The mol% G + C of the DNA is: 45.7 (HPLC), 44.0 (T_m).

Type strain: ATCC 27801, DSM 1534.

GenBank accession number (16S rRNA): L40809.

2. ***Thiomicrospira chilensis*** Brinkhoff, Muyzer, Wirsén and Kuever 1999a, 878^{VP}
chi.len' sis. M.L. adj. *chilensis* from Chile, a country of South America.

Cells are rod-shaped and motile, but during late exponential growth, cells have a strong tendency to clump and to form aggregates. Strong sulfur precipitation in liquid batch cultures as *T. crunogena*. See Table BXII.γ.50 for further details. Found in sediment samples of *Thioploca* mats obtained from the Chilean coast shelf. Isolated from a sample that was filtered through a 0.45-μm pore-size filter before inoculation (Brinkhoff and Muyzer, 1997).

The mol% G + C of the DNA is: 49.9 (HPLC).

Type strain: ATCC 700858, DSM 12352.

GenBank accession number (16S rRNA): AF013975.

3. ***Thiomicrospira crunogena*** Jannasch, Wirsén, Nelson, and Robertson 1985, 422^{VP}
cru.no.ge' na. Gr. n. *crunos* spring; Gr. v. *genomai* to be generated; Gr. part. adj. *crunogena* spring-born.

T. crunogena can withstand relatively high concentrations of sulfide, continuing to fix CO₂ at concentrations up to 800 μM. Sulfide is toxic at 2000 μM. Strong precipitation of sulfur in liquid batch cultures. Barotolerant; at 250 atm, the CO₂ fixation rate is 80% of that obtained at 1 atm. Growth does not occur at 400 atm or above. Cells are motile and vibrio shaped. For further physiological and biochemical characteristics see Table BXII.γ.50.

Isolated from a marine hydrothermal vent system (21°N, 109°W, East Pacific Rise) at a depth of 2600 m.

The mol% G + C of the DNA is: 44.2. (HPLC), 42.0 (T_m).

Type strain: ATCC 35932, DSM 12353, LMD 84.00.

GenBank accession number (16S rRNA): L40810.

Additional Remarks: Two further described *Thiomicrospira* strains belong to this species as determined by 16S rRNA gene sequence analysis and DNA–DNA hybridization (Wirsén et al., 1998): *Thiomicrospira* sp. str. L-12 (Ruby and Jannasch, 1982) and *Thiomicrospira* sp. str. MA-3 (Wirsén et al., 1993). All three strains of *T. crunogena*, but especially strain MA-3, can grow on sulfidic minerals (Eberhard et al., 1995). They also show robust growth properties compared to the other species of the genus *Thiomicrospira*. *Thiomicrospira crunogena* str. L-12 was isolated from mussel periostracum from the Galapagos Rift hydrothermal vent site and deposited under DSM 12346. The 16S rRNA gene sequence is available from GenBank under accession number L01576. *Thiomicrospira crunogena* str. MA-3 was isolated from a polymetal sulfide rock from the Mid-Atlantic Ridge TAG hydrothermal vent site and deposited under ATCC 700270 (DSM 12354). The 16S rRNA gene sequence is available from GenBank under accession number AF069959.

4. ***Thiomicrospira frisia*** Brinkhoff, Muyzer, Wirsén, and Kuever 1999b, 391^{VP}
fr'i.si.a. M.L. fem. adj. *frisica* pertaining to Frisia, a coastal region in northwestern Germany and the Netherlands from where the organism was obtained.

When thiosulfate is used as the primary energy source very small amounts of sulfur and sulfite are produced. Compared to most other *Thiomicrospira* spp., sulfur precipitation in liquid medium is less obvious. Cells are rod-shaped and motile, but during late exponential growth cells have a strong tendency to clump and to form aggregates. See Table BXII.γ.50 for more details on the physiology. See Fig. BXII.γ.90A for details of the morphology.

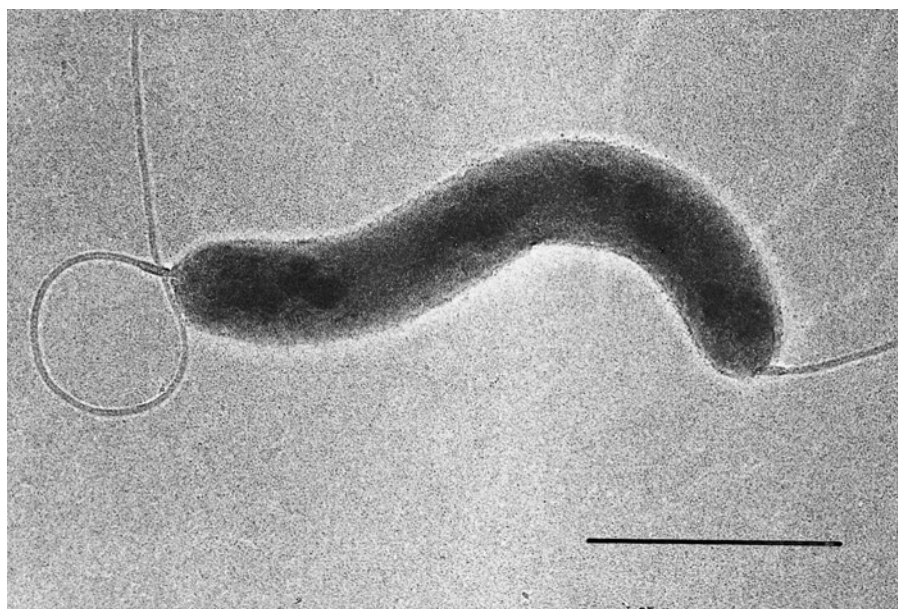


FIGURE BXII.γ.89. Electron micrograph of a *T. pelophila* cell, which was platinum-shadowed to show the shape and the flagellation. Bar = 0.5 μm.

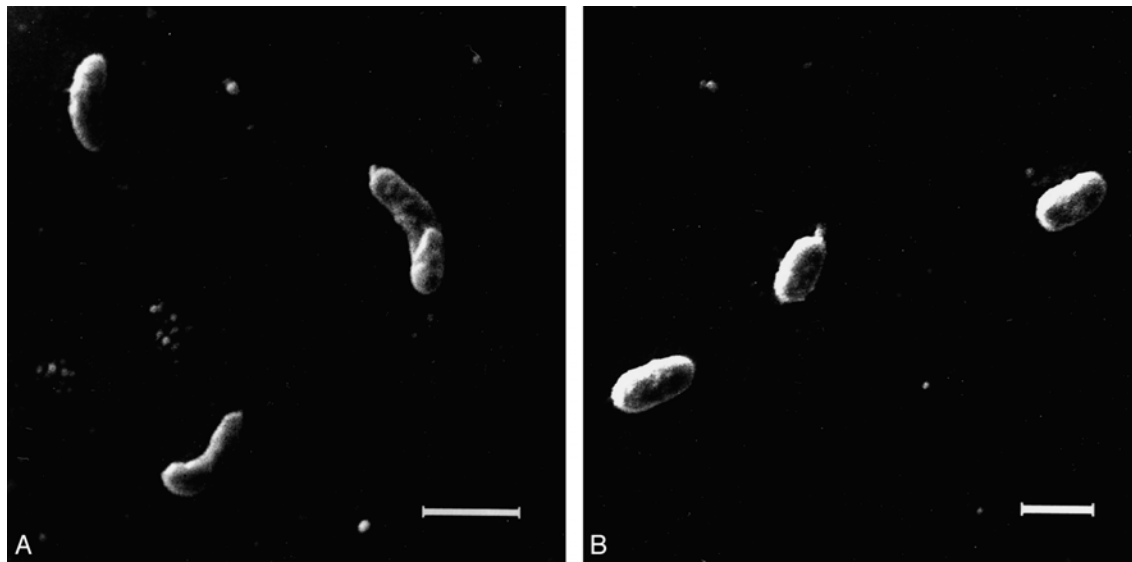


FIGURE BXII.γ.90. Electron micrographs of *T. frisia* and *T. kuenenii* to demonstrate the two morphologies found for *Thiomicrospira* spp. A, morphology of *T. kuenenii*, which is typical for most *Thiomicrospira* spp.; B, rod-like morphology of *T. frisia*. Bars = 1 μ m. (Reproduced with permission from T. Brinkhoff et al., International Journal of Systematic Bacteriology 49: 385–392, 1999, ©International Union of Microbiological Societies.)

Isolated from a marine intertidal mud flat of the German Wadden Sea.

The mol% G + C of the DNA is: 39.6 (HPLC).

Type strain: ATCC 700878, DSM 12351.

GenBank accession number (16S rRNA): AF013974.

5. *Thiomicrospira kuenenii* Brinkhoff, Muyzer, Wirsén, and Kuever 1999b, 390^{VP}

kue.nen'i.i. M.L. gen. n. *kuenenii* of Kuenen; named after J.G. Kuenen, a Dutch microbiologist of the Delft School of Microbiology, who proposed the genus *Thiomicrospira*.

When thiosulfate is used as the primary energy source very small amounts of sulfur and sulfite are produced. Compared to most other *Thiomicrospira* spp., sulfur precipitation in liquid medium is less obvious. See also Table BXII.γ.50 and Fig. BXII.γ.90B. Isolated from a marine intertidal mud flat of the German Wadden Sea.

The mol% G + C of the DNA is: 42.4 (HPLC).

Type strain: ATCC 700877, JB-A1, DSM 12350.

GenBank accession number (16S rRNA): AF013978.

6. *Thiomicrospira thyasirae* (Wood and Kelly 1989) Wood and Kelly 1995, 418^{VP} (Effective publication: Wood and Kelly 1993, 47) (*Thiobacillus thyasiris* Wood and Kelly 1989, 165.) *thy.a.si'rae*. M.L. adj. *Thyasira* a genus of the bivalve mollusk family of the *Thyasiridae*.

Morphologically and physiologically nearly identical to *T. pelophila*. 16S rRNA gene sequences of *T. thyasirae* and *T. pelophila* are almost identical (one mismatch in 1424 bp) (Brinkhoff and Muyzer, 1997). DNA–DNA hybridization data will be required to decide whether *T. thyasirae* can be retained as a separate species. Some biochemical and physiological properties of a culture obtained from the DSMZ vary to the original description (Table BXII.γ.50). Cells exhibit pleomorphism in response to environmental conditions.

Isolated from the gills of the mussel *Thyasira flexuosa* collected from Plymouth Sound, U.K.

The mol% G + C of the DNA is: 45.6 (HPLC), 52 (T_m).

Type strain: ATCC 51452, DSM 5322.

GenBank accession number (16S rRNA): AF016046.

Other Organisms

1. *Thiomicrospira* sp. strain MA2–6 Muyzer, Teske, Wirsén and Jannasch 1995.

Vibrio-shaped isolate from a marine hydrothermal vent site at the Mid-Atlantic Ridge, (23°2' N, 44°57' W).

Deposited strain: not publicly available

GenBank accession number (16S rRNA): L40811.

Additional Remarks:

2. *Thiomicrospira* sp. strain SL-1 Brinkhoff and Muyzer 1997.

Vibrio-shaped cells isolated from a microbial mat of the hypersaline Solar Lake, Sinai, Egypt. Physiological properties similar to *T. crunogena*, but probably a different species

(Kuever, unpublished). Isolated from a sample that was filtered through a 0.45- μ m pore-size filter before inoculation.

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF013971.

Additional Remarks:

3. *Thiomicrospira* sp. strain JB-B2 Brinkhoff and Muyzer 1997.

Vibrio-shaped cells, isolated from a water sample overlying a marine intertidal mud flat of the Jadebusen Bay, Germany. High similarity (99%) of the 16S rRNA gene sequence to the sequence of *T. crunogena* and therefore probably a strain of this species (Wirsén et al., 1998).

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF013972.

Additional Remarks:

4. *Thiomicrospira* sp. strain Milos T-1 Brinkhoff, Sievert, Kuever and Muyzer 1999c.

Vibrio-shaped isolate obtained from sediment of a marine shallow-water hydrothermal vent off the island of Milos, Greece.

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF082327.

Additional Remarks:

5. *Thiomicrospira* sp. strain Art-3 Brinkhoff and Muyzer 1997.

Rod-shaped isolate obtained from sediment of a saline spring at the city of Artern, Germany. This organism is morphologically and physiologically similar to *T. frisia* (Kuever, unpublished).

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF013973.

Additional Remarks:

6. *Thiomicrospira* sp. strain Milos T-2 Brinkhoff, Sievert, Kuever and Muyzer 1999c.

Rod-shaped isolate obtained from the sediment surface of a marine shallow water hydrothermal vent off the island of Milos, Greece.

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF082328.

Additional Remarks:

7. *Thiomicrospira* sp. strain JB-AF1 Brinkhoff and Muyzer 1997.

Isolated from a sample obtained from a marine intertidal mud flat of the Jadebusen Bay, Germany. The sample was

filtrated through a 0.22- μ m pore-size filter before inoculation. Morphologically and by 16S rRNA gene sequence identical to *T. pelophila*.

Deposited strain: not publicly available

GenBank accession number (16S rRNA): AF013976.

Additional Remarks:

Species Incertae Sedis

1. *Thiomicrospira denitrificans* Timmer-ten Hoor 1975, 344^{AL} *de.ni.tri'fi.cans*. L. prep. *de* away from; L. n. *nitrum* soda; M.L. n. *nitrum* nitrate; M.L. v. *denitrifico* denitrify; M.L. part. adj. *denitrificans* denitrifying.

Colonies are brown, 1–2 mm in diameter. Sulfur is not deposited. Chemolithotrophic, obtaining energy from the oxidation of reduced sulfur compounds such as sulfide and thiosulfate. CO₂, fixed via the Calvin cycle, is the primary source of carbon. Ammonium salts serve as nitrogen source. This species is microaerophilic, only able to use oxygen as an electron acceptor when pO₂ is <0.5%. In practice, aerobic growth has been attained only in an oxygen-limited chemostat. Nitrate or nitrite can serve as electron acceptors, being reduced to molecular nitrogen. The dissimilatory nitrate reductase is constitutive. Although isolated from marine samples, this species does not have a requirement for NaCl. See Table BXII.γ.50 for further physiological details.

Found in mud of marine tidal flats where sulfide is produced.

The mol% G + C of the DNA is: 36.0 (*T_m*).

Type strain: ATCC 33889, DSM 1251.

GenBank accession number (16S rRNA): L40808.

Family III. Francisellaceae fam. nov.

ANDERS B. SJÖSTEDT

Fran.ci.sel.la'ce.ae. M.L. fem. n. *Francisella* type genus of the family; *-aceae* ending for the family *Francisella*.

Short, rod-shaped or coccoid cells 0.2–0.7 × 0.2–1.7 μ m. Gram negative, faintly staining. Endospores are not produced. **Non-motile. Aerobic.** Produce acid from D-glucose. **Cysteine is required for, or enhances, growth.** Weakly catalase positive. **H₂S is produced in cysteine-supplemented medium. Unique fatty acid composition;** long-chain saturated and monounsaturated C₁₈ to C₂₆ acids, relatively large amounts of saturated even-chain acids (C_{10:0}, C_{14:0}, and C_{16:0}), and two long-chain hydroxy acids (C_{16:0} 3OH and C_{18:0} 3OH). Susceptible to tetracycline and chloramphenicol. Intracellular parasites, transmissible to a wide range of animal species, including man, and the etiological agents of the zoonotic disease tularemia. Most commonly isolated from rodents, lagomorphs, blood-sucking insects, and ticks. The family belongs to the *Gamma*proteobacteria. Signatures of *Francisella* 16S rRNA gene sequences are listed in Table BXII.γ.51. A gene encoding a 17-kDa lipoprotein of *F. tularensis* is conserved in all francisellae investigated.

The mol% G + C of the DNA is: 30–36.

Type genus: **Francisella** Dorofe'ev 1947, 176.

FURTHER DESCRIPTIVE INFORMATION

The family comprises closely related organisms within the single genus *Francisella*. The two recognized species, *F. tularensis* and *F. philomiragia*, show a 16S rRNA gene sequence similarity of $\geq 98.3\%$ (Forsman et al., 1994). The family is circumscribed and readily distinguishable from other families based on a unique set of phenotypic characteristics that include a coccoidal morphology, Gram negativity, acid but no gas production from a limited number of carbohydrates, requirement for cysteine, and a unique fatty acid composition. There is no DNA relatedness between francisellae and members of other genera showing minor serological cross-reactivity (Ritter and Gerloff, 1966). This has been substantiated by phylogenetic analysis based on comparison of 16S rRNA gene sequences, demonstrating that the family is a member of the *Gamma*proteobacteria (Forsman et al., 1994). It is most closely related to, but still relatively distant from, the family *Piscirickettsiaceae*.

F. tularensis shows slow growth and usually requires cysteine

TABLE BXII.γ.51. 16S rRNA sequence signatures for the genus *Francisella*^a

Position ^b	<i>Francisella</i> ^c	<i>Bacteria</i> ^d
151	C	A (G)
336	A	G
402	U	C
849	C	U (G)
889	C	G
1280	A	U (C)
1290	G	C (U)
1453	A	G
1455	U	A (G)

^aAdapted from Forsman et al. (1994).

^bPositions numbered according to the nomenclature of *E. coli*.

^cBases found in the species of *Francisella*, including *W. persica* and the symbionts isolated from *D. andersoni* and *O. moubata*.

^dBases found in other *Bacteria*. Letter in parentheses denote bases found in less than one-fourth of the aligned strains of *Bacteria*.

or cystine for growth. The supplements enhance, but are not required for growth of isolates of *F. tularensis* subsp. *novicida* and *F. philomiragia*. Cystine heart agar supplemented with hemoglobin, glucose cystine heart agar, or Mueller–Hinton agar supplemented with hemoglobin and IsoVitalX are commonly used for cultivation of *F. tularensis*. Although cultivation from clinical samples of tularemia patients is often avoided due to the contagiousness of *F. tularensis*, isolation is usually possible from most types

of relevant clinical specimens, e.g., wound samples, lymph node aspirates, and blood, after prolonged cultivation.

Colony-type variants of *F. tularensis*, including those associated with virulence and immunogenicity, have been described (Eigelsbach et al., 1951). Possibly, these variants can be attributed to a reversible phase variation in their lipopolysaccharide (LPS) (Cowley et al., 1996). The LPS of *F. tularensis* has been characterized and contains a unique somatic antigen (Vinogradov et al., 1991). Extracts of *F. tularensis* strains contain phosphoantigens, pyrophosphoesters bound to derivatives of uracil monophosphate or thymidine monophosphate (Poquet et al., 1998). A gene encoding a 17-kDa lipoprotein of the live vaccine strain, *F. tularensis* strain LVS, is conserved in *Francisella* species (Sjöstedt et al., 1992) and has been used both for diagnostic and taxonomic purposes (Sjöstedt et al., 1997; Niebylski et al., 1997). Francisellae are widely distributed in nature. Members of *F. tularensis* are the etiologic agents of tularemia, a zoonotic disease of humans. Tularemia is transmitted by direct contact with infected animals, through contaminated water or food, or by vectors such as biting insects or ticks within endemic regions over the northern hemisphere. The disease shows epidemic outbreaks, both in humans and in animals. The bacterium is highly virulent for humans and numerous animal hosts, in particular, rodents, hares, and rabbits. Clinical manifestations in humans depend on the means of transmission. *F. tularensis* subsp. *novicida* and *F. philomiragia* have been linked to water-borne transmission and can cause disease in immunocompromised individuals.

Genus I. Francisella Dorofe'ev 1947, 176^{AL}

ANDERS B. SJÖSTEDT

Fran.ci.sel'la. M.L. dim. ending -ella; M.L. fem. n. *Francisella* named after Edward Francis, American bacteriologist, who extensively studied the etiologic agent and pathogenesis of tularemia and is credited with naming the disease.

Short, rod-shaped or coccoid cells, 0.2–0.7 × 0.2 μm (*Francisella tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica*) or 0.7 × 1.7 μm (*F. philomiragia* and *F. tularensis* subsp. *novicida*) when examined during active growth, pleomorphism occurs thereafter. Gram negative, faintly staining. Endospores are not produced. **Nonmotile. Aerobic**, growth enhanced under microaerophilic conditions. Colonies are distinct, convex, pale white, and reach maximum size in 3–4 days. **Cysteine (or cystine) is required for growth** (*F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica*) or enhances growth (*F. philomiragia* and *F. tularensis* subsp. *novicida*). Weakly catalase positive. *F. tularensis* is oxidase negative and *F. philomiragia* oxidase positive (Kovacs modification). **H₂S is produced** in cysteine-supplemented medium. Strains from both species, with the exception of *F. tularensis* subsp. *mediasiatica*, produce acid but no gas from D-glucose and maltose. Urease negative and no reduction of nitrate. **Strains are characterized by a unique fatty acid composition**; long-chain saturated and monosaturated C₁₈ to C₂₆ acids, relatively large amounts of saturated even-chain acids (C_{10:0}, C_{14:0}, and C_{16:0}), and two long-chain hydroxy acids (C_{16:0} 3OH and C_{18:0} 3OH).

Strains are susceptible to quinolones, aminoglycosides, tetracycline, and chloramphenicol. β-Lactamase positive. *F. tularensis* is a **facultative intracellular bacterium** and *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica* **cause the zoonotic disease tularemia** in humans and

animals, predominantly rodents and lagomorphs. Also *F. tularensis* subsp. *novicida* may cause a tularemia-like disease. *F. philomiragia* has been associated with septicemic and pneumonic disease in humans with immunocompromised conditions.

16S rDNA sequence analysis has demonstrated that the genus belongs to the *Gammaproteobacteria*. *Francisella* strains show a high degree of 16S rRNA gene sequence similarity, ≥98.3%, and *F. tularensis* strains >99.8%. 16S rRNA gene sequence signatures of *Francisella* are shown in Table BXII.γ.51. Phylogenetically related to *Wolbachia persica* and two other endosymbionts of ticks. DNA relatedness of *F. tularensis* strains, including isolates of *F. tularensis* subsp. *novicida*, is 75–98% (65°C, hydroxyapatite method) to the type strain *F. tularensis* ATCC 6223^T. Members of *F. philomiragia* show 72–83% relatedness to the type strain *F. philomiragia* ATCC 25015^T.

Type species: Francisella tularensis (McCoy and Chapin 1912) Dorofe'ev 1947, 176 (*Bacterium tularense* McCoy and Chapin 1912, 61.)

FURTHER DESCRIPTIVE INFORMATION

Francisella, the only recognized genus in “Francisellaceae”, belongs to the *Gammaproteobacteria*. The most closely related organisms are the obligate intracellular bacterium *W. persica*, first isolated from the tick *Argas persicus (arbores)*, and two other endosymbionts of ticks (Fig. BXII.γ.91). *Francisella* shows the closest relationship, albeit a rather distant one, to *Piscirickettsiaceae*. The

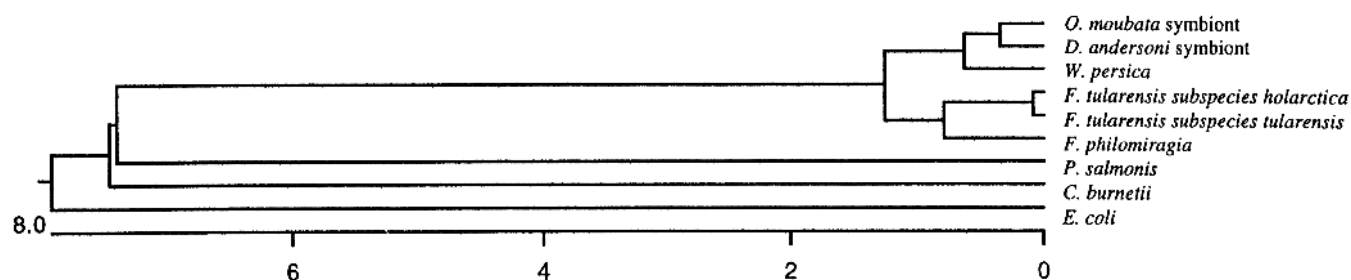


FIGURE BXII.γ.91. Evolutionary distance tree showing relationships among the recognized members of *Francisellaceae*, *Wolbachia persica*, and symbionts from the ticks *Dermacentor andersoni* and *Ornithodoros moubata*. *Escherichia coli* and *Coxiella burnetii* are included for further comparison to other members of the *Gamma*proteobacteria. The scale represents the number of nucleotide substitutions calculated against the number of compared nucleotide sites.

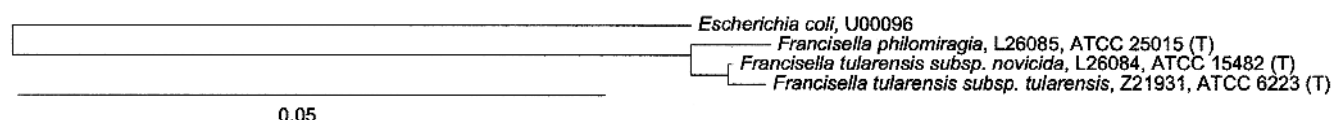


FIGURE BXII.γ.92. Phylogenetic relationships of the genus *Francisella*. The distances in the tree were calculated using 1101 positions (the least-squares method, Jukes-Cantor model). (Courtesy T. Lilburn of the Ribosomal Database Project.)

type species, *F. tularensis*, is the source of virtually all knowledge of the genus (see Fig. BXII.γ.92).

Cells of *Francisella* typically appear as short, rod-shaped or coccoid cells. They are small, singly occurring, nonmotile, and nonsporulating. Cells of *F. tularensis* are $0.2\text{--}0.7 \times 0.2 \mu\text{m}$ and *F. tularensis* subsp. *novicida* and *F. philomiragia* are approximately $0.7 \times 1.7 \mu\text{m}$ (Fig. BXII.γ.93). This morphology occurs typically during the logarithmic growth phase in a liquid medium. In infected tissue and during stationary growth phase, the organisms show pleomorphism. In those situations, cells often elongate, filamentous cells form, filament fragmentation occurs, and a small number of cells not larger than 300 nm are demonstrable (Ribi and Shepard, 1955). Minute forms have been shown to be infectious for mice and pass through membranes with 600 nm average pore diameter (Foshay and Hesselbrock, 1945). *Francisella* cells are faintly stained in tissue. The characteristic morphology is most readily demonstrable with a polychrome (Giemsa) stain.

Virulent strains of *F. tularensis* are surrounded by a relatively thick electron-transparent capsule that can be removed in hypertonic solutions; loss of capsule is accompanied by loss of virulence but viability is unaffected (Hood, 1977). Underlying the capsule is an outer layer surrounding the cell wall and a delicate double-layered cell wall. The lipid concentration of the capsule and cell wall is unusually high for Gram-negative bacteria. The composition is also characteristic for the genus, long-chain saturated and monosaturated C_{18} to C_{26} acids, relatively large amounts of saturated even-chain acids ($C_{10:0}$, $C_{14:0}$, and $C_{16:0}$), and two long-chain hydroxy acids ($C_{16:0} 3\text{OH}$ and $C_{18:0} 3\text{OH}$) (Jantzen et al., 1979; Hollis et al., 1989). Members of the genus also contain a ubiquinone with eight isoprene units as the major isoprenol (Hollis et al., 1989). The live vaccine strain of *F. tularensis*, LVS, grown in a chemically defined medium, was found to have a lipid content of 21% (dry wt) and to contain phosphatidylethanolamine (76%) and phosphatidylglycerol (24%) as its two major phospholipids (Anderson and Bhatti, 1986). As opposed to other Gram-negative bacteria, all members of the genus seem to lack monounsaturated 16-carbon acid and most

strains lack dodecanoic acid but possess small amounts (1–2%) of a 16-carbon aldehyde (Hollis et al., 1989).

Francisella species are strictly aerobic. *F. tularensis* subsp. *holarctica*, *F. tularensis* subsp. *mediasiatica*, and *F. tularensis* subsp. *tularensis* grow more slowly and less abundantly than do isolates of *F. tularensis* subsp. *novicida* and *F. philomiragia*. *F. tularensis* is usually described as a fastidious organism since clinical isolates grow slowly and usually require cysteine or cystine for growth. These supplements enhance but are not required for growth of isolates of *F. tularensis* subsp. *novicida* and *F. philomiragia*. Cystine heart blood agar supplemented with 1% hemoglobin¹ provides excellent conditions for isolation of *F. tularensis* (Payne and Morton, 1992). Other suitable media are Mueller–Hinton agar (Baker et al., 1985)² supplemented with 2% IsoVitalX and 0.025% ferric pyrophosphate, and glucose cysteine heart blood agar³. Glucose cysteine agar (BBL) and cystine heart agar (Difco) are two commercially produced solid media intended for cultivation of *F. tularensis*.

1. Cystine heart blood agar with hemoglobin. Suspend 102 g of cystine heart agar (Difco) in 1.0 liter of distilled water, then mix thoroughly. Heat with frequent agitation and boil for 1 min. Dispense into tubes and sterilize by autoclaving at 118–121°C for 15 min. For larger quantities, autoclave at the same temperature for 30 min. Cool to 45–48°C. Aseptically add 10 ml of hemoglobin and 50 ml of citrated bovine blood. Mix thoroughly and pour into plates. Incubate at 37°C for 24 h before use to decrease surface moisture and to test for sterility. Cystine heart agar without supplements also supports growth of *F. tularensis*.

2. Modified Mueller–Hinton II agar. Suspend 38 g of Mueller–Hinton II agar (BBL) in 1.0 liter of distilled water and mix thoroughly. Heat with frequent agitation and boil for 1 min to completely dissolve the powder. Sterilize by autoclaving at 121°C for 15 min. Aseptically add 0.1% glucose, 2% IsoVitalX (BBL) which contains cysteine, and 0.025% ferric pyrophosphate. The pH is adjusted to 7.0 with sterile 1 N NaOH. Incubate at 37°C for 24 h before use to decrease surface moisture and to test for sterility.

3. Glucose cysteine heart blood agar. Suspend 58 g of glucose cysteine heart agar (BBL) in 1.0 liter of distilled water, then proceed as indicated in the preceding footnote. After autoclaving and cooling, aseptically add 50 ml of citrated bovine, rabbit, or sheep blood. Glucose cysteine heart agar without supplements also supports growth of *F. tularensis*.



FIGURE BXII.γ.93. Electron micrographs of (A) the live vaccine strain (ATCC 29684) of *F. tularensis*, (B) the type strain of *F. tularensis* subsp. *novicida* (ATCC 15482), and (C) the type strain of *F. philomiragia* (ATCC 25015) ($\times 40,000$).

There are also several reports documenting isolation of the organism on media lacking cysteine, e.g., enriched chocolate agar, Thayer–Martin agar, and Mueller–Hinton agar and in blood culture bottles (Provenza et al., 1986; Reary and Klotz, 1988; Bernard et al., 1994). In fact, cysteine enhanced, but was not

critically required for, growth of a number of field isolates on agar bases supplemented with 5% blood (Payne and Morton, 1992). The low-virulence *F. tularensis* type strain, ATCC 6223, is distinctly more fastidious than virulent strains (Eigelsbach and McGann, 1984) and, in contrast to virulent strains, it may not grow on trypticase soy blood agar or brain–heart infusion blood agar (Payne and Morton, 1992). Modified Mueller–Hinton broth (Baker et al., 1985), T-medium (Tresselt and Ward, 1964), or Chamberlain’s chemically defined medium (Chamberlain, 1965) can be used for liquid cultivation of *Francisella*. Provided safety precautions can be met with cultivation from samples of tularemia patients, isolation is usually possible from most types of relevant clinical specimens, e.g., wound samples, lymph node aspirates, and blood. However, the success of cultivation is dependent on rapid transportation and only some 20% of wound samples from patients with ulceroglandular tularemia yielded growth of *F. tularensis* after an average transportation time of 3–4 days (Sjöstedt et al., 1997).

Degradation of a limited number of carbohydrates occurs under suitable conditions. Acid but no gas is produced. Reactions are characteristically slow. All subspecies of *F. tularensis* except *F. tularensis* subsp. *mediasiatica* form acid from D-glucose and maltose whereas degradation of sucrose is a distinguishing feature of *F. tularensis* subsp. *novicida*. *F. tularensis* subsp. *tularensis* differs from subsp. *holarctica* by its capacity to utilize glycerol and the possession of a citrulline ureidase pathway (Marchette and Nicholes, 1961). In addition, the subsp. *mediasiatica* ferments glycerol and citrulline (Olsufjev and Meshcheryakova, 1983). *Francisella* isolates produce H_2S in media supplemented with cysteine whereas the capability to form H_2S in triple sugar iron agar is confined to *F. philomiragia*. Most isolates of *F. philomiragia* and *F. tularensis* subsp. *novicida*, but not of other subspecies of *F. tularensis*, grow in nutrient broth containing 6% NaCl. Generally, strains within each subspecies of *F. tularensis*, as well as *F. philomiragia*, show a high degree of homogeneity in their biochemical profile. However, strains of *F. tularensis* subsp. *novicida* vary in degradation of maltose and glycerol. Moreover, in contrast to an absolute majority of other strains of *F. tularensis*, *F. tularensis* subsp. *novicida* strains lack a growth requirement for cysteine. *F. philomiragia* also lacks a cysteine requirement and degrades sucrose and, in addition, is distinguished by its clearing of blood agar and by being oxidase positive (Kovacs modification). Optimal growth occurs at 37°C for *F. tularensis* and at 25°C or 37°C for *F. philomiragia* strains.

Analysis of DNA relatedness strongly indicated the presence of two species, *F. tularensis* and *F. philomiragia*, within the genus *Francisella* (Hollis et al., 1989). All data were based on relative DNA reassociation at 65°C. The type strain *F. tularensis* ATCC 6223^T showed close to 90% DNA relatedness to strains of *F. tularensis* subsp. *holarctica*, *F. tularensis* subsp. *tularensis*, and *F. tularensis* subsp. *novicida*, thereby supporting the assignment of the former species *F. novicida* as a subspecies of *F. tularensis*. The calculated divergence was <2.5%. The DNA relatedness of the type strain of *F. philomiragia* ATCC 25015^T was, on average, 77% to 13 strains showing the phenotypic characteristics of the species *F. philomiragia*, with a maximum divergence of 3.5%. The type strain of *F. philomiragia* showed 39% average DNA relatedness to the *F. tularensis* type strain and seven other *F. tularensis* strains, including the reference strain of *F. tularensis* subsp. *novicida* and two *novicida*-like strains.

The 16S rDNA sequence similarities of strains within the genus *Francisella* (Forsman et al., 1994) corroborated the differ-

entiation of the genus into the species *F. tularensis* and *F. philomiragia*. In spite of clear phenotypic distinctions between the two species, the 16S rDNA sequence similarity between *F. philomiragia* and *F. tularensis* strains is high, $\geq 98.3\%$. The 16S rDNA sequence analysis strengthened the argument for assigning subsp. *novicida* as a subspecies of *F. tularensis*, the degree of 16S rDNA similarity being $\geq 99.8\%$ (Forsman et al., 1994).

By PCR amplification of the rRNA gene cluster from a large number of *Francisella* strains, combined with endonuclease digestion of the resulting amplicons with *Hae*III, four patterns were demonstrable: one for *F. philomiragia*, one for *F. tularensis* subsp. *novicida*, and two for other strains of *F. tularensis* (Ibrahim et al., 1996).

A spontaneous mutant of *F. tularensis* strain LVS expressing an antigenically distinct LPS has been described (Cowley et al., 1996). This phase variation has been linked to a capacity to switch to increased induction of nitric oxide (NO) and suppression of microbial intracellular growth in rat macrophages. When the LPS returns to its original antigenic form, NO production is reduced and intracellular growth is restored (Cowley et al., 1996). A capsule-deficient mutant of *F. tularensis* strain LVS has been described. In contrast to the encapsulated vaccine strain, the capsule-deficient variant is avirulent in mice and susceptible to the bactericidal effect of nonimmune human serum (Sandström et al., 1988). A cryptic plasmid has been isolated and characterized from *F. tularensis* subsp. *novicida* (Pavlov et al., 1994) and another from *F. tularensis* subsp. *holarctica* (Zakharenko et al., 1985).

Conventional serologic procedures indicate that most *F. tularensis* strains are qualitatively similar in antigenic composition, with the exception of some strains of *F. tularensis* subsp. *novicida*. *F. tularensis* shows a minor antigenic relationship to *Brucella* (Saslaw and Carlisle, 1961) and *Yersinia pestis* (Larson et al., 1951). Immunization with the attenuated vaccine strain *F. tularensis* strain LVS protects mammals against all subspecies of *F. tularensis*. A live *F. tularensis* subsp. *novicida* vaccine protects against homologous infection and against *F. tularensis* strains of moderate virulence (Claflin and Larson, 1972). Commercial *F. tularensis* antiserum agglutinates *F. tularensis* strains of *F. tularensis* subsp. *tularensis* and *F. tularensis* subsp. *holarctica*, but not all strains of *F. tularensis* subsp. *novicida* and not strains of *F. philomiragia* (Claridge et al., 1996).

The lipopolysaccharide (LPS) of *F. tularensis* contains a unique O-antigen, though related to that of some species of *Pseudomonas aeruginosa* and *Shigella flexneri* (Vinogradov et al., 1991). It induced production of tumor necrosis factor- α , interleukin-1, and NO, but only at concentrations 100–1000 times higher than *E. coli* LPS (Sandström et al., 1992; Ancuta et al., 1996). Colony type variants of *F. tularensis*, including those associated with virulence and immunogenicity, have been described (Eigelsbach et al., 1951). Possibly, these variants can be attributed to phase variation in LPS as described above (Cowley et al., 1996).

A number of *F. tularensis* proteins have been cloned and characterized, among them a heat-labile outer membrane protein, a 17-kDa outer membrane lipoprotein, a cytoplasmic 23-kDa protein, an acid phosphatase, a triosephosphate isomerase, a putative ABC transporter protein, and the GroE and DnaK homologs (Nano, 1988; Sjöstedt et al., 1990; Mdululi et al., 1994; Zuber et al., 1995b; Reilly et al., 1996; Ericsson et al., 1997; Golovliov et al., 1997; Keeling and Doolittle, 1997). The gene encoding the 17-kDa lipoprotein of *F. tularensis* strain LVS is conserved in *Francisella* spp. although the immunoreactive proteins in *F. philomiragia* and *F. tularensis* subsp. *novicida* strains display a slightly dif-

ferent M_r (Sjöstedt et al., 1992). Primers specific to this gene can be used to amplify a DNA fragment from the tick endosymbiont designated DAS (Niebylski et al., 1997) and from *W. persica* (Forsman et al., 1994).

Extracts of *F. tularensis* strains have been found to contain so-called phosphoantigens, a pyrophosphoester bound to derivatives of uracil monophosphate or thymidine monophosphate, and potent inducers of V γ 9V δ 2 T cells in humans. Tularemia, but not vaccination with *F. tularensis* strain LVS, resulted in prominent expansion of V γ 9V δ 2 cells (Poquet et al., 1998).

F. tularensis shows pronounced sensitivity to aminoglycosides; streptomycin, gentamicin, and kanamycin, and also tetracyclines, chloramphenicol, and quinolones. Aminoglycosides and tetracyclines have a well-documented effect on tularemia. Notably, administration of doxycycline is recommended for at least 3 weeks (Tärnvik et al., 1997). β -Lactamase activity has been demonstrated in all *F. tularensis* isolates tested, except those from mid-Asia (Pavlovich and Mishankin, 1992). Antibiotic testing of 15 Scandinavian strains showed that all were resistant to β -lactams but had a mixed susceptibility/resistance pattern to macrolides. All strains were susceptible to gentamicin, chloramphenicol, doxycycline, and four quinolones (Scheel et al., 1993). Similar antimicrobial susceptibility was found among *F. philomiragia* strains (Hollis et al., 1989). Since quinolones show low MIC values, have good intracellular penetration, and proved to afford complete recovery in a small number of tularemia patients (Syrjälä et al., 1991; Scheel et al., 1992), they are promising agents for treatment.

F. tularensis is widely distributed in nature and is found throughout the Northern Hemisphere. Large outbreaks, especially in parts of the continental USA, the southern part of the former USSR, and northern Scandinavia, have been reported. Four subspecies, *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, *F. tularensis* subsp. *mediasiatica*, and *F. tularensis* subsp. *novicida* (Olsufjev and Meshcheryakova, 1983; Hollis et al., 1989), are recognized. All have been associated with human disease, although they differ in virulence for humans and rabbits. Tularemia is transmitted by direct contact with infected animals, through contaminated water or food, or by vectors such as biting insects or ticks. Airborne transmission also occurs, especially during processing of agricultural products. The disease is often epidemic, both in humans and in animals, and clinical manifestations depend on the type of reservoir involved and the means of transmission. The bacterium has been isolated from approximately 250 wildlife species, many of which allow transmission of tularemia to man (Olsufjev, 1974).

F. tularensis subsp. *tularensis* has been isolated only in North America and is the predominant form on the continent. It is highly virulent for humans and numerous animal hosts, including domestic rabbits. Most isolates derive from ticks and rabbits. *F. tularensis* subsp. *holarctica* is spread throughout the Northern Hemisphere and causes a milder form of disease. The subspecies seems to be predominantly associated with aquatic rodents, e.g., muskrats and beaver in North America, and ground voles in the former USSR. In Europe, tularemia is most frequently observed in hares and rodents. *F. tularensis* subsp. *mediasiatica*, which has only been isolated in the Central Asian republics of the former USSR, is often found in species of *Lepus* and *Gerbillinae* and in ticks. *F. tularensis* subsp. *holarctica* and *F. tularensis* subsp. *mediasiatica* are significantly less virulent for rabbits than is *F. tularensis* subsp. *tularensis*. *F. tularensis* subsp. *tularensis* is highly infectious and even in humans, the infectious dose is 10 bacteria after

subcutaneous injection and 25 organisms when given by aerosol (McCrum, 1961; Saslaw et al., 1961). Human-to-human transmission is extremely rare.

F. tularensis subsp. *novicida* and *F. philomiragia* are often closely linked to water-borne transmission. The reference strain of subsp. *novicida* was isolated from water in Utah (Larson et al., 1955; Owen et al., 1964) and in some cases of human disease, the suspected source has been natural water (Clarridge et al., 1996). Five isolates of *F. philomiragia*, including the type strain, were isolated from a marshy area in Utah (Jensen et al., 1969). Five out of 13 patients with a disease caused by *F. philomiragia* had a history of near-drowning (Wenger et al., 1989).

Human tularemia is an acute, febrile, zoonotic disease. The clinical picture and severity vary considerably depending on the route of infection and the virulence of the etiological organism. The incubation period is usually 3–6 days but may vary from 1 to 14 days. Before the advent of antibiotic therapy, tularemia in untreated patients in North America was associated with a mortality ranging from 5 to 30% (Dienst, 1963; Evans et al., 1985). Due to the inherently lower virulence of *F. tularensis* subsp. *holartica* and *F. tularensis* subsp. *mediasiatica*, virtually no mortality has been recorded in Eurasia.

F. tularensis subsp. *novicida* is generally less virulent than the other subspecies and the type strain is pathogenic only for mice, guinea pigs, and hamsters, while rabbits and rats are resistant. A few cases of human disease caused by *novicida*-like strains have been reported, including one case resembling glandular tularemia, one case of typhoidal tularemia, and two cases of pneumonia in compromised patients (Hollis et al., 1989; Clarridge et al., 1996). *F. philomiragia* may cause severe disease with pneumonia and/or septicemia. A majority of cases has occurred in immunocompromised individuals with a history of granulomatous disease or near drowning (Wenger et al., 1989).

The laboratory diagnosis of tularemia relies mainly on serology. Agglutination techniques or enzyme-linked immunoassays are routinely used (Carlsson et al., 1975; Viljanen et al., 1983; Koskela and Salminen, 1984). The most common antigen is purified LPS (Carlsson et al., 1975). Seroconversion occurs normally 6–10 days after onset of disease (Koskela and Salminen, 1984). Lymphocyte stimulation tests have also been performed for diagnostic purposes (Tärnvik and Löfgren, 1975; Syrjälä et al., 1984) although not on a routine basis. In a study of 200 cases of tularemia, more than 20% of the patients showed a positive lymphocyte stimulation test during the first week of infection and virtually 100% within 2 weeks (Syrjälä et al., 1984). Cultivation can be used for identification, but due to the contagiousness of the organism, it is usually performed only at reference laboratories. Identification of *F. tularensis* in clinical specimens has been based on direct or indirect fluorescent antibody techniques (Prochazka, 1966; Karlsson and Söderlind, 1973), although not generally included as part of routine diagnostic procedures on human specimens. It is a preferred method for establishing the diagnosis on postmortem material from deceased animals (Mörner, 1981).

Antigen detection in urine and RNA hybridization on a wound specimen have been attempted as diagnostic measures (Sandström et al., 1986; Forsman et al., 1990). PCR has been applied for sensitive and specific detection of *F. tularensis* in blood from infected mice (Long et al., 1993) and for wound specimens from tularemia patients (Sjöstedt et al., 1997). The latter study showed that PCR-based detection is a promising alternative for early diagnosis of ulceroglandular tularemia. Based on the specific sig-

natures of 16S rRNA gene sequences of *Francisella* (Table BXII.γ.51), PCR-based identification of *Francisella* strains at the genus level, and differentiation at the species level, have been developed (Forsman et al., 1994).

ENRICHMENT AND ISOLATION PROCEDURES

Isolation of *F. tularensis* may be attempted from any tissue or specimen. Due to the contagiousness of the organism, all isolation procedures should be performed by trained staff working under biosafety level three conditions using efficient exhaust cabinets. Many laboratories rely only on serologic diagnosis of suspected cases of tularemia.

Detailed procedures for collection and storage of clinical specimens from patients with suspected tularemia have been described (Mollaret, 1991). Procedures for testing water samples or collection fluids from impinged air samples for the presence of *F. tularensis* have been described by Eigelsbach and McGann (1981). Successful isolation will depend on precautions taken to maintain the viability of *F. tularensis* organisms and to prevent overgrowth by normal flora. All specimens should be kept at a temperature of 10°C or lower and procedures for isolation performed as soon as possible.

Recommended media for isolation of *F. tularensis* include cystine heart blood agar supplemented with 1% hemoglobin, Mueller–Hinton agar supplemented with 2% IsoVitalax and 0.025% ferric pyrophosphate, and glucose cysteine heart blood agar. In addition, the commercially available media, glucose cysteine agar (BBL) and cystine heart agar (Difco), allow isolation of the organism from most types of clinical specimens.

Direct culture on any of the recommended media has the advantage of being more rapid and less hazardous than inoculation of susceptible laboratory animals. If specimens are from normally nonsterile sites, incorporation of antibiotics (penicillin, polymyxin B, and cycloheximide) in the medium is recommended (Berdal and Söderlund, 1977; Pavlovich and Mishankin, 1987). If adequate facilities are available for inoculating and housing infectious animals and isolation by direct culture is unsuccessful, materials can be inoculated intraperitoneally into guinea pigs or mice. Inoculation of only a few viable cells of *F. tularensis* usually results in death within 1 week. Moribund animals display gross enlargement of spleen, often 4–5 times normal size, and numerous minute, gray foci of necrosis. The presence of *F. tularensis* is readily confirmed from spleen, liver, and lung of the animals.

MAINTENANCE PROCEDURES

The organism can be maintained in the laboratory by transfer on the same media used for isolation. Attenuation has been observed after repeated passage and for preservation of original phenotypes of strains, plating from original isolates is recommended. Viability of isolates during long-term storage is improved by the addition of stabilizers. Recommended conditions are storage at –70°C or lyophilization in the presence of stabilizers such as 20% sucrose, 2.6% gelatin, 0.2% agar (Faibich, 1959), or 50% skim milk.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

General characteristics Francisellae typically appear as small, coccobacillary, faintly stained Gram-negative organisms demonstrable after 2–4 days of cultivation of clinical specimens.

Media *Francisella* subspecies grow on the commercially available media, glucose cysteine agar, cystine heart agar, and

Mueller–Hinton agar, the latter with IsoVitalX and ferric pyrophosphate as supplements. The etiological agents of tularemia, *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica*, do not grow on standard blood agar, nutrient agar, or other commonly used media lacking cysteine.

Direct identification Direct fluorescent microscopy is rarely used for diagnosis of tularemia in human specimens nowadays, but is a preferred method for establishing the diagnosis on post-mortem material from deceased animals and has shown to be sensitive and specific (Mörner, 1981). A slide agglutination test based on commercial antiserum (Difco) can be used for rapid, tentative identification of *F. tularensis*.

Serology ELISA (enzyme-linked immunosorbent assay) is the standard serological test used to diagnose tularemia. Seroconversion commonly occurs within 6–10 days after onset of disease (Koskela and Salminen, 1984).

Pathology The postmortem findings of tularemia in hares and rodents are characterized by focal necrosis in liver, spleen, and bone marrow (Mörner et al., 1988). In humans, abscess necrosis of lymph nodes with or without epithelioid cell reaction is observed within one to two weeks of infection, and later caseous necrosis occurs (Sutinen and Syrjälä, 1986). These changes closely resemble those observed in tuberculosis.

Gas-liquid chromatography (GLC) GLC analysis of cellular fatty acids identifies the unique fatty acid profile characteristic of *Francisella* strains (Jantzen et al., 1979; Hollis et al., 1989). The method has been, thus far, completely specific in identifying isolates of *Francisella* at the genus level.

Molecular diagnosis Based on *Francisella*-specific 16S rRNA gene sequence signatures, PCR-based methods allowing identification of *Francisella* strains at the genus level and differentiation at the species level have been developed (Forsman et al., 1994). Another molecular target is a gene encoding a 17-kDa lipoprotein from *F. tularensis* strain LVS (Sjöstedt et al., 1990). This gene is conserved in *F. tularensis* and *F. philomiragia* isolates (Sjöstedt et al., 1992). Primer sequences are indicated in Table BXII.γ.52. Multiplex PCR-based identification of the 17-kDa lipoprotein gene and the *Francisella* 16S rRNA gene sequence has been applied on wound specimens from tularemia patients (Sjöstedt et al., 1997).

DIFFERENTIATION OF THE GENUS *FRANCISELLA* FROM OTHER GENERA

Table BXII.γ.53 indicates the characteristics useful to distinguish *Francisella* from other genera of clinically relevant, small, Gram-negative, nonsporeforming, coccobacillary bacteria pathogenic for man and other mammals. Based on DNA relatedness and 16S rRNA gene sequence analysis, no close relatives of *Francisella* are recognized (Ritter and Gerloff, 1966; Forsman et al., 1994).

The requirement of *F. tularensis* strains for cysteine in growth media is a unique characteristic that distinguishes it from virtually all other clinically relevant species. It grows on media developed for isolation of legionellae. Unlike *Legionella* spp., francisellae produce acid from carbohydrates, and do not contain branched-chain fatty acids. The less fastidious species of *Francisella*, *F. philomiragia* and *F. tularensis* subsp. *novicida*, lack the requirement for cysteine and may be more difficult to differentiate from the Gram-negative genera *Pasteurella*, *Brucella*, and *Yersinia*. The latter two genera may show weak agglutination with *F. tularensis* antiserum. In contrast to the case of *Pasteurella*, *Brucella*, and *Yersinia*, a capsule is not easily demonstrable in isolates of *Francisella* and they show a weak Gram stain (Table BXII.γ.53). Other characteristic, distinguishing features of *Francisella* include weak catalase production and no production of NH₃ in liquid media. The primers indicated in Table BXII.γ.52 only give amplicons from francisellae after PCR-based amplification.

TAXONOMIC COMMENTS

Circumscription As indicated in the previous edition of this *Manual*, the taxonomic position of the genus *Francisella* long remained uncertain (Eigelsbach and McGann, 1984). The highly infectious nature of these bacteria has been one of the reasons for slow development of taxonomic classification. After originally being designated “*Bacterium*”, it was subsequently placed in the genus *Pasteurella*, based on serological studies. However, DNA hybridization studies indicated that the genus was not closely related to *Pasteurella*, *Yersinia*, or the coliforms (Ritter and Gerloff, 1966). These early findings using DNA hybridization were corroborated when 16S rDNA sequence analysis demonstrated that although the genus groups within the *Gammaproteobacteria*, it shows no close relationship to other characterized genera (Forsman et al., 1994). *Francisella* is the only genus described in “*Francisellaceae*”. The closest relationship is to the intracellular bacterium *W. persica* and, outside “*Francisellaceae*”, to members of *Piscirickettsiaceae*, in particular the intracellular bacterium *Piscirickettsia salmonis*, the etiological agent of a septicemic disease in fish (Fryer et al., 1992). The cellular fatty acid composition of *Francisella* and the unusually high lipid content of the cell wall are traits distinctly different from those of other Gram-negative bacteria (Hood, 1977).

In 1959, Olsufiev et al. (1959) recognized the presence of two etiological agents of tularemia varying in virulence. The designation *F. tularensis* biovar *tularensis* was proposed for the organism responsible for the severe form of human illness in North America, and *F. tularensis* biovar *palaeartica* for the less virulent organism encountered in Europe, Asia, and North America. In 1970, Olsufiev proposed that the biovars adopt the status of subspecies and suggested the designation *F. tularensis* subsp. *holarctica* for the less virulent form (Olsufiev, 1970). Moreover, the designation *F. tularensis* subsp. *mediasiatica* for strains from the Cen-

TABLE BXII.γ.52. Nucleic acid composition of primers used in the PCR-based identification of *Francisella* isolates at the genus and species level^a

Designation	Sequence (5′–3′)	Direction	Fragment size
F11 ^b	TACCAGTTGGAAACGACTGT	Forward	1141 bp
F5	CCTTTTGTGAGTTTCGCTCC	Reverse	
TUL4-435 ^b	GCTGTATCATCATTTAATAAACTGCTG	Forward	428 bp
TUL4-863	TTGGGAAGCTTGATCATGGCACT	Reverse	

^aAdapted from Forsman et al. (1994) and Sjöstedt et al. (1997).

^bF11 and F5 target the 16S rRNA gene. TUL4-435 and TUL4-863 target a 17-kDa lipoprotein gene. All primers anneal at 65°C.

TABLE BXII.γ.53. Differential characteristics of the genus *Francisella* and other genera of clinically relevant, small, Gram-negative, nonsporeforming coccobacillary bacteria^a

Characteristic	<i>Francisella</i>	<i>Brucella</i>	<i>Pasteurella</i>	<i>Yersinia</i>
Mol% G + C of DNA	33–36	55–58	40–45	46–50
Cells: 0.25–0.5 μm	+ ^b	—	d	—
Capsule easily demonstrated	—	—	+	+
Gram stain: weak counterstain	+	+	—	—
Strictly aerobic	+	+	—	—
CO ₂ enhances growth	d	+	d	—
Optimum temperature, °C	37	37	37	25
Cysteine/cystine enhance growth	d ^c	—	—	—
Chains occur in liquid media	—	+	d	d
Carbohydrates are dissimilated	+ ^d	—	+	+
Acid production from sucrose	—	—	+	d
Catalase	+ ^e	+	+	+
Oxidase	d ^f	d	+	—
NH ₃ produced in liquid media	—	+	+ ^e	+
Sodium ricinoleate solubility	+	—	+	+
Penicillin-sensitive <i>in vitro</i>	—	—	d	d
Arthropod vectors can occur	+	—	—	+
Agglutination of <i>F. tularensis</i> antiserum	d ^g	d, weak	—	d, weak
Amplicons by use of <i>Francisella</i> -specific primers ^h	+	—	—	—

^aAdapted from Eigelsbach and McGann (1984). For symbols see standard definitions.

^bCell size of tularemia-causing species in infected tissue.

^cAbsolute growth requirement for *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica* and enhance growth of *F. tularensis* subsp. *novicida* and *F. philomiragia*.

^dAcid but no gas produced. Weak reaction and some also delayed, 3–7 d.

^eWeak reaction.

^f*F. tularensis* negative, *F. philomiragia* positive (Kovacs modification).

^gNo agglutination of strains of *F. philomiragia* and some of *F. tularensis* subsp. *novicida*.

^hPrimer sequences specific to 16S rDNA and a gene encoding a 17-kDa lipoprotein of *F. tularensis* are given in Table BXII.γ.52.

tral Asian republics of the former USSR was proposed. Biochemical properties, virulence, and geographical origin of strains formed the basis for this differentiation. Subdivisions of *F. tularensis* subsp. *holarctica* were also proposed (Olsufjev and Meshcheryakova, 1983). Strains were distinguished with regard to susceptibility to erythromycin and designated biovar I Ery^S and biovar II Ery^R, and Japanese strains were designated *F. tularensis* subsp. *holarctica* biovar japonica Rodionova. However, no, or very few, other phenotypic attributes support the subdivision of the subspecies. The designations type A and type B have been proposed for *F. tularensis* subsp. *tularensis* and *F. tularensis* subsp. *holarctica*, respectively (Jellison, 1974).

The original strain of *F. tularensis* subsp. *novicida* was isolated from a water sample in Utah in 1951, classified in the genus *Pasteurella* in 1955 (Larson et al., 1955), and transferred to the genus *Francisella*, as *F. novicida*, in 1959 (Olsufjev et al., 1959). In 1989, it was suggested as a biogroup of *F. tularensis*, based on DNA relatedness and biochemical characteristics (Hollis et al., 1989); the 16S rDNA sequences of *F. tularensis* subsp. *novicida* and *F. tularensis* subsp. *tularensis* are also very similar (Fig. BXII.γ.92). Members of *F. tularensis* subsp. *novicida* have less fastidious metabolic requirements than members of the other three subspecies. Despite differences in growth requirements and some metabolic reactions, the placement of *F. novicida* as a subspecies of *F. tularensis* is strongly supported. Analysis of fatty acid profiles showed *F. novicida* to be similar to members of the species *F. tularensis*; the two species also shared more than 75% DNA relatedness (Hollis et al., 1989). Moreover, an almost complete identity in 16S rDNA sequences between the two species has been demonstrated (Forsman et al., 1994). Although the type strain is considered nonpathogenic for humans and shows low viru-

lence for mice, *novicida*-like isolates have been identified as the etiological agents of severe diseases in man (Hollis et al., 1989; Clarridge et al., 1996). The relationship between *F. tularensis* subsp. *novicida* strains and other members of *F. tularensis* is also supported by the fact that *F. tularensis* subsp. *novicida* could be transformed with DNA from *F. tularensis* strain LVS (Anthony et al., 1991). In summary, *F. tularensis* subsp. *novicida* demonstrates heterogeneity with regard to virulence and metabolic characteristics but displays the characteristic fatty acid profile of *F. tularensis*, >70% DNA relatedness, and almost complete 16S rRNA gene sequence similarity to other members of the species.

F. philomiragia was originally designated *Yersinia philomiragia* (Jensen et al., 1969) but subsequent analysis showed no relationship to other members of the genus *Yersinia* (Ursing et al., 1980b) and in the latest version of the *Manual*, it was considered a species *incertae sedis* misclassified in that genus (Bercovier and Mollaret, 1984). In 1989, transfer to the genus *Francisella* based on DNA relatedness and fatty acid analysis was proposed, resulting in the designation *F. philomiragia* comb. nov. (Hollis et al., 1989). The species shows a high 16S rRNA gene sequence similarity to, and the unique fatty acid profile typical of, strains of *F. tularensis*, but is distinguished from most *F. tularensis* strains by its ability to degrade certain carbohydrates and by having no requirement of cysteine for growth.

Anticipated changes The species *W. persica* described under the section “species *incertae sedis*” and the two other symbionts of ticks described in the section “Other organisms” all show the unique 16S rRNA gene sequence signatures of *Francisella*. Thus, they should be included as members of *Francisella* in the future (Fig. BXII.γ.91). At present, insufficient data for determining their relationship at the species level exist.

ACKNOWLEDGMENTS

I thank Arne Tärnvik and Gunnar Sandström for critically reviewing this manuscript, Thomas Svensson for help with sequence analyses, Lenore Johansson for taking electron micrographs, and Stig Granström for advice on agar media. The literature search for this chapter was concluded on 30 April, 1998.

FURTHER READING

Jellison, W.L. 1974. Tularemia in North America 1930–1974, University of Montana Foundation, Missoula.
Penn, R. L. 1995. *Francisella tularensis* (Tularemia). In Mandell, Bennett

and Dolin (Editors), Mandell, Douglas and Bennett's Principles and Practice of Infectious Diseases, Churchill Livingstone, New York. 2060–2068.

Pollitzer, R. 1967. History and incidence of tularemia in the Soviet Union. A review, The Institute of Contemporary Russian Studies, Fordham University, New York.

Tärnvik, A. 1989. Nature of protective immunity to *Francisella tularensis*. Rev. Infect. Dis. 11: 440–451.

Tärnvik, A., M. Ericsson, G. Sandström and A. Sjöstedt. 1992. *Francisella tularensis* — a model for studies of the immune response to intracellular bacteria in man [editorial]. Immunology. 76: 349–354.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *FRANCISELLA*

Differential characteristics of the species and subspecies of *Francisella* are presented in Tables BXII.γ.54 and BXII.γ.55.

List of species of the genus *Francisella*

1. ***Francisella tularensis*** (McCoy and Chapin 1912) Dorofe'ev 1947, 176^{AL} (*Bacterium tularense* McCoy and Chapin 1912, 61.)
tu.la.ren'sis. M.L. adj. *tularensis* tularensis pertaining to Tulare County, California, where the disease was first described in rodents.

The characteristics of relevance for identification are as listed in Table BXII.γ.54 and other characteristics are given in Table BXII.γ.55. Four subspecies, *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, *F. tularensis* subsp. *mediasiatica*, and *F. tularensis* subsp. *novicida*, are recognized; these are differentiated by the characteristics listed in Tables BXII.γ.54 and BXII.γ.55. Other differences are related to geographical origin of strains of the subspecies.

The mol% G + C of the DNA is: 34 (T_m , Bd).

Type strain: B38, ATCC 6223.

GenBank accession number (16S rRNA): Z21931, Z21932.

Additional Remarks: Strain B38 is an avirulent strain with more fastidious growth requirements and a greater tendency towards variation in colony type than is found in virulent strains.

- a. ***Francisella tularensis* subsp. *tularensis*** (McCoy and Chapin 1912) Dorofe'ev 1947, 176^{AL} (*Bacterium tularense* McCoy and Chapin 1912, 61.)

Characteristics distinguishing this subspecies from the other subspecies are indicated in Tables BXII.γ.54 and BXII.γ.55. Highest virulence of all subspecies, most readily demonstrable in higher mammals and man. Produce acid but no gas from maltose, D-glucose, and glycerol. No degradation of lactose and sucrose. Possession of a citrulline ureidase pathway. Isolates have been found only in North America.

TABLE BXII.γ.54. Characteristics of diagnostic value in identifying the genus *Francisella* and its species and subspecies^{a,b}

Characteristics	<i>F. tularensis</i> subsp. <i>tularensis</i>	<i>F. tularensis</i> subsp. <i>holarctica</i>	<i>F. tularensis</i> subsp. <i>mediasiatica</i>	<i>F. tularensis</i> subsp. <i>novicida</i>	<i>F. philomiragia</i>
Size	<0.5 μm	<0.5 μm	<0.5 μm	<1.5 μm	<1.5 μm
Capsule ^c	y	y	y	n	nt
Gram stain	w ^d	w	w	w	w
Growth on MacConkey agar	—	—	—	d ^e	d
Cysteine required for growth ^f	+	+	+	—	—
H ₂ S production in cysteine-supplemented medium	+	+	+	+	+
β-lactamase	+	+	—	+	+
Acid production from: ^g					
Maltose	+	+	—	d	+
Lactose	—	—	—	—	—
Sucrose	—	—	—	+	+
D-Glucose	+	+	—	+	d
Glycerol	+	—	+	d	—
Citrulline ureidase pathway	+	—	+	+	nt
Agglutination of <i>F. tularensis</i> antiserum	+	+	+	D	—
% 16S rRNA gene sequence similarity to ATCC 6223 ^h	≥99.8	≥99.8	≥99.8	≥99.8	≥98.3
Presence of <i>F. tularensis</i> 17-kDa lipoprotein gene	+	+	+	+	+

^aFor symbols see standard definitions; nt, not tested.

^bData from Eigelsbach and McGann (1984), Hollis et al. (1989), and Clarridge et al. (1996).

^cPresence of capsule associated with virulence; y, yes; n, no.

^dw, weak Gram-negative reaction.

^eA majority of strains reportedly grow under these conditions.

^fA few strains of *F. tularensis* subsp. *tularensis* and *F. tularensis* subsp. *holarctica* lack the cysteine requirement.

^gCysteine agar (carbohydrate base) used for assaying for *F. tularensis* subsp. *tularensis*, *F. tularensis* subsp. *holarctica*, and *F. tularensis* subsp. *mediasiatica* and O/F medium for *F. tularensis* subsp. *novicida* and *F. philomiragia*. Strains show weak reactions and some also delayed, 3–7 d.

^hRefers to similarities of partial sequences, 390–450 bp in length.

TABLE BXII.γ.55. Other characteristics of *F. philomiragia* and the subspecies of *F. tularensis*^{a,b}

Characteristics	<i>F. tularensis</i> subsp. <i>tularensis</i>	<i>F. tularensis</i> subsp. <i>holarctica</i>	<i>F. tularensis</i> subsp. <i>mediasiatica</i>	<i>F. tularensis</i> subsp. <i>novicida</i>	<i>F. philomiragia</i>
Mol% G + C of DNA	33–36	33–36	33–36	34	33–34
Aerobic, microaerophilic	+	+	+	+	+
CO ₂ enhances growth	+	+	+	+	+
Optimum temperature, °C	37	37	37	37	25 or 37
Growth in nutrient broth, 0% NaCl	—	—	—	—	—
Growth in nutrient broth, 6% NaCl	—	—	—	d ^c	d ^c
Catalase	+w	+w	+w	+w	+w
Oxidase (Kovacs modification)	—	—	—	—	+
Indole	—	—	—	—	+
Urease	—	—	—	—	—
Nitrate reduction	—	—	—	—	—
H ₂ S slant, Triple Sugar Iron	—	—	—	—	+
Gelatin hydrolysis	—	—	—	—	+
Motility	—	—	—	—	—
LD ₅₀ rabbits	<10 ¹	>10 ⁶	>10 ⁶	>10 ⁶	nt
Median infectious dose for mice, <10 ³	+	+	+	—	nt

^aFor symbols, see standard definitions; nt, not tested.

^bData from Eigelsbach and McGann (1984), Hollis et al., (1989), and Clarridge et al. (1996).

^cA majority of strains reportedly grow under these conditions.

The mol% G + C of the DNA is: 34. (*T_m*).

Type strain: GIEM Schu, ATCC 6223.

GenBank accession number (16S rRNA): Z21931, Z21932.

Additional Remarks: Virulent strains, such as SCHU, are available only from individual investigators.

- b. ***Francisella tularensis* subsp. *holarctica*** (ex Olsufiev, Emelyanova and Dunaeva 1959) Olsufjev and Meshcheryakova 1983, 872^{VP}
hol.arcticus. Gr. prep. *holos* whole; Gr. n. *actos* the great and little bear constellations; M.L. adj. *holarctica* generally distributed in the arctic regions.

Characteristics distinguishing this subspecies from the other subspecies are indicated in Tables BXII.γ.54 and BXII.γ.55. Isolates show lower virulence than isolates of *F. tularensis* subsp. *tularensis*. Minimum infectious dose <10³ for mice and LD₅₀ >10⁶ for rabbits. Produce acid but no gas from maltose and D-glucose. No degradation of lactose, glycerol, and sucrose. Isolates have been found on all continents throughout the Northern Hemisphere.

The mol% G + C of the DNA is: 34 (*T_m*).

Type strain: GIEM 503.

Additional Remarks: Strain GIEM 503 is the attenuated live vaccine strain.

- c. ***Francisella tularensis* subsp. *mediasiatica*** (Aikimbaev 1966) Olsufjev and Meshcheryakova 1983, 872^{VP}
med.asiaticus. Gr. adj. *media* middle; Gr. adj. *asiatica* Asian; M.L. adj. *mediasiatica* pertaining to mid-Asia.

Characteristics distinguishing this subspecies from the other subspecies are indicated in Tables BXII.γ.54 and BXII.γ.55. Isolates show lower virulence than isolates of *F. tularensis* subsp. *tularensis*. Minimum infectious dose <10³ for mice and LD₅₀ >10⁶ for rabbits. Produce acid but no gas from glycerol. No degradation of lactose, sucrose, maltose, and D-glucose. Isolates have been found only in the Central Asian republics of the former USSR, in particular in the valleys and deltas of the rivers Amu-Darja, Ili, and Tshu.

The mol% G + C of the DNA is: unknown.

Type strain: 543.

Additional Remarks: Strain 543 is available from the Gamaleya Institute of Epidemiology and Microbiology, Academy of Medical Sciences, Moscow, Russia (alternative designation FSC147 is available at the Institute of Microbiology, FOA, Umeå, Sweden). It was isolated in 1965 in the Alma-Ata region, Kazakhstan.

- d. ***Francisella tularensis* subsp. *novicida* subsp. nov.** (Larson, Wicht and Jellison 1955) (*Francisella novicida* (Larson, Wicht and Jellison 1955) Olsufjev, Emelyanova and Dunaeva 1959, 146^{AL}; *Pasteurella novicida* Larson, Wicht and Jellison 1955, 253.)
no.vici.da. L. adj. *novus* new; L. v. suff. *-cida* from L. v. *caedo* to cut, kill; M. L.n. *novicida* new killer.

Characteristics distinguishing this subspecies from the other subspecies are indicated in Tables BXII.γ.54 and BXII.γ.55. It was originally considered a separate species, *F. novicida* (Olsufjev et al. 1959). However, Hollis et al. (1989) showed by DNA relatedness studies that isolates of this species were indistinguishable from the type strain of *F. tularensis* and proposed that it be designated a biogroup of *F. tularensis*. It displays the characteristic fatty acid profile of *F. tularensis*, >70% DNA relatedness, and almost complete 16S rRNA gene sequence similarity to other subspecies of *F. tularensis*. The assignment of *F. novicida* as a distinct subspecies is supported by its less fastidious growth requirements, including lack of cysteine requirement, degradation of sucrose, unique morphology, and lower virulence as compared to the other subspecies of *F. tularensis*.

F. tularensis subsp. *novicida* is often closely linked to water-borne transmission. The type strain of *F. tularensis* subsp. *novicida* was isolated from water in Utah (Larson et al., 1955) and in some cases of human disease, the suspected source has been natural water (Clarridge et al., 1996). Isolates show lower virulence than those of other subspecies and there are only a few documented cases of human disease caused by isolates of the subspecies (Hollis et al., 1989; Clarridge et al., 1996). Unlike

other subspecies, growth is enhanced by, but not absolutely dependent on, the presence of cysteine in media. Another characteristic distinguishing the subspecies is growth in nutrient broth containing 6% NaCl. Produce acid but no gas from sucrose and D-glucose, but no degradation of lactose. Some isolates produce acid from glycerol and maltose. Not all isolates are agglutinated by commercially available antiserum against *F. tularensis*. Isolates have been found only in North America.

The mol% G + C of the DNA is: 34 (T_m).

Type strain: ATCC 15482 (type strain).

GenBank accession number (16S rRNA): L26084.

Additional Remarks: The type strain was isolated from a water sample from Ogden Bay, Utah in 1951 (patient died Larson et al., 1955).

2. **Francisella philomiragia** (Jensen, Owen and Jellison 1969, 1237) Hollis, Weaver, Steigerwalt, Wenger, Moss, and Brenner 1990, 105^{VP} (Effective publication: Hollis, Weaver, Steigerwalt, Wenger, Moss and Brenner 1989, 1601) (*Yersinia philomiragia* Jensen, Owen and Jellison 1969, 1237.) *phi.lo.mi.ra'gi.a.* Gr. adj. *philos* loving; M.L. n. *miragia* plural of Latinized English word mirage; *philomiragia* loving mirages, because of the mirages observed in the area where the first isolations of this species were made.

The characteristics are listed in Tables BXII.γ.54 and BXII.γ.55. The name *Y. philomiragia* was proposed by Jensen et al. (1969) for a small Gram-negative bacillus isolated in

1959 from a dying muskrat found in a marshy area in Utah. The species description was based on this strain and four other strains isolated from water samples in the same area in 1960. Because of morphological resemblance to *Y. pestis* and some degree of DNA relatedness to *Y. pestis* (Ritter and Gerloff, 1966), the organism was assigned to the genus *Yersinia*. Later studies indicated, however, no significant DNA relatedness between *Y. philomiragia* and other *Yersinia* species, other members of the *Enterobacteriaceae*, or *Pasteurella multocida* (Ursing et al., 1980b). Hollis et al. (1989) reviewed data on 14 *Y. philomiragia* strains isolated from human specimens. The biochemical characteristics, fatty acid composition, and DNA relatedness of these strains and the five original strains demonstrated a close relationship at the species level and supported placement of this species in the genus *Francisella*.

F. philomiragia has been isolated from water, muskrats, and humans in North America and Europe. It rarely causes human disease, but serious infections may occur. Wenger et al. (1989) reviewed case reports of 14 patients infected with *F. philomiragia*. A majority of patients belonged to two risk groups. Patients with chronic granulomatous disease and near-drowning victims. Pneumonia and fever-bacteremia were reported in most patients and one patient died.

The mol% G + C of the DNA is: 34 (T_m).

Type strain: ATCC 25015, DSM 7535.

Additional Remarks: The type strain is the original isolate from a muskrat in Utah (Jensen et al., 1969).

Other Organisms

1. *Dermacentor andersoni* symbiont Niebylski, Peacock, Fischer, Porcella, and Schwan 1997, 3933 and *Ornithodoros moubata* symbiont Noda, Munderloh, and Kurti 1997, 3926.

By analysis of their 16S rDNA sequences, two endosymbionts of ticks have been found to be phylogenetically related to members of the genus *Francisella*. An endosymbiont (DAS) present in >90% of isolates of the wood tick *D. andersoni* from Montana (Niebylski et al., 1997) and one found in the ovaries and Malpighian tubules from the African soft tick *O. moubata* (Noda et al., 1997) show ≥97.2% 16S rRNA gene sequence similarity to *F. tularensis* and *F. philomiragia*, and ≥99.4% similarity to *W. persica*. The DAS organism has been shown to cross-react in fluorescent antibody tests with *W. persica* (Burgdorfer et al., 1973). It can be cultivated in chicken embryo cell cultures and in Vero cells (Niebylski et al., 1997), and grows well in guinea pigs and golden hamsters, killing the animals when high inocula are given (Burgdorfer et al., 1973). The data generated so far showing the immunological cross-reactivity and almost complete identity with regard to 16S rRNA gene sequence (Fig. BXII.γ.91) demonstrate that the *D. andersoni* symbiont and *W. persica* are closely related. However, due to lack of DNA relatedness data, it cannot be concluded that they belong to the same species. Moreover, the 16S rRNA gene sequences of the *D. andersoni* and *O. moubata* endosymbionts show high similarity, 99.3%, to those of *Francisella* isolates and a parsimony analysis demonstrates a close relationship (Fig. BXII.γ.91). The two endosymbionts and *W. persica* all harbor the 16S rRNA sequence signatures of *Francisella* (Table BXII.γ.51). Thus, the data available at present

support their membership in *Francisella* though, due to the lack of DNA relatedness data, their exact relationship at the species level cannot be determined.

Species Incertae Sedis

1. **Wolbachia persica** Suitor and Weiss 1961, 105^{AL} *per'si.ca.* L. fem. adj. *persica* from the specific epithet of the reputed host tick, *Argas persicus* (later reclassified as *A. arboreus*).

The name *W. persica* was proposed by Suitor and Weiss (1961) for a coccoid, Gram-negative bacillus first isolated from the tick *Argas persicus* feeding on the buff-backed heron, *Bubulcus ibis*, in Egypt. Because of morphological resemblance, Gram-negative staining, and an arthropod association, the organism was assigned to the genus *Wolbachia*. In 1989, an analysis of 16S rRNA gene sequences of members of the *Rickettsiae* demonstrated that *W. persica* had been misclassified and instead was a member of the *Gammaproteobacteria* (Weisburg et al., 1989). Subsequent analysis indicated that it has a close relationship to members of the genus *Francisella* (Forsman et al., 1994).

W. persica can be cultivated in the yolk sac of chicken embryos, where it grows profusely and kills the embryos in 5–10 days, depending on the dose of inoculum (Burgdorfer et al., 1973). It also grows well in several types of mammalian cells or insect cell cultures. An antigenically related organism was isolated from the hard tick *Dermacentor andersoni* collected in Montana (Burgdorfer et al., 1973) and further characterized by Niebylski et al. (1997) (see Other Organisms).

W. persica has been isolated by removing and triturating

the Malpighian tubules from ticks (Suitor and Weiss, 1961). After resuspension in a salt solution, this mixture was inoculated into chicken embryos via the yolk sac. Dying embryos are examined for the presence of coccoid bodies, which are most clearly visualized by Giemsa staining following Carnoy's fixation. For the production of seed and other tests, the isolates are passaged in eggs. *W. persica* metabolizes D-glucose, as well as serine, glutamine, and other substrates, including glutamate (Weiss et al., 1962). Erythromycin, chloramphenicol, tetracycline, and *p*-aminobenzoic acid inhibit growth in chicken embryos. The viability of a suspension of *W. persica* in sucrose, phosphate buffer, glutamate solution ([Bovarnick et al., 1950]) was retained for a decade at -70°C .

W. persica displays the unique 16S rRNA sequence sig-

natures of *Francisella* (Table BXII.γ.51), supporting its placement in the genus *Francisella*. The 16S rRNA gene sequence shows high similarity to those of two symbionts of *D. andersoni* and *Ornithodoros moubata*, and the three organisms are located on the same branch of the phylogenetic tree, showing their close relationship (Fig. BXII.γ.51). However, the lack of DNA relatedness data hinders determination of the relationship between *W. persica*, the two tick endosymbionts, and *Francisella* spp. at the species level.

The mol% G + C of the DNA is: 30 (T_m , Bd) (Kingsbury and Weiss, 1968).

Type strain: ATCC VR-331.

GenBank accession number (16S rRNA): M21292.

ATCC VR-331 was the first *Argas persicus* (*A. arboreus*) isolate (Suitor and Weiss, 1961).

Order VI. Legionellales ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Le.gi.on.el.la'les. M.L. n. *Legionella* type genus of the order; -ales ending to denote order;
M.L. fem. n. *Legionellales* the *Legionella* order.

The order *Legionellales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the order contains the families *Legionellaceae* and *Coxiellaceae*.

Facultative (*Legionellaceae*) and obligate (*Coxiellaceae*) intra-

cellular parasites infecting protozoa, invertebrates, animals, and man.

Type genus: **Legionella** Brenner, Steigerwalt and McDade 1979, 658.

Family I. Legionellaceae Brenner, Steigerwalt and McDade 1979, 658^{AL}

WASHINGTON C. WINN, JR.

Le.gi.on.el.la'ce.ae. M.L. n. *Legionella* type genus of the family; -aceae ending to denote family;
M.L. fem. pl. n. *Legionellaceae* the *Legionella* family.

Rods $0.3\text{--}0.9 \times 2\text{--}20 \mu\text{m}$ or more. Do not form endospores or microcysts. Not encapsulated. Not acid-fast in culture. **Gram negative.** Most species are motile by one, two, or more straight or curved polar or lateral flagella; nonmotile strains of motile species are occasionally seen and motility may be lost *in vitro*. **Aerobic.** L-cysteine-HCl and iron salts are required for growth. Catalase positive; oxidase negative or weakly positive. Nitrates are not reduced. Urease negative. Gelatin is liquefied by most species. **Chemoorganotrophic**, using amino acids as carbon and energy sources. **Carbohydrates are neither fermented nor oxidized** (Tables BXII.γ.56 and BXII.γ.57). **Branched chain fatty acids** predominate in whole cell hydrolysates (Table BXII.γ.58). All species contain ubiquinones with 9–14 isoprene units in the side chains (Table BXII.γ.59). Isolated from surface water, mud or moist soil, thermally polluted lakes and streams, and potable water systems. Associated with free-living environmental amoebae. There is no known soil or animal source. Pathogenic or potentially pathogenic for humans. Isolated from sputum, bronchoalveolar lavage washings, lung tissue, pleural fluid, pericardial fluid, blood, heart valves, abscesses, and prosthetic devices.

Growth of stock cultures obtained from humans occurs at temperatures between 25 and 43°C , but not at 50°C . The optimum temperature is $36^{\circ} \pm 1^{\circ}\text{C}$ (Weaver and Feeley, 1979; Thacker et al., 1981); however, in nature, the optimum growth temperature may be 45°C or higher (Fliermans et al., 1981; Or-

ison et al., 1981). The optimum pH for growth of legionellae is 6.8–7.0 (Feeley et al., 1978).

The mol% G + C of the DNA is: 38–52.

Type genus: **Legionella** Brenner, Steigerwalt and McDade 1979, 658.

FURTHER DESCRIPTIVE INFORMATION

Analysis of highly conserved sequences of 16S RNA has established that the family *Legionellaceae* forms a subline within the *Gamma*proteobacteria (Fry et al., 1991b; Hookey et al., 1995, 1996; Birtles et al., 1996; Riffard et al., 1998); the two closest relatives are *Coxiella burnetii* and *Wolbachia persica*. The relationships among strains, as determined by 16S rRNA analysis, support the proposal that a single genus, *Legionella*, be included in the family, *Legionellaceae*. Over the succeeding years, usage among the microbiology community has established the precedence of the single genus approach.

The family contains one genus, *Legionella*, with more than 40 species. The number will undoubtedly continue to expand, because a number of unnamed species have been detected by DNA–DNA hybridization (D.J. Brenner, personal communication). All species tested grow in free-living amoebae, and at least one species has been isolated only in amoebae (Rowbotham, 1983). The family is circumscribed and readily distinguishable from other families on the basis of a unique set of phenotypic characteristics

the Malpighian tubules from ticks (Suitor and Weiss, 1961). After resuspension in a salt solution, this mixture was inoculated into chicken embryos via the yolk sac. Dying embryos are examined for the presence of coccoid bodies, which are most clearly visualized by Giemsa staining following Carnoy's fixation. For the production of seed and other tests, the isolates are passaged in eggs. *W. persica* metabolizes D-glucose, as well as serine, glutamine, and other substrates, including glutamate (Weiss et al., 1962). Erythromycin, chloramphenicol, tetracycline, and *p*-aminobenzoic acid inhibit growth in chicken embryos. The viability of a suspension of *W. persica* in sucrose, phosphate buffer, glutamate solution ([Bovarnick et al., 1950]) was retained for a decade at -70°C .

W. persica displays the unique 16S rRNA sequence sig-

natures of *Francisella* (Table BXII.γ.51), supporting its placement in the genus *Francisella*. The 16S rRNA gene sequence shows high similarity to those of two symbionts of *D. andersoni* and *Ornithodoros moubata*, and the three organisms are located on the same branch of the phylogenetic tree, showing their close relationship (Fig. BXII.γ.51). However, the lack of DNA relatedness data hinders determination of the relationship between *W. persica*, the two tick endosymbionts, and *Francisella* spp. at the species level.

The mol% G + C of the DNA is: 30 (T_m , Bd) (Kingsbury and Weiss, 1968).

Type strain: ATCC VR-331.

GenBank accession number (16S rRNA): M21292.

ATCC VR-331 was the first *Argas persicus* (*A. arboretus*) isolate (Suitor and Weiss, 1961).

Order VI. Legionellales ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Le.gi.on.el.la'les. M.L. n. *Legionella* type genus of the order; -ales ending to denote order;
M.L. fem. n. *Legionellales* the *Legionella* order.

The order *Legionellales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the order contains the families *Legionellaceae* and *Coxiellaceae*.

Facultative (*Legionellaceae*) and obligate (*Coxiellaceae*) intra-

cellular parasites infecting protozoa, invertebrates, animals, and man.

Type genus: **Legionella** Brenner, Steigerwalt and McDade 1979, 658.

Family I. Legionellaceae Brenner, Steigerwalt and McDade 1979, 658^{AL}

WASHINGTON C. WINN, JR.

Le.gi.on.el.la'ce.ae. M.L. n. *Legionella* type genus of the family; -aceae ending to denote family;
M.L. fem. pl. n. *Legionellaceae* the *Legionella* family.

Rods $0.3\text{--}0.9 \times 2\text{--}20 \mu\text{m}$ or more. Do not form endospores or microcysts. Not encapsulated. Not acid-fast in culture. **Gram negative.** Most species are motile by one, two, or more straight or curved polar or lateral flagella; nonmotile strains of motile species are occasionally seen and motility may be lost *in vitro*. **Aerobic.** L-cysteine-HCl and iron salts are required for growth. Catalase positive; oxidase negative or weakly positive. Nitrates are not reduced. Urease negative. Gelatin is liquefied by most species. **Chemoorganotrophic**, using amino acids as carbon and energy sources. **Carbohydrates are neither fermented nor oxidized** (Tables BXII.γ.56 and BXII.γ.57). **Branched chain fatty acids** predominate in whole cell hydrolysates (Table BXII.γ.58). All species contain ubiquinones with 9–14 isoprene units in the side chains (Table BXII.γ.59). Isolated from surface water, mud or moist soil, thermally polluted lakes and streams, and potable water systems. Associated with free-living environmental amoebae. There is no known soil or animal source. Pathogenic or potentially pathogenic for humans. Isolated from sputum, bronchoalveolar lavage washings, lung tissue, pleural fluid, pericardial fluid, blood, heart valves, abscesses, and prosthetic devices.

Growth of stock cultures obtained from humans occurs at temperatures between 25 and 43°C , but not at 50°C . The optimum temperature is $36^{\circ} \pm 1^{\circ}\text{C}$ (Weaver and Feeley, 1979; Thacker et al., 1981); however, in nature, the optimum growth temperature may be 45°C or higher (Fliermans et al., 1981; Or-

ison et al., 1981). The optimum pH for growth of legionellae is 6.8–7.0 (Feeley et al., 1978).

The mol% G + C of the DNA is: 38–52.

Type genus: **Legionella** Brenner, Steigerwalt and McDade 1979, 658.

FURTHER DESCRIPTIVE INFORMATION

Analysis of highly conserved sequences of 16S RNA has established that the family *Legionellaceae* forms a subline within the *Gamma*proteobacteria (Fry et al., 1991b; Hookey et al., 1995, 1996; Birtles et al., 1996; Riffard et al., 1998); the two closest relatives are *Coxiella burnetii* and *Wolbachia persica*. The relationships among strains, as determined by 16S rRNA analysis, support the proposal that a single genus, *Legionella*, be included in the family, *Legionellaceae*. Over the succeeding years, usage among the microbiology community has established the precedence of the single genus approach.

The family contains one genus, *Legionella*, with more than 40 species. The number will undoubtedly continue to expand, because a number of unnamed species have been detected by DNA–DNA hybridization (D.J. Brenner, personal communication). All species tested grow in free-living amoebae, and at least one species has been isolated only in amoebae (Rowbotham, 1983). The family is circumscribed and readily distinguishable from other families on the basis of a unique set of phenotypic characteristics

TABLE BXII.γ.56. Selected characteristics of *Legionella* species isolated from humans^a

Characteristic	<i>L. anisa</i>	<i>L. birminghamsensis</i>	<i>L. bozemanii</i>	<i>L. cincinnatiensis</i>	<i>L. dumoffii</i>	<i>L. feeleii</i>	<i>L. gormanii</i>	<i>L. hackeliae</i>	<i>L. jordanis</i>	<i>L. lansingensis</i>	<i>L. longbeachae</i>
Gram reaction	—	—	—	—	—	—	—	—	—	—	—
Acid-fastness ^c	—	—	—	—	—	—	—	—	—	—	—
Growth on:											
BCYE-α agar	+	+	+	+	+	+	+	+	+	+	+
Blood agar with trypticase soy base	—	—	—	—	—	—	—	—	—	—	—
Growth requirement for L-cysteine	+	+	+	+	+	+	+	+	+	+	+
Browning of tyrosine-containing media	+	—	+	+	+	+	+	+	+	—	+
Acid from carbohydrates	—	—	—	—	—	—	—	—	—	—	—
Nitrate to nitrite	—	—	—	—	—	—	—	—	—	—	—
Hippurate hydrolysis	—	—	—	—	—	±	—	—	—	—	—
Gelatin liquefaction	+	+	+	+	+	±	+	+	+	—	+
β-lactamase production	+	+	±	+	+	—	+	+	+	—	±
Catalase	+	+	+	+	+	+	+	+	+	+	+
Oxidase	+	±	±	—	—	—	—	+	+	+	+
Motility ⁱ	+	+	+	+	+	+	+	+	+	+	+
Autofluorescence ^j	BW	—	BW	—	BW	—	BW	—	—	—	—

^aCharacteristics based on varying numbers of strains. NA, not available; w, weak.^bAmoeba-associated pathogens.^cModified decolorization.^dMay be acid-fast in tissue.^eOn initial isolation; may adapt to growth without cysteine.^fOne strain negative.^gOn initial isolation; may be + after several transfers on BCYE agar.^hA few strains negative.ⁱOne polar flagellum.^jBW, blue-white; R, red.

(continued)

that include Gram negativity, nonfermentative metabolism, requirement for L-cysteine and iron salts, and predominantly branched-chain cellular fatty acids. Relatively large amounts of branched chain and hydroxy-substituted branched chain fatty acids are found in flavobacteria, but these organisms were shown by DNA–DNA hybridization to be unrelated to legionellae. DNA–DNA hybridization was also used to rule out relatedness between legionellae and members of other families that resemble *Legionellaceae* in some phenotypic characteristics (Brenner et al., 1979).

Legionellae in formalin-fixed, paraffin-embedded tissue may not be demonstrable with the staining techniques usually employed histologically. The simplest means for demonstrating legionellae in tissue is to prepare imprint smears of the fresh cut surface of the tissue, after which the smears are stained by Gram's method. If sufficient numbers of organisms are present, they may be seen in tissue sections using one of the variations of the tissue Gram stain; e.g., the Brown and Hopps stain, or even hematoxylin and eosin. One of the silver impregnation stains (Warthin-Starry, Steiner, or Dieterle), designed originally for spirochetes, provides a more sensitive means to demonstrate legionellae in paraffin-fixed tissue (Chandler et al., 1977; van Orden and Greer, 1977). The Gimenez stain, designed for rickettsiae, has been used to color *Legionella* cells in secretions and imprint smears from unfixed tissue (Greer et al., 1980). The Gimenez method employs carbol fuchsin, and the sensitivity of Gram's stain may be improved without overstaining inflammatory cells by adding 0.05%

carbol fuchsin to the saffranin counterstain (W.C. Winn, Jr., personal observation). All of these histochemical stains, which have varying degrees of sensitivity for detecting bacteria, are completely nonspecific; they provide information about the morphology of the bacterial cells at the light microscopic level.

Bacterial cells have vacuoles that stain with Sudan black B and may contain poly-β-hydroxybutyrate (Chandler et al., 1979). Diaminopimelic acid has been demonstrated in the cell wall (Guerant et al., 1979). Some species show a blue-white or red auto-fluorescence under long wavelength (365 nm) ultraviolet light (Tables BXII.γ.56 and BXII.γ.57), and many strains of approximately half of the recognized species produce a water-soluble, brown pigment on tyrosine-containing media (Baine et al., 1978).

The term legionellosis is used by many to indicate all infections caused by legionellae. Strains of approximately 50% of the recognized species have been implicated in human disease, usually pneumonia. All species should be considered potential pathogens if the immunologic, inflammatory, and/or nonspecific defenses of the patient are sufficiently compromised. Pneumonia caused by *L. pneumophila* is commonly referred to as Legionnaires' disease (Fraser et al., 1977) and pneumonia due to *L. micdadei* has been referred to as Pittsburgh pneumonia (Pasculle et al., 1980). Several *Legionella* species, including *L. pneumophila*, *L. micdadei*, *L. feeleii*, and *L. anisa*, can also cause a mild, non-pneumonic, febrile disease termed Pontiac fever (Glick et al., 1978; Herwaldt et al., 1984a; Goldberg et al., 1989; Fields et al., 1990).

TABLE BXII.γ.56. (cont.)

Characteristic	<i>L. lytic</i> ^b	<i>L. maceachernii</i>	<i>L. micdadei</i>	<i>L. oakridgensis</i>	<i>L. parisiensis</i>	<i>L. pneumophila pneumophila</i>	<i>L. pneumophila fraseri</i>	<i>L. pneumophila pascullei</i>	<i>L. saintelensis</i>	<i>L. tucsonensis</i>	<i>L. wadsworthii</i>
Gram reaction	NA	—	—	—	—	—	—	—	—	—	—
Acid-fastness ^c	NA	—	— ^d	—	—	—	—	—	—	—	—
Growth on:											
BCYE-α agar	NA	+	+	+	+	+	+	+	+	+	+
Blood agar with trypticase soy base	NA	—	—	—	—	—	—	—	—	—	—
Growth requirement for L-cysteine	NA	+	+	+	+	+	+	+	+	+	+
Browning of tyrosine-containing media		+	— ^g	+	+	+	+	+	+	—	—
Acid from carbohydrates		—	—	—	—	—	—	—	—	—	—
Nitrate to nitrite		—	—	—	—	—	—	—	—	—	—
Hippurate hydrolysis		—	—	—	—	+	± ^h	+	—	—	—
Gelatin liquefaction		+	—	+	+	+	+	+	+	+	+
β-lactamase production		—	—	+	+	+	+	+	+	+	+
Catalase		+	+	+	+	+	+	+	+	+	+
Oxidase		+	+	±	+	+	+	+	+	—	—
Motility ⁱ		+	+	—	+	+	+	+	+	+	—
Autofluorescence ^j		—	—	—	BW	—	—	—	—	BW	—

^aCharacteristics based on varying numbers of strains. NA, not available; w, weak.

^bAmoeba-associated pathogens.

^cModified decolorization.

^dMay be acid-fast in tissue.

^eOn initial isolation; may adapt to growth without cysteine.

^fOne strain negative.

^gOn initial isolation; may be + after several transfers on BCYE agar.

^hA few strains negative.

ⁱOne polar flagellum.

^jBW, blue-white; R, red.

Genus I. *Legionella* Brenner, Steigerwalt and McDade 1979, 658^{AL}

WASHINGTON C. WINN, JR.

Le.gi.on.el'la. M.L. n. *legio* legion or army; M.L. dim. ending *-ella*; M.L. fem. n. *Legionella* small legion or army.

As the only genus in the family, the definition of *Legionella* is identical to that of *Legionellaceae*. **Aerobic. L-cysteine-HCl and iron salts are required for growth.** Oxidase negative or weakly positive. Nitrates are not reduced. Urease negative. Gelatin is liquefied by most species. Chemoorganotrophic, using amino acids as carbon and energy sources. **Carbohydrates are neither fermented nor oxidized** (Tables BXII.γ.56 and BXII.γ.57). **Branched chain fatty acids** predominate in whole cell hydrolysate (Table BXII.γ.58). All species contain ubiquinones with 9–14 isoprene units in the side chains (Table BXII.γ.59).

The mol% G + C of the DNA is: 38–52.

Type species: *Legionella pneumophila* Brenner, Steigerwalt and McDade 1979, 658.

FURTHER DESCRIPTIVE INFORMATION

Analysis of highly conserved sequences of 16S rRNA has established that the genus *Legionella* forms a subline within the *Gammaproteobacteria* (Fry et al., 1991b; Birtles et al., 1996); the two

closest relatives are *Coxiella burnetii* and *Wolbachia persica*. The relationships among strains, as determined by 16S rRNA analysis, support the proposal that a single genus, *Legionella*, be included in the family *Legionellaceae* (Fry et al., 1991a). Genetic relationships among the species of *Legionella* indicate subgroups within some species (Fry et al., 1991b; Hookey et al., 1995, 1996; Birtles et al., 1996; Riffard et al., 1998) and subspecies within *L. pneumophila* (Brenner et al., 1988c). As a generalization, the species that exhibit blue-white autofluorescence under long-wavelength ultraviolet light (Wood's Lamp) tend to be most closely related to each other (Hookey et al., 1996), and the two species that fluoresce red are close relatives. Interestingly, *L. wadsworthii* appears in the blue-white fluorescent clade, although it does not exhibit autofluorescence. Hookey et al. (1996) suggest that *L. wadsworthii* may be descended from an autofluorescent ancestor. By analysis of 16S rRNA, Hookey et al. (1995) were able to distinguish all but seven *Legionella* species from each other. The indistinguishable species were *L. erythra* from *L. rubrilucens* (red-

TABLE BXII.γ.57. Selected characteristics of *Legionella* species that have been isolated only from the environment^a

Characteristic	<i>L. brunensis</i>	<i>L. cherrii</i>	<i>L. erythra</i>	<i>L. fairfieldensis</i>	<i>L. geestiana</i>	<i>L. gratiana</i>	<i>L. israelensis</i>	<i>L. jamestownensis</i>	<i>L. londrinensis</i>	<i>L. moravica</i>	<i>L. nautarum</i>	<i>L. parisiensis</i>
Gram reaction	—	—	—	—	—	—	—	—	—	—	—	—
Acid-fastness ^b	—	—	—	—	—	—	—	—	—	—	—	—
Growth on:												
BCYE-α agar	+	+	+	+	+	+	+	+	+	+	+	+
Blood agar with trypticase soy base	—	—	—	—	—	—	—	—	—	—	—	—
Growth requirement for L-cysteine	+	+	+	+	+	+	+	+	+	+	+	+
Browning of tyrosine-containing media	w	+	+	—	w	—	—	+	+	w	—	+
Acid from carbohydrates	—	—	—	—	—	—	—	—	—	—	—	—
Nitrate to nitrite	—	—	—	—	—	—	—	—	—	—	—	—
Hippurate hydrolysis	—	—	—	—	w	—	—	—	— or w	—	—	—
Gelatin liquefaction	+	+	+	—	+	+	w	+	+	+	—	+
β-lactamase production	+	+	+	—	—	+	NA	+	+	+	+	+
Catalase	+	+	+	+	+	+	+	+	+	+	+	+
Oxidase	— or w	—	+	w	—	+	—	—	—	— or w	+	+
Motility ^c	+	+	+	+	+	+	+	+	+	+	+	+
Autofluorescence ^d	—	BW	R	—	—	—	—	—	—	—	—	BW

^aCharacteristics based on varying numbers of strains. In some cases a species is represented by a single isolate. NA, not available; w, weak.^bModified decolorization.^cOne polar flagellum.^dBW, blue-white; R, red.^eAutofluorescence may be lost on subculture.

(continued)

fluorescing species), *L. anisa* or *L. cherrii* from *L. tucsonensis* (blue-white fluorescing species), and *L. quateirensis* from *L. shakespearei*. Riffard et al. (1998) have grouped closely related species into three phylogenetic groups, based on intergenic 16S-23S ribosomal spacer PCR analysis (Table BXII.γ.60).

Legionellae are rod shaped or filamentous, 0.3–0.9 × 2–20 μm or more in length. The filaments are found after growth on agar media, less commonly in yolk sac material, and rarely in human lung or guinea pig tissue (Chandler et al., 1979; Weaver and Feeley, 1979). Electron microscopic examination reveals filamentous nucleoids, ribosomes, and Sudan black B-staining vacuoles thought to contain poly-β-hydroxybutyrate granules (Chandler et al., 1979). Cells are enclosed by a double envelope, composed of two three-layered unit membranes; a peptidoglycan layer has been demonstrated for *L. pneumophila*, and for *L. micdadei* (Flesher et al., 1979; Pasculle et al., 1979; Gress et al., 1980). Cell division occurs by a pinching process (Chandler et al., 1979; Keel et al., 1979).

One, two, or occasionally more flagella occur per cell. The flagella are curved or straight and have a polar or lateral arrangement (Chandler et al., 1980).

Legionellae are not acid-fast by the Ziehl-Neelsen procedure for mycobacteria (Hébert et al., 1980a) after they have been isolated on agar. *L. micdadei*, however, may appear acid-fast in tissue; modified acid-fast stains that use weak acids as decolorizing agents may be required to demonstrate the acid-fastness (Pasculle et al., 1979).

Legionellae were first isolated in guinea pigs and in embryonated hens' eggs (Tatlock, 1944; Jackson et al., 1952; Bozeman et al., 1968; McDade et al., 1979). With the advent of supportive culture media, inoculation of animals and eggs has become obsolete. It has been suggested, however, that preincubation of

environmental samples with protozoa increases the recovery of isolates (Rowbotham, 1983; Sanden et al., 1992). Legionellae do not grow on commonly employed media such as standard blood agar or nutrient agar. Mueller-Hinton agar, supplemented with hemoglobin and IsoVitalX or GC base with similar supplements, was used to cultivate the first isolates of *L. pneumophila* (Centers for Disease Control, 1977). In one case, the agar was a commercially prepared chocolate agar (Dumoff, 1979). A buffered charcoal yeast extract (BCYE-α) agar supplemented with 0.1% α-ketoglutarate is now preferred for isolation of *L. pneumophila* (Feeley et al., 1979; Pasculle et al., 1980). All species grow on BCYE-α agar. The medium is buffered at pH 6.9, the optimum for growth of *Legionella* spp., by ACES buffer. The activated charcoal serves as a scavenger for toxic free radicals that are generated by the interaction of light with the medium (Hoffman et al., 1983). Yeast extract serves as a rich growth medium. Variations on BCYE-α agar have been developed (Wadowsky and Yee, 1981), and some have proven particularly useful for isolating legionellae from environmental specimens (Edelstein, 1982). Some investigators have found that addition of 1.0% bovine serum albumin enhances the growth of *L. micdadei* and some strains of *L. bozemanii* (Morrill et al., 1990). Antibiotics may be added to suppress the growth of indigenous human or environmental flora (Edelstein and Finegold, 1979; Edelstein, 1981). It is essential to include a nonselective BCYE-α agar, because both vancomycin and cefamandole, two commonly used agents for suppressing Gram-positive bacteria, will inhibit various *Legionella* species (Lee et al., 1993b). *L. pneumophila* grew better than other species. Strains of 25 of the 28 species tested were recovered on media that contained vancomycin. In contrast, cefamandole completely inhibited strains of 11 species and inhibited growth of strains from an additional 8 species. Several complex and semisynthetic broth

TABLE BXII.γ.57. (cont.)

Characteristic	<i>L. parisiensis</i>	<i>L. quateensis</i>	<i>L. quintivanii</i>	<i>L. rubrilucens</i>	<i>L. santitricus</i>	<i>L. shakespearei</i>	<i>L. spirilis</i>	<i>L. steigerwaltii</i>	<i>L. taurinensis</i>	<i>L. waltersii</i>	<i>L. worleensis</i>	Genomospices 1
Gram reaction	—	—	—	—	—	—	—	—	—	—	—	—
Acid-fastness ^b	—	—	—	—	—	—	—	—	NA	—	—	NA
Growth on:												
BCYE-α agar	+	+	+	+	+	+	+	+	+	+	+	+
Blood agar with trypticase soy base	—	—	—	—	—	—	—	—	—	—	—	—
Growth requirement for L-cysteine	+	+	+	+	+	+	+	+	+	+	+	+
Browning of tyrosine-containing media	+	+	+	+	+	+	+	+	— (96%)	—	+	+
Acid from carbohydrates	—	—	—	—	—	—	—	—	—	—	—	—
Nitrate to nitrite	—	—	—	—	—	—	—	—	—	—	—	—
Hippurate hydrolysis	—	—	—	—	—	—	+	(w)	— (90%)	+	—	—
Gelatin liquefaction	+	+	+	+	+	+	+	+	+	+	+	+
β-lactamase production	+	+	—	+	+	+	+	+	+	+	+	—
Catalase	+	+	+	+	+	+	+	+	+	+	—	+
Oxidase	+	—	—	—	+	w	+	—	+	+	—	—
Motility ^c	+	+	+	+	+	+	+	+	+	+	+	+
Autofluorescence ^d	BW	—	—	R	—	—	—	BW	R (67%) / — ^e	—	—	—

^aCharacteristics based on varying numbers of strains. In some cases a species is represented by a single isolate. NA, not available; w, weak.

^bModified decolorization.

^cOne polar flagellum.

^dBW, blue-white; R, red.

^eAutofluorescence may be lost on subculture.

media are available for physiologic studies of *L. pneumophila* (Saito et al., 1981), but have not proven useful for isolation of clinical or environmental strains.

Media containing dyes that have some differential value have been described (Vickers et al., 1981), based on the appearance of isolated colonies. Data are not available for more recently isolated species.

Several species exhibit a blue-white or red autofluorescence, and many species produce a diffusible, brown pigment on tyrosine-containing agar (Tables BXII.γ.56 and BXII.γ.57).

Growth of stock cultures obtained from humans occurs at temperatures between 25°C and 43°C, but not at 50°C. The optimum temperature is 36° ± 1°C (Weaver and Feeley, 1979; Thacker et al., 1981); however, in nature, the optimum growth temperature may be 45°C or higher (Fliermans et al., 1981; Orison et al., 1981). The optimum pH for growth of legionellae is 6.8–7.0 (Feeley et al., 1978).

CO₂ does not appear to stimulate the growth of legionellae, with the exception of *L. gormanii*. Because of this exception, and also the buffering effect of CO₂, incubation in an atmosphere of air + 2.5% CO₂, or in a candle jar, has been recommended for all legionellae; however, CO₂ may slightly inhibit growth in BCYE (Pasculle et al., 1980).

Pinpoint colonies appear on BCYE-α agar¹ in about 3 d after

primary culture; the colony diameter reaches 3–4 mm after 5–7 d of incubation at 36° ± 1°C. Colonies are gray, glistening, convex, and circular with an entire edge. With a dissecting microscope one can visualize iridescence in varying colors of red, blue, and green; there is often a faceted or “cut-glass” appearance of the colonies when viewed with reflected light.

Carbohydrates are neither fermented nor oxidized by legionellae (Weaver and Feeley, 1979; Riley and Weaver, personal communication). Amino acids can presumably serve as the carbon source (Warren and Miller, 1979; George et al., 1980).

Legionellae are nutritionally fastidious. They require iron and L-cysteine for growth. As indicated previously, soluble ferric pyrophosphate, ferric nitrate, or hemoglobin can provide the iron, and IsoVitalX (BBL) can serve as a source of the L-cysteine. The following amino acids are essential for growth: cysteine, serine, methionine, arginine, valine, leucine, isoleucine, and threonine; no vitamin or cofactor requirements have been demonstrated (George et al., 1980). Chemically defined and semi-synthetic liquid media have been developed for legionellae (Pine et al., 1979; Warren and Miller, 1979; Ristoph et al., 1980).

One or more cryptic plasmids are present in many, but not all, strains of many species (Knudson and Mikesell, 1980). Neither pathogenicity nor antibiotic resistance, nor any other cell

1. BCYE agar: add 10.0 g of ACES buffer (N-2-acetamido-2-aminoethane-sulfonic acid; pKa = 6.9 at 20°C; available from Sigma Chemical Co., St. Louis, MO) to 500 ml of distilled water and dissolve by heating in a water bath at 45–50°C. Mix this solution with 440 ml of distilled water to which 40 ml of 1.0 N KOH has been added. Add the following ingredients: activated charcoal (Norit SG, available through Sigma Chemical Co., Cat. No. C5510 [acid-washed with phosphoric and sulfuric acids]), 2.0 g; yeast extract, 10.0 g; and agar (Difco or Oxoid), 17.0 g. Dissolve by boiling. Autoclave 15 min at 121°C. Cool to 50°C. Aseptically add L-cysteine-HCl·H₂O solution (0.4 g in 10 ml of distilled water, sterilized by filtration), followed by ferric pyrophosphate solution (0.25 g in 10 ml of distilled water, sterilized by filtration). (Soluble ferric pyrophosphate is available on request from

Director, Biological Products Div., Centers for Disease Control and Prevention, Atlanta, GA 30333. It must be kept dry and stored in the dark until used. It is not usable if the color changes from green to yellow or brown. Do not heat over 60°C to dissolve the ferric pyrophosphate; a 50°C water bath is satisfactory.) The pH of the final solid medium should be 6.9 ± 0.05 at room temperature. Since reagents vary, each laboratory must determine the amount of KOH required. Do this by holding the bulk medium at 50°C while pouring one plate and checking its pH. When necessary, adjust the bulk medium with either 1.0 N KOH or 1.0 N HCl. Note the pKa of ACES buffer is influenced by temperature (0.02/°C); consequently, this must be considered with all pH determinations. Dispense 20 ml portions of the complete medium into 10- × 100-mm plastic Petri dishes; swirl the medium between pouring plates to keep charcoal particles suspended.

TABLE BXII.7.58. Major fatty acids of *Legionella* species (% total fatty acids)^{a,b}

Fatty acids	<i>L. adelaidensis</i> *	<i>L. anisa</i>	<i>L. birminghamsensis</i> *	<i>L. bozemanii</i>	<i>L. brunensis</i>	<i>L. chervii</i>	<i>L. cincinnatiensis</i> *	<i>L. dumoffii</i>	<i>L. erythra (ATCC 35303)</i>	<i>L. erythra</i>	<i>L. farfieldensis</i> *	<i>L. felii</i>	<i>L. gestiana</i>	<i>L. gormanii</i>	<i>L. gratiana</i>	<i>L. hackeliae</i>	<i>L. israelensis</i>	<i>L. jamesoniensis</i>	<i>L. jordanis</i>	<i>L. lansingensis</i> *	<i>L. londiniensis</i>	<i>L. longbeachae</i>	<i>L. lytica</i>
C _{14:0} iso	7	6	—	3	2	5	34	2	tr	2	14	3	tr	4	2	2	2	2	2	NA	tr	5	NA
C _{15:0} iso	6	1	—	tr	4	tr	—	tr	1	1	2	tr	16	—	NA	4	1	1	2	5	tr	—	NA
C _{15:0} ante	48	24	100	29	100	27	—	30	8	12	15	18	9	22	NA	33	27	30	46	NA	5	15	NA
C _{16:1} iso A	100	NA	—	NA	NA	NA	—	NA	—	—	NA	2	0	NA	NA	NA	NA	NA	NA	NA	tr	—	NA
C _{16:1} iso B	—	—	—	—	NA	—	100	14	4	—	NA	—	—	—	NA	—	1	3	1	NA	NA	—	NA
C _{16:0}	44	23	46	14	14	29	—	8	38	24	100	16	2	17	14	15	22	21	18	40	11	25	NA
C _{16:1} o7c	—	9	—	9	NA	9	—	1	—	27	NA	22	—	11	NA	13	7	4	4	NA	NA	24	NA
Un16	—	1	—	1	NA	tr	—	7	—	14	NA	—	—	1	NA	—	—	tr	—	NA	NA	1	NA
C _{16:0}	—	8	—	11	NA	6	NA	—	26	14	NA	16	18	11	30	6	7	3	1	NA	15	9	NA
C _{16:1}	NA	NA	NA	NA	NA	NA	NA	NA	—	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	19	NA	NA
C _{17:1} ante	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	1	3	1	12	—	—	NA
C _{17:0} iso	3	1	—	1	3	1	—	—	1	1	—	tr	9	1	3	2	1	1	2	9	0	tr	NA
C _{17:0} ante	14	7	54	11	62	12	36	17	10	10	9	7	7	9	—	14	17	21	19	100	23	8	NA
C _{17:0} cyc	—	9	—	9	—	3	—	8	—	—	—	tr	—	10	—	3	5	5	2	—	4	tr	NA
C _{18:0} iso	—	NA	—	NA	0	NA	—	NA	—	—	—	tr	4	NA	4	NA	NA	NA	NA	—	8	1	NA
C _{18:0}	—	1	—	2	—	1	—	1	4	2	—	1	—	2	—	1	1	1	tr	—	—	—	NA
C _{19:0} ante	—	NA	—	—	—	NA	—	—	—	—	—	—	—	NA	—	—	NA	NA	—	—	—	—	NA
C _{21:0}	—	—	—	—	—	—	—	—	tr	tr	—	1	—	—	—	—	2	1	—	—	—	—	NA
C _{14:1}	4	—	—	—	0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NA
C _{14:0}	12	—	—	—	3	—	10	—	—	—	2	—	—	—	—	—	—	—	—	—	—	—	NA
C _{15:0}	16	—	—	—	3	—	8	—	—	—	5	—	—	—	—	—	—	—	—	5	—	—	NA
C _{15:1}	9	—	—	—	1	—	—	—	—	—	6	—	—	—	—	—	—	—	—	—	—	—	NA
C _{16:0}	44	—	43	—	30	—	—	—	—	—	33	—	—	—	—	—	—	—	—	20	—	—	NA
C _{16:1}	100	—	37	—	22	—	—	—	—	—	51	—	—	—	—	—	—	—	—	12	—	—	NA
C _{17:0}	3	—	—	—	1	—	7	—	—	—	4	—	—	—	—	—	—	—	—	—	—	—	NA
C _{18:0}	12	—	—	—	3	—	7	—	—	—	8	—	—	—	—	—	—	—	—	—	—	—	NA
C _{20:0}	12	—	—	—	5	—	5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NA
Lambert and Moss group (Lambert, Moss, 1989)	—	AI5/16C	—	AI5	—	AI5/16C	—	AI5	16C	16C	—	16C	—	AI5/16C	—	AI5	AI5/16C	AI5/16C	AI5	—	—	16C	—

^aRelative quantities among nonhydroxy fatty acids.^bNA, not available; tr, trace.^cAmoeba-associated strains.

(continued)

TABLE BXII.7.58. (cont.)

Fatty acids	<i>L. maceachernii</i>	<i>L. micdadei</i>	<i>L. moravica*</i>	<i>L. nautarum</i>	<i>L. oakridgensis</i>	<i>L. parvius</i>	<i>L. pneumophila pneumophila</i>	<i>L. pneumophila fraseri</i>	<i>L. pneumophila pascuelli</i>	<i>L. quaternensis</i>	<i>L. quinivannii*</i>	<i>L. rubrilucens</i>	<i>L. saintthelensis</i>	<i>L. sancticrucis</i>	<i>L. shakspereae</i>	<i>L. spiritensis</i>	<i>L. steigervallii</i>	<i>L. tucsonensis*</i>	<i>L. taurinensis</i>	<i>L. wadsworthii</i>	<i>L. wallersti</i>	<i>L. worshleensis</i>	Genomospecies 1
C _{14:0} iso	tr	tr	10	2	tr	3	5	NA	NA	3	8	2	5	6	9	2	2	8	3	1	13	4	2
C _{15:0} iso	—	1	1	0	—	tr	—	NA	NA	NA	NA	tr	tr	—	2	tr	tr	NA	—	tr	tr	tr	1
C _{15:0} ante	28	32	45	19	2	23	13	NA	NA	8	100	12	20	10	10	20	23	100	1–20	37	9	12	23
C _{16:1} iso A	NA	NA	NA	tr	—	NA	—	NA	NA	tr	NA	—	—	—	NA	—	NA	NA	—	NA	NA	1	18
C _{16:1} iso B	3	2	NA	—	2	—	3	NA	NA	—	—	—	—	—	—	5	—	NA	—	—	—	—	NA
C _{16:0} iso	11	12	39	9	26	17	32	NA	NA	20	94	23	22	30	29	28	15	77	>20	10	17	9	21
C _{16:1} o7c	14	8	NA	NA	14	7	16	NA	NA	NA	NA	27	25	22	NA	15	8	NA	—	8	NA	NA	NA
Un16	—	—	NA	NA	—	2	—	NA	NA	NA	NA	—	—	tr	NA	—	1	NA	—	6	NA	NA	NA
C _{16:0}	6	6	NA	19	10	9	7	NA	NA	7	NA	15	7	10	NA	5	18	NA	10–20	1	NA	16	NA
C _{16:1}	NA	NA	NA	27	NA	NA	NA	NA	NA	35	NA	NA	NA	NA	NA	NA	NA	NA	10–20	NA	NA	35	NA
C _{17:1} ante	5	5	—	—	1	—	tr	NA	NA	0	5	—	—	—	1	3	—	—	—	—	—	0	1
C _{17:0} iso	—	tr	0	0	—	1	tr	NA	NA	4	—	tr	10	4	3	tr	8	49	—	1	NA	5	12
C _{17:0} ante	22	21	12	13	5	9	8	NA	NA	—	—	9	1	8	—	16	14	—	10	19	2	—	—
C _{17:0} cyc	—	1	—	—	12	10	2	NA	NA	—	—	—	1	—	—	1	NA	—	—	6	—	9	—
C _{18:0} iso	NA	NA	3	2	2	NA	—	NA	NA	10	—	tr	tr	1	—	tr	3	—	—	1	—	—	—
C _{18:0}	tr	tr	—	—	13	2	3	NA	NA	—	—	2	1	1	—	1	NA	—	—	2	—	—	—
C _{19:0} ante	—	—	—	—	1	NA	—	NA	NA	—	—	—	—	—	—	tr	—	—	—	—	—	—	—
C _{21:0}	2	2	—	—	—	—	1	NA	NA	—	—	—	—	—	—	1	—	—	—	—	—	—	—
C _{14:1}	—	—	1	—	—	—	—	NA	NA	—	6	—	—	—	—	—	—	—	—	—	2	—	tr
C _{14:0}	—	—	8	—	—	—	—	NA	NA	—	9	—	—	—	4	—	—	—	—	—	2	—	tr
C _{15:0}	—	—	7	—	—	—	—	NA	NA	—	10	—	—	—	5	—	—	—	—	—	6	—	3
C _{15:1}	—	—	2	—	—	—	—	NA	NA	—	—	—	—	—	—	—	—	—	—	—	6	—	4
C _{16:0}	—	—	85	—	—	—	—	NA	NA	—	79	—	—	—	11	—	—	41	NA	—	16	—	10
C _{16:1}	—	—	100	—	—	—	—	NA	NA	—	95	—	—	—	16	—	—	78	NA	—	19	—	—
C _{17:0}	—	—	3	—	—	—	—	NA	NA	—	16	—	—	—	4	—	—	3	NA	—	2	—	2
C _{18:0}	—	—	35	—	—	—	—	NA	NA	—	7	—	—	—	1	—	—	4	NA	—	2	—	tr
C _{20:0}	—	—	11	—	—	—	—	NA	NA	—	6	—	—	—	—	—	—	—	NA	—	3	—	1
Lambert and Moss group (Lambert, Moss, 1989)	AI5	AI5	—	—	16C	AI5/16C	16C	—	—	—	—	16C	16C	16C	—	16C	AI5/16C	—	—	AI5	—	—	—

*Relative quantities among nonhydroxy fatty acids.

bNA, not available; tr, trace.

cAmoeba-associated strains.

TABLE BXII.γ.59. Ubiquinones of *Legionella* species^{a,b}

<i>Legionella</i> species	Q-9	Q-10	Q-11	Q-12	Q-13	Q-14	Ubiquinone group
<i>L. adelaidensis</i>	2	40	100	12	—	—	—
<i>L. anisa</i>	2–3	4	2–3	3	tr–1	—	B
<i>L. birminghamensis</i>	NA	NA	NA	NA	NA	NA	—
<i>L. bozemanii</i>	2–3	4	3–4	4	tr–1	—	B
<i>L. brunensis</i>	NA	NA	NA	NA	NA	NA	—
<i>L. cherrii</i>	2	3	3–4	4	1	—	B
<i>L. cincinnatiensis</i>	NA	NA	NA	NA	NA	NA	—
<i>L. dumoffii</i>	2–3	3–4	3–4	4	1	—	B
<i>L. erythra</i>	—	—	1–2	4	1–2	—	A
<i>L. fairfieldensis</i>	NA	NA	NA	NA	NA	NA	—
<i>L. feeleyi</i>	—	—	—	tr	4	2–3	E
<i>L. geestiana</i>	NA	NA	1	4	39	52	—
<i>L. gormanii</i>	3	4	4	4	—	—	B
<i>L. gratiana</i>	1.7	4	1.9	2.2	tr	NA	—
<i>L. hackeliae</i>	—	—	—	2	4	tr–1	D
<i>L. israelensis</i>	—	tr	1–2	4	3–4	—	D
<i>L. jamestowniensis</i>	—	tr	1–2	4	4	tr	D
<i>L. jordanis</i>	—	tr	tr	3–4	4	tr	D
<i>L. lansingensis</i>	—	—	1X	1X	4X	1X	—
<i>L. londiniensis</i>	2	24	55	16	2	tr	—
<i>L. longbeachae</i>	3–4	4	3–4	3–4	tr	—	B
<i>L. lytica</i> ; amoeba-associated strains	NA	NA	NA	NA	NA	NA	NA
<i>L. maceachernii</i>	—	—	—	tr–1	4	1	D
<i>L. micdadei</i>	—	—	tr–1	3	4	tr	D
<i>L. moravica</i>	NA	NA	NA	NA	NA	NA	—
<i>L. nautarum</i>	NA	NA	NA	18	76	6	—
<i>L. oakridgensis</i>	2	4	1–2	tr	—	—	C
<i>L. parisiensis</i>	1–2	3–4	3	4	1	—	B
<i>L. pneumophila</i> subsp. <i>pneumophila</i>	—	—	1–2	4	1–2	—	A
<i>L. pneumophila</i> subsp. <i>fraseri</i>	NA	NA	NA	NA	NA	NA	—
<i>L. pneumophila</i> subsp. <i>pascullei</i>	NA	NA	NA	NA	NA	NA	—
<i>L. quateirensis</i>	NA	tr	2	21	69	8	—
<i>L. quinlivanii</i>	NA	NA	NA	NA	NA	NA	—
<i>L. rubrilucens</i>	—	—	1–2	4	1–2	—	A
<i>L. sainthelensi</i>	3	4	3–4	3–4	tr	—	B
<i>L. santicrucis</i>	1–2	4	3–4	3–4	tr	1	B
<i>L. shakespearei</i>	—	1X	1X	3X	1X	—	—
<i>L. spiritensis</i>	—	—	—	tr–1	4	1	D
<i>L. steigerwaltii</i>	1	2	2–3	4	—	—	B
<i>L. tucsonensis</i>	NA	NA	NA	NA	NA	NA	—
<i>L. taurinensis</i>	—	—	1X	5X	1X	—	—
<i>L. wadsworthii</i>	2	4	2–3	tr–1	—	—	C
<i>L. waltersii</i>	NA	NA	NA	Major	NA	NA	—
<i>L. worsteiensis</i>	NA	1	4	26	62	4	—
Genomospecies 1	—	—	—	—	Major	—	—

^aRelative proportion of ubiquinones^bSymbols: tr, trace; nX, relative proportion of ubiquinone, —, not present.

function was correlated with the presence of any plasmid (Mike-sell et al., 1981; Aye, Wachsmuth, and Feeley, personal communication).

Motile strains of all *Legionella* species possess cross-reacting H antigens (Lewallen et al., 1979; Hébert et al., 1980d). O antigens, which define the serogroups, are primarily defined by the characteristics of cellular lipopolysaccharide (Ciesielski et al., 1986; Otten et al., 1986). They are largely unique for each species, although some cross-reactions have been demonstrated (Table BXII.γ.60). More than one O-antigen group is present in several species (Table BXII.γ.61).

Legionella species are susceptible *in vitro* to a variety of antimicrobial agents, including erythromycin, rifampin, and newer macrolides, such as azithromycin and clarithromycin, which are effective *in vivo*, as measured in experimental animals or in clinical studies (Hamedani et al., 1991; Donowitz and Earnhardt, 1993; Stamler et al., 1994). Quinolones have been used successfully to treat *Legionella* infections (Saito et al., 1986; Unertl et al.,

1989), but clinical failures have also been described (Kurz et al., 1988). Tetracycline, doxycycline, and minocycline are also effective *in vitro* and possibly *in vivo*. There are no prospective, random, placebo-controlled studies of antimicrobial therapy for *Legionella* infection. Retrospective analysis of two early epidemics (Philadelphia in 1976 and Burlington, Vermont, in 1977) documented the efficacy of erythromycin and, less conclusively, tetracycline (Fraser et al., 1977; Broome et al., 1979). Other antibiotics that are not effective against *Legionella* include aminoglycosides, sulfamethoxazole-trimethoprim, chloramphenicol, and a variety of cephalosporins (Thornsberry et al., 1978).

Many species of *Legionella* produce some level of β-lactamase activity when assayed with a chromogenic cephalosporin test (Tables BXII.γ.56 and BXII.γ.57) (Hébert et al., 1980b). The β-lactamase is assumed to be chromosomal rather than of plasmid origin and is more active on cephalosporins than on penicillins (Thornsberry and Kirven, 1978). The correlation between *in vitro* susceptibility and clinical efficacy of antibiotics for legi-

TABLE BXII.γ.60. Cross-reactions among *Legionella* species and with other bacteria

Species (antigen)	Method	Cross-reacts with antiserum to:	Absorbed by:	Reference
<i>L. adelaidensis</i>	Agglutination (unabsorbed neat)	<i>L. hackeliae</i> , serogroup 1 and <i>L. dumoffii</i> (1+); no reaction with <i>L. adelaidensis</i> antiserum at optimal dilution	None	Benson et al., 1991a
<i>L. anisa</i>	Direct immunofluorescence	<i>L. longbeachae</i> , serogroup 2 and <i>L. bozemanii</i> , serogroup 2	None	Bornstein et al., 1989a
<i>L. anisa</i>	Direct immunofluorescence	<i>L. gratiana</i> (+ to ++)	NA	Bornstein et al., 1989a
<i>L. anisa</i>	Slide agglutination	<i>L. gratiana</i> (+++)	Removed by absorption with <i>L. anisa</i> cells	Bornstein et al., 1989a
<i>L. anisa</i>	Slide agglutination	<i>L. tucsonensis</i> (+++)	Cross-reaction removed by absorption of antiserum with <i>L. santacrucis</i> and <i>L. anisa</i> cells	Thacker et al., 1989a
<i>L. bozemanii</i>	Agglutination	<i>L. longbeachae</i> , serogroup 2; <i>L. jordanis</i>	NA	Pelaz et al., 1987
<i>L. bozemanii</i>	Microagglutination	<i>L. jordanis</i>	NA	Pelaz et al., 1987
<i>L. bozemanii</i>	Immunofluorescence	<i>L. jordanis</i>	NA	Pelaz et al., 1987
<i>L. bozemanii</i> , serogroup 2	Direct immunofluorescence	<i>L. gratiana</i> (+ to ++)	NA	Bornstein et al., 1989a
<i>L. bozemanii</i> , serogroup 2	Slide agglutination	<i>L. gratiana</i> (+++)	Removed by absorption with <i>L. anisa</i> cells	Bornstein et al., 1989a
<i>L. cincinnatiensis</i>	Slide agglutination	<i>L. sainthelensi</i> (2+)	NA	Thacker et al., 1988a
<i>L. cincinnatiensis</i>	Direct immunofluorescence	<i>L. longbeachae</i> , <i>L. dumoffii</i>	NA	Jernigan et al., 1994
<i>L. dumoffii</i>	Agglutination	<i>L. pneumophila</i> , serogroup 7	NA	Pelaz et al., 1987
<i>L. erythra</i> , serogroup 2	Indirect immunofluorescence	<i>L. rubrilucens</i>	Cross-reaction partially removed by absorption with <i>L. erythra</i> , serogroup 2	Saunders et al., 1992
<i>L. geestiana</i>	Slide agglutination	<i>L. jamestowniensis</i> (+++)	Cross-reaction removed by absorption with <i>L. jamestowniensis</i> cells	Dennis et al., 1993
<i>L. gratiana</i>	Direct immunofluorescence	<i>L. bozemanii</i> , serogroup 2 (+ +); <i>L. longbeachae</i> , serogroup 2 (+ +); <i>L. anisa</i> (+ +); <i>L. longbeachae</i> , serogroup 1 (+); <i>L. bozemanii</i> , serogroup 1 (+)	NA	Bornstein et al., 1989a
<i>L. israelensis</i>	Slide agglutination	<i>L. wadsworthii</i> (+++ to ++++)	Removed by absorption with <i>L. wadsworthii</i> cells	Bercovier et al., 1986
<i>L. jordanis</i>	Agglutination, microagglutination, and immunofluorescence	<i>L. bozemanii</i>	NA	Pelaz et al., 1987
<i>L. jordanis</i>	Slide agglutination	<i>L. bozemanii</i> , serogroups 1 and 2 (+++ +), <i>L. parisiensis</i> (+++ +), <i>L. anisa</i> (+++ +), <i>L. longbeachae</i> , serogroups 1 (+ +) and 2 (+)	Adsorption of antisera with <i>L. jordanis</i> removed cross-reactions	Thacker et al., 1988b
<i>L. jordanis</i>	Slide agglutination	<i>L. cincinnatiensis</i> (1+)	Removed by <i>L. santacrucis</i> and <i>L. sainthelensi</i> cells	Thacker et al., 1988b
<i>L. jordanis</i>	Direct immunofluorescence	<i>L. bozemanii</i>	Reduced by adsorption with <i>L. bozemanii</i> cells	Cherry et al., 1982a
<i>L. jordanis</i>	Slide agglutination	<i>L. cincinnatiensis</i> (+)	Cross-reaction removed by absorption with <i>L. santacrucis</i> and <i>L. sainthelensi</i> cells	Thacker et al., 1988b
<i>L. lansingensis</i>	Slide agglutination	<i>L. pneumophila</i> , serogroup 3	Cross-reaction absorbed with <i>L. pneumophila</i> , serogroup 6 cells	Thacker et al., 1992
<i>L. longbeachae</i> , serogroup 1	Agglutination	<i>L. longbeachae</i> , serogroup 2	NA	Pelaz et al., 1987
<i>L. longbeachae</i> , serogroup 1	Microagglutination	<i>L. longbeachae</i> , serogroup 2	NA	Pelaz et al., 1987
<i>L. longbeachae</i> , serogroup 1	Immunofluorescence	<i>L. longbeachae</i> , serogroup 2	NA	Pelaz et al., 1987

(continued)

TABLE BXII.γ-60. (cont.)

Species (antigen)	Method	Cross-reacts with antiserum to:	Absorbed by:	Reference
<i>L. longbeachae</i> , serogroup 2	Agglutination, microagglutination, and immunofluorescence	<i>L. longbeachae</i> , serogroup 1; <i>L. jordanis</i>	NA	Pelaz et al., 1987
<i>L. longbeachae</i> , serogroup 2	Immunodiffusion	<i>L. longbeachae</i> , serogroup 1	NA	Pelaz et al., 1987
<i>L. longbeachae</i> , serogroups 1 and 2	Direct immunofluorescence	<i>L. gratiana</i> (+ to ++)	NA	Bornstein et al., 1989a
<i>L. maceachernii</i>	Direct immunofluorescence	Undiluted antiserum <i>L. micdadei</i> (+++)	NA	Brenner et al., 1985
<i>L. oakridgensis</i>	Direct immunofluorescence	At 1:8 dilution <i>L. sainthelensi</i> (+++ to ++++); at working dilution <i>L. sainthelensi</i> (++)	Absorption of antisera to <i>L. sainthelensi</i> with <i>L. longbeachae</i> or <i>L. oakridgensis</i> cells removed reactivity with the absorbing antigen	Campbell et al., 1984a
<i>L. parisiensis</i>	Direct immunofluorescence	<i>L. bozemanii</i> , serogroups 1 and 2	Not removed by absorption with s	Lo Presti et al., 1997
<i>L. parisiensis</i>	Direct immunofluorescence	<i>L. bozemanii</i> (+); <i>L. micdadei</i> (+); <i>L. jordanis</i> (++) with undilute antiserum <i>L. bozemanii</i> and <i>L. jordanis</i> (++++)	NA	Brenner et al., 1985
<i>L. pneumophila</i> , serogroup 1	Agglutination / immunofluorescence	<i>L. pneumophila</i> , serogroup 7	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 2	Agglutination	<i>L. pneumophila</i> , serogroup 3	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 2	Microagglutination	<i>L. jordanis</i>	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 2	Immunofluorescence	<i>L. pneumophila</i> , serogroups 3 and 6	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 3	Agglutination	<i>L. pneumophila</i> , serogroup 4	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 3	Microagglutination	<i>L. pneumophila</i> , serogroups 2 and 6	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 3	Immunodiffusion	<i>L. pneumophila</i> , serogroup 2	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 3	Immunofluorescence	<i>L. pneumophila</i> , serogroups 2 and 6	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 4	Slide agglutination	<i>L. worsleiensis</i> (++++)	Cross-reaction to serogroup 4 removed by absorption with serogroup 4 cells	Dennis et al., 1993
<i>L. pneumophila</i> , serogroup 4 (Los Angeles 1 strain)	Agglutination	<i>L. pneumophila</i> , serogroup 8	NA	Pelaz et al., 1987
s, serogroup 4 (Los Angeles 1 strain)	Immunodiffusion	<i>L. pneumophila</i> , serogroups 5 and 8	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 4 (Los Angeles 1 strain)	Immunofluorescence	<i>L. pneumophila</i> , serogroups 5 and 8	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 5 (Dallas 1-E strain)	Immunodiffusion	<i>L. pneumophila</i> , serogroup 8	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 6	Agglutination	<i>L. pneumophila</i> , serogroup 2	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 6	Immunodiffusion and immunofluorescence	serogroup 3	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 7	Agglutination	<i>L. pneumophila</i> , serogroup 1	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 7	Immunofluorescence	<i>L. dumoffii</i>	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 8	Agglutination	<i>L. pneumophila</i> , serogroups 3 and 4	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 8	Microagglutination	<i>L. pneumophila</i> , serogroup 5	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 8	Immunodiffusion	<i>L. pneumophila</i> , serogroup 4	NA	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 8	Immunofluorescence	<i>L. pneumophila</i> , serogroups 4 and 5	Na	Pelaz et al., 1987
<i>L. pneumophila</i> , serogroup 8	Slide agglutination	<i>L. worsleiensis</i> (++)	NA	Dennis et al., 1993

(continued)

TABLE BXII.γ.60. (cont.)

Species (antigen)	Method	Cross-reacts with antiserum to:	Absorbed by:	Reference
<i>L. pneumophila</i> , serogroups 4, 10, 13	Slide agglutination	<i>L. quateirensis</i> (+ + + +)	Cross-reactions removed by absorption with <i>L. pneumophila</i> , serogroup 13	Dennis et al., 1993
<i>L. quateirensis</i>	Slide agglutination	<i>L. pneumophila</i> , serogroups 4, 5, 8, 10, 13, 14; <i>L. worsleiensis</i>	Cross-reactions removed by absorption with <i>L. quateirensis</i> cells	Dennis et al., 1993
<i>L. quateirensis</i>	Slide agglutination	<i>L. worsleiensis</i> (+ + + +)	Cross-reaction removed by absorption with <i>L. quateirensis</i> cells	Dennis et al., 1993
<i>L. quinlivanii</i> , serogroup 2	Slide agglutination	Unnamed genomospecies 1	Cross-reaction not removed by absorption	Benson et al., 1996
<i>L. rubrilucens</i>	Indirect immunofluorescence	<i>L. erythra</i> , serogroup 2	Cross-reaction removed by absorption with <i>L. rubrilucens</i> or <i>L. erythra</i> , serogroup 2 cells	Saunders et al., 1992
<i>L. sainthelensi</i>	Slide agglutination	<i>L. cincinnatiensis</i> (2+)	Removed by <i>L. santicrucis</i> and <i>L. sainthelensi</i> cells	Thacker et al., 1988a
<i>L. sainthelensi</i>	Direct immunofluorescence	At 1:8 dilution: <i>L. longbeachae</i> , serogroups 1 (+ + + to + + + +) and 2 (+ +); <i>L. oakridgensis</i> (+ + + to + + + +); at working dilution: <i>L. longbeachae</i> , serogroup 2 (+ +) and <i>L. oakridgensis</i> (+ +)	Absorption of antisera to <i>L. longbeachae</i> , serogroups 1 and 2 or <i>L. oakridgensis</i> with <i>L. sainthelensi</i> abolishes or reduces reactivity with <i>L. sainthelensi</i> and <i>L. longbeachae</i> , serogroups 1 and 2	Campbell et al., 1984a
<i>L. sainthelensi</i>	Slide agglutination	<i>L. cincinnatiensis</i> (+ +)	Cross-reactions removed by absorption with <i>L. sainthelensi</i> and <i>L. santicrucis</i> cells	Thacker et al., 1988a
<i>L. santicrucis</i>	Slide agglutination	<i>L. cincinnatiensis</i> (4+)	Removed by <i>L. santicrucis</i> and <i>L. sainthelensi</i> cells	Thacker et al., 1988a
<i>L. santicrucis</i>	Direct immunofluorescence	<i>L. longbeachae</i> , serogroup 1 (+ + +); with undiluted antiserum <i>L. sainthelensi</i> (+ +)	NA	Brenner et al., 1985
<i>L. santicrucis</i>	Slide agglutination	<i>L. tucsonensis</i> (+ + + +)	Cross-reaction removed by absorption of antiserum with <i>L. santicrucis</i> and <i>L. anisa</i> cells	Thacker et al., 1989a
<i>L. taurinensis</i>	Direct immunofluorescence	<i>L. spiritensis</i> , serogroup 1 (+ + + to + + + +)	NA	Lo Presti et al., 1999
<i>L. tucsonensis</i>	Slide agglutination	<i>L. gratiana</i> (+ +)	Removed by absorption with <i>L. anisa</i> cells	Bornstein et al., 1989a
<i>L. tucsonensis</i>	Slide agglutination	<i>L. longbeachae</i> , serogroup 1 (+ + + +); <i>L. longbeachae</i> , serogroup 2 (+); <i>L. cincinnatiensis</i> (+)	Cross-reactions with <i>L. longbeachae</i> , serogroup 1 removed by absorption	Thacker et al., 1989a
<i>L. wadsworthii</i>	Agglutination, microagglutination	<i>L. oakridgensis</i>	NA	Pelaz et al., 1987
<i>L. worsleiensis</i>	Slide agglutination	<i>L. pneumophila</i> , serogroups 4, 5, 8, 10; <i>L. quateirensis</i>	Cross-reactions with <i>L. pneumophila</i> , serogroup 4 and <i>L. quateirensis</i> not removed by absorption	Dennis et al., 1993
<i>L. worsleiensis</i>	Slide agglutination	<i>L. quateirensis</i> (+ + + +)	Cross-reactions removed by absorption with <i>L. worsleiensis</i> cells	Dennis et al., 1993
Genomospecies 1	Slide agglutination	<i>L. quinlivanii</i> , serogroup 2 (+)	NA	Benson et al., 1996

onellae is poor (Kirby et al., 1978). Clinical response and survival, however, do correlate with the timely administration of effective antibiotics (Heath et al., 1996). Resistance to penicillins and cephalosporins may be partially related to β -lactamase activity,

although infections caused by *L. micdadei*, which does not produce β -lactamase, do not respond to penicillins or cephalosporins. The most likely explanation for the poor correlation between *in vitro* and *in vivo* results is the accumulation of antibiotics

TABLE BXII.γ.61. Serogroups of *Legionella* species^a

Species	Serogroup	Type strain	Source of isolate	Reference
<i>L. pneumophila</i>	1	Philadelphia 1	Human lung	McKinney et al., 1979a
	2	Togus 1	Human lung	McKinney et al., 1979a, b
	3	Bloomington 2	Creek water	McKinney et al., 1979a
	4	Los Angeles 1	Human lung	McKinney et al., 1979a
	5	Dallas 1-E	Cooling tower water	England et al., 1980
	6	Chicago 2	Human lung	McKinney et al., 1979b
	7	Chicago 8	Environment	Bibb et al., 1983
	8	Concord 3	Human lung	Bissett et al., 1983
	9	N-23-G1-C2	Human lung	Edelstein et al., 1984
	10	Leiden 1	Human lung	Meenhorst et al., 1985
	11	797-PA-H	Human lung	Thacker et al., 1986
	12	570-CO-H	Human lung	Thacker et al., 1987
	13	82A3105	Human lung	Lindquist et al., 1988
	14		Human lung	Benson et al., 1988
<i>L. bozemanii</i>	Lansing 3	Lansing 3	Human lung	Brenner et al., 1988c
	JENA-1	JENA-1	Hot water system	Luck et al., 1995
<i>L. erythra</i>	1	WIGA	Human lung	Brenner et al., 1980c
	2	Toronto 3	Human lung	Tang et al., 1984
<i>L. feeleii</i>	1	SE-32A-C8	Cooling tower water	Brenner et al., 1985
	2	LC217	Water	Saunders et al., 1992
<i>L. hackeliae</i>	1	WO-44C	Coolant water	Herwaldt et al., 1984a
	2	691XWI-H	Human lung	Thacker et al., 1985
<i>L. longbeachae</i>	1	Lansing 2	Human lung	Brenner et al., 1985
	2	798-PA-H	Human lung	Wilkinson et al., 1985b
<i>L. quinlivanii</i>	1	Long Beach 4	Respiratory secretions	McKinney et al., 1981
	2	Tucker 1	Human lung	Bibb et al., 1981
<i>L. sainthelensi</i>	1	1442-AUSE	Air Conditioning System	Benson et al., 1989
	2	LC870	Cooling tower water	Birtles et al., 1991
<i>L. spiritensis</i>	1	Mt. St. Helens 4	Water	Campbell et al., 1984a
	2	1489-CA-H	Human bronchial washing	Benson et al., 1990a
	1	Mt. St. Helens 9	Water	Brenner et al., 1985
	2	ML76	Environment	Harrison et al., 1988

^aNA, not available.

within phagocytic cells. The most effective antibiotics are those that are effective *in vitro* and are also concentrated in macrophages.

Erythromycin is the drug of choice in treating human infection and should be given for a 3-week course (Kirby et al., 1978). Although there are few data, it has been suggested that rifampin should be used in combination with erythromycin when patients respond poorly to erythromycin alone or when serious disease is present (Cordes and Fraser, 1980).

All *Legionella* species are potentially pathogenic for humans. Antibodies to *Legionella* spp. are found in several species of domestic animals, but definitive evidence of active infection is lacking. Isolation of the bacterium has been accomplished only rarely and in a setting where autopsy contamination by water cannot be excluded completely (Boldur et al., 1987). Guinea pigs are easily infected by *Legionella* spp., and lethal infection often results (McDade et al., 1977; Winn et al., 1982). Rhesus monkeys and marmosets develop a fibrinopurulent pneumonia after infection with an aerosol of *L. pneumophila* (Baskerville et al., 1983). Mice and rats are variably resistant to the development of active infection, depending on the strain of rodent (Yamamoto et al., 1991b).

Disease caused by any *Legionella* species is often termed legionellosis. The two main types of legionellosis caused by *Legionella* are *Legionella* pneumonia (LP) (Fraser et al., 1977) and Pontiac fever (Glick et al., 1978). Most clinical, epidemiologic, and diagnostic data pertain to the most commonly isolated species, *L. pneumophila*; the limited data available for other species suggest that they follow the same pattern. The pneumonia has a 2- to 10-d incubation period after a presumed airborne ex-

posure to the organism. There is no proof of person-to-person spread. The earliest symptoms include malaise, myalgia, headache, and often a nonproductive cough. Rapidly rising fever and chills usually appear within a day, at which time there are definite clinical and radiologic indications of pneumonia. Other common symptoms are chest pain, abdominal pain, vomiting, diarrhea, and mental confusion. Pathologically there is focal, lobular pneumonia, which may coalesce and even produce a lobar distribution (Winn and Myerowitz, 1981). The inflammatory response usually consists of a mixture of polymorphonuclear neutrophils and macrophages. Treatment with antibiotics is necessary; untreated, the fatality rate is 15% or higher.

Subclinical and mild cases may not be recognized unless they are a part of a larger epidemic. Smoking, preexisting respiratory illness, and conditions where the patient is immunocompromised are predisposing factors. LP occurs most often in men above the age of 50. It has been estimated that there are 8000–18,000 cases of LP in the United States each year, many not recognized and/or reported (Marston et al., 1994). The disease occurs worldwide. LP has been caused by approximately half of the recognized species, and could potentially be caused by any if the patient is sufficiently immunocompromised.

Legionella pneumonia occurs as sporadic or epidemic disease, which may be acquired in the community (Storch et al., 1979) or as a nosocomial infection (England and Fraser, 1981). Remarkable differences in the incidence of LP in similar patient populations presumably relate to the frequency and quantity of virulent strains in the local environment (Goldstein et al., 1982). Subclinical disease occurs, even in severely immunocompromised patients (Jacobs et al., 1990). Mixed infections with multiple

Legionella species (Dowling et al., 1983) occur. Differing serologic types have also been isolated from human lung (Joly and Winn, 1984), although it has been suggested that strains with differing antigenic characteristics may derive from a common parent strain (Harrison et al., 1990).

Legionella spp. also produce acute, nonpneumonic, febrile illness accompanied by chills, headache, and myalgia (Pontiac fever). It has a high attack rate and a short incubation period of hours to several days. It is, by definition, an epidemic disease. Pontiac fever is not accompanied by clinically detectable disease of the lower respiratory tract. It is self-limiting and not fatal. Pontiac fever has been caused by *L. pneumophila* (Glick et al., 1978), *L. feeleyi* (Herwaldt et al., 1984a), *L. micdadei* (Goldberg et al., 1989), and *L. anisa* (Fensterheib et al., 1990). Severe *Legionella* pneumonia and nonpneumonic disease (differing from Pontiac fever in the sporadic nature of the infection) have occurred after exposure to the same environmental source (Girard et al., 1982).

All species of *Legionella* have been isolated from the environment. Natural environmental isolates have come from streams or lakes, sometimes heated by natural or industrial thermal source. *L. pneumophila* has been identified from moist soil adjacent to a body of water (Fliermans et al., 1981), and *L. longbeachae* has been recovered from moist potting soil (Steele et al., 1990). Although a variety of environmental cofactors have been described, it is becoming increasingly clear that the association of legionellae with a variety of free-living protozoa is the most important factor (Rowbotham, 1986; Fields, 1993). Isolation of strains may be accomplished only by coculture with amoebae, both for human (Rowbotham, 1983) and environmental (Sanden et al., 1992) sources.

There have been no isolates from dry soil. Urban environmental isolates were made from water collected from air conditioning cooling towers and evaporative condensers (Orrison et al., 1981). More recently, colonization of potable water systems by *Legionella* spp. has been frequently demonstrated, with isolates being more common on the hot water than the cold water side. The sediment of hot water heaters are a particularly productive source for isolates, especially if the water temperature has been lowered to minimize scalding and reduce heating costs. Water heaters in which the heating elements are remote from the bottom of the tank where the sediment accumulates are more likely to yield isolates than units with heating elements in the bottom of the tanks (Alary and Joly, 1991). It is interesting that the portions of municipal water systems that have been colonized are all peripheral; contamination of central water distribution facilities, where chlorination is performed, has not been demonstrated. Dissipation of chlorine as water flows through the system may explain this phenomenon.

Not surprisingly, epidemic disease can usually be traced to exposure to a contaminated water source. Sporadic cases are likely related to similar exposures, but investigation is more difficult. The contaminated water may be either environmental surface water or potable water. Factors that favor production of disease include a host who is compromised by immunosuppression or chronic cardiopulmonary disease, a vehicle for generating aerosols (which may be quite local), and undoubtedly the inoculum and virulence of the bacteria. The most vulnerable patients are in health care centers, where epidemic disease has been concentrated. Epidemics have also been associated, however, with hotels, cruise ships, and even grocery stores.

The primary hazard to researchers working with *Legionella*

species is exposure to aerosols, although self-inflicted wounds are a potential hazard. In the absence of large-scale aerosols, as might be produced in an experimental situation, the usual precautions for class II pathogens are sufficient. Organisms isolated on agar plates can be safely studied at the bacteriology bench. There are no documented occupational infections in standard diagnostic bacteriology laboratories.

ENRICHMENT AND ISOLATION PROCEDURES

The diagnosis of *Legionella* infection can be made by three methods, which are in decreasing order of importance, isolation of the organism in culture, direct identification of antigens or nucleic acid in tissues or body fluids, and demonstration of an immune response (Zuravleff et al., 1983).

Human specimens Lung tissue, lung aspirates, pleural fluid, transtracheal aspirate or sputum, and bronchoalveolar lavage fluid are the specimens usually cultured when *Legionella* is suspected (Edelstein et al., 1980). Samples that cannot be processed within 2 h of collection can be stored on wet ice or refrigerated for 2 d. Samples that will not be cultured within a reasonable time (2–5 d) should be frozen at -70°C . Ten percent tissue suspensions in distilled water or bacteriologic broth, body fluids, or sputum should be inoculated onto media designed to grow legionellae, such as BCYE- α . If indigenous flora is likely to be present, inoculate media with and without addition of antibiotics for selective inhibition of nonlegionellae. Media with selective antibiotics should never be inoculated without an accompanying noninhibitory plate.

Inhibition of *Legionella* growth, presumably by antibiotics or toxic compounds in the specimen, may be documented as a zone of no growth where the specimen was placed on the agar, in conjunction with heavy growth in adjacent areas, where the inoculum had been spread over the surface of the agar. Homogenization and acid treatment of respiratory secretions increased the number of isolates from sputum, but not bronchoalveolar lavage fluid in one study (Buesching et al., 1983), but this finding has not been reported by other investigators. Brief heat treatment of specimens also selectively inhibits contaminating flora, but *Legionella* spp. may also be inhibited (Edelstein et al., 1982b).

Blood cultures may be positive in severely ill patients who have high-level bacteremia (Rihs et al., 1985), but the sensitivity in mild to moderate cases is not known. Continuous-reading blood culture instruments do not register growth when *Legionella* spp. are present; staining of fluid with acridine orange or blind subculture onto BCYE- α agar may be required to document the presence of bacteria. *In vitro* experiments with spiked blood samples suggest that the lysis-centrifugation method, a convenient means of culturing blood for fastidious pathogens, may be used to recover legionellae from blood efficiently, but clinical confirmation is lacking.

Each batch of media must be tested for pH and sensitivity for primary isolation of legionellae. The pH of solid media is determined with a surface electrode or by emulsifying the agar from one plate in distilled water. The quality of media from various manufacturers can vary (Lee et al., 1993b). If media are commercially prepared, the manufacturer should supply documentation of successful quality control. If the media are prepared locally or if the sensitivity of media for primary isolation is in doubt, a control should be prepared from a freshly isolated strain that has been minimally passed on BCYE- α agar. After *in vitro* passage on non-charcoal-containing agar, strains with altered growth characteristics emerge. These strains will grow well on

batches of media that do not support the growth of bacteria in fluid or tissue; a false sense of security as to the adequacy of the media may result (Keathley and Winn, 1985). Media without antibiotics, stored refrigerated in the dark, in sealed, plastic containers remain stable for at least 2 months. Users should follow manufacturers' instructions for storage and outdates of commercially prepared media.

Media for isolation of *Legionella* spp. should be incubated at 35–37°C. Incubation in an atmosphere of 2–5% CO₂ provides marginal enhancement of isolation for some *Legionella* spp., but higher concentrations of CO₂ may be inhibitory. Plates should be held for at least 5–7 d before discarding. Colonies are usually evident after incubation for 48–72 h. Use of a stereomicroscope will facilitate detection of colonies and allow assessment of fine features of colonial morphology. *Legionella* spp. produce smooth, entire colonies that exhibit a mottled, cut-glass appearance when viewed with a stereomicroscope under reflected light. Red, blue, or green iridescence may be visible early with a stereomicroscope and later with the naked eye in areas of heavy growth. *L. pneumophila* produces sticky colonies that "string" when a portion of the colony is lifted with an inoculating needle or loop.

Environmental samples A comprehensive manual for culture of environmental specimens is available from the Centers for Disease Control and Prevention (CDC) (Centers for Disease Control and Prevention, 1992). Environmental, nonpotable water samples do not require concentration. Potable water samples must be concentrated either by filtration, as recommended by CDC, or by centrifugation (Brindle et al., 1987). Collection of at least 1 l of potable water is desirable, but as many as 10 l may be required if small numbers of legionellae are present. For filtration the water should be passed through a 0.2-μm polycarbonate filter, after which the filter is folded to the outside, placed in 10 ml of water, vortexed, and subcultured onto solid media. Serial dilutions may be performed to quantify the bacterial load. Plates are incubated for 7 d in 2.5–5.0% CO₂. Samples from faucets and showers may be taken with polyester swabs that are immersed in 3–5 ml of water; the faucet aerator or shower head should be removed if possible. Samples of sediment from hot water heaters should be plated on solid agar. Detailed instructions for recovering legionellae from air samples have been provided by CDC (Centers for Disease Control and Prevention, 1992).

Environmental samples, especially those from nonpotable water, are frequently contaminated with a variety of bacterial species. A variation on BCYE agar that includes glycine has proved useful for culturing environmental specimens (Wadowsky and Yee, 1981). Addition of selective antibiotics provides an additional competitive advantage for *Legionella* spp. Antibiotic combinations have included cefamandole or vancomycin for inhibition of Gram-positive flora, polymyxin B for Gram-negative flora, and anisomycin for yeast. Cefamandole may inhibit many isolates of *Legionella*, so selective plates should not be used alone. As has been recommended for sputum, brief acidification of the sample to pH 2.0 for 5 min followed by neutralization may provide additional suppression of contaminating environmental flora (Centers for Disease Control and Prevention, 1992). Heat enrichment of environmental specimens that were negative for legionellae on primary culture may yield as many as 30% additional isolates, putatively because the bacteria multiply in free-living amoebae in the samples. Tubes of water are incubated in closed tubes at 35°C, followed by periodic subculture onto agar.

If necessary, the aliquots for subculture may be treated with acid before testing.

MAINTENANCE PROCEDURES

Stock cultures on BCYE-α agar slants are recommended for maintenance of stock cultures. Inoculated slants are incubated for 2 d at 35°C and then stored in the dark at room temperature. Under these conditions, *L. pneumophila* cultures remain viable for 1–3 months (Hébert, 1980). Other species remain viable for variable periods from 4 months to 1 year. Lyophilization or storage at –50°C or lower is recommended for long-term maintenance (Feeley and Weaver, personal communication); a convenient and effective method for storage of isolates is to excise a piece of agar containing isolated colonies, after which the agar is frozen in a sterile container at –50°C or less (W.C. Winn, Jr., personal observation). Virulence and plasmids are often lost on lyophilization or subculture on non-charcoal-containing agar (McDade and Shepard, 1979). This conversion to avirulence is a one-way phenomenon and does not appear to occur if one of the charcoal agars, such as BCYE-α agar, is used for maintenance (Catrenich and Johnson, 1988).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Direct Detection of Antigen in Specimens

Human specimens The first method used to detect *Legionella* antigen in clinical specimens was direct immunofluorescence, using polyclonal antisera. Commercially available polyclonal conjugates are available for some species and serotypes. Cross-reactions have been demonstrated with *Pseudomonas aeruginosa* and *Bacteroides fragilis* (Edelstein and Edelstein, 1989), *Bordetella pertussis* (Benson et al., 1987), *Francisella tularensis* (Roy et al., 1989), and with spores of *Bacillus cereus* (Flournoy et al., 1988). A commercially available monoclonal antibody to an antigen that is specific for the species *L. pneumophila* provides cleaner staining and fewer cross-reactions than the polyclonal antisera, but monoclonal reagents are not available for other species (Edelstein et al., 1985; Tenover et al., 1985). The direct immunofluorescence test has been discontinued in many laboratories because of low sensitivity (Edelstein et al., 1980; Buesching et al., 1983). Although the specificity is high, false-positive results cause unacceptably low predictive values if the prevalence of disease is very low. In addition, legionellae in water used to make reagents may cause clusters of false-positive tests. Radioimmunoassays and enzyme immunoassays have been developed for detection of *L. pneumophila* serogroup 1 antigen in urine. A commercially available enzyme immunoassay, using polyclonal antiserum, (BINAX, Portland, Maine) has a sensitivity of 70–80% and a specificity of 90% (Kazandjian et al., 1997). *L. pneumophila* antigen may be excreted for weeks or months, so that detection of antigen does not always indicate current disease (Kohler et al., 1984).

Environmental specimens Initially direct immunofluorescence was commonly used for detection of legionellae in water samples. Many specimens that were positive by immunofluorescence could not be confirmed by culture, probably reflecting both insensitivity of early culture techniques, fastidious or nonreplicating legionellae, and/or immunologic crossreactions. The preferred method for testing environmental samples is culture.

Direct Detection of Nucleic Acid in Specimens

Human specimens A direct nucleic acid probe for *L. pneumophila* was marketed briefly, but withdrawn because of unac-

ceptably frequent false-positive reactions. There are no commercially available molecular amplification tests for clinical specimens, but amplification procedures that are approved for environmental specimens have been used successfully to detect legionellae in respiratory secretions (Kessler et al., 1993).

Environmental specimens Nucleic acid amplification, using the polymerase chain reaction, has been used successfully (Jaulhac et al., 1992), and an amplification kit is commercially available, although not approved for use with human specimens (Kessler et al., 1993; Oshiro et al., 1994).

Serologic Diagnosis

Human infections The indirect immunofluorescent antibody (IFA) test is the test of choice for serologic diagnosis of human infections. Although other methods have been used, appreciable experience has been accumulated only with the IFA test. Either heat-killed or formalin-inactivated bacterial antigen may be employed (Wilkinson and Brake, 1982). It is important to use antiglobulin conjugates that are active against both IgG and IgM (Wilkinson et al., 1979a). If the criterion for a positive test is a fourfold increase in paired sera to a titer of at least 1:128, the sensitivity of the test is 70–80% (Edelstein et al., 1980), and the specificity is approximately 99% (Wilkinson et al., 1981). The details of IFA are given by Wilkinson et al. (1979b). Sera should be examined for at least 6 weeks to demonstrate seroconversion; some patients do not seroconvert for 1 year or longer (Edelstein et al., 1980). A titer of 1:256 in a single serum has been used as presumptive evidence of *Legionella* infection during an epidemic, but the diagnosis of sporadic cases cannot be made with a single serum specimen. An elevated IgM titer suggests recent infection (Zimmerman et al., 1982), but these titers may persist for as long as a year in some patients (Zimmerman et al., 1982). Antisera are available for certain serogroups of *L. pneumophila* and for selected other species. Cross-reactions among *Legionella* serogroups and species make it impossible to assign a precise diagnosis with certainty (Wilkinson et al., 1983a). In addition, cross-reactions have been described with other bacterial species, including *Campylobacter* spp. (Boswell and Kudesia, 1992), *Pseudomonas* spp. (Edelstein et al., 1980; Klein, 1980; Collins et al., 1984), and *Bacteroides fragilis* (Edelstein et al., 1980). Under the best of circumstances, therefore, serologic diagnosis is only presumptive.

Animals Serologic studies have demonstrated anti-*Legionella* antibodies in a variety of animals (Collins et al., 1982). In the absence of bacterial isolates, it is difficult to interpret the serologic data.

Molecular Epidemiology of *Legionella* Infections It is not possible to ascribe an etiologic role to a *Legionella* isolate, simply because the same species has been recovered from human and environmental sources. An exhaustive investigation of all potential environmental sites must be undertaken. The likelihood of an etiologic association is increased by application of a variety of subtyping tools: monoclonal antibody typing (Joly and Winn, 1984; Gomez-Lus et al., 1993), ribotyping (Schoonmaker et al., 1992; Gomez-Lus et al., 1993), restriction endonuclease treatment with pulsed field electrophoresis (Saunders et al., 1991; Schoonmaker et al., 1992), polymerase chain reaction, arbitrarily primed polymerase chain reaction (Gomez-Lus et al., 1993; Ledesma et al., 1995), and alloenzyme analysis (Edelstein et al., 1986; Tompkins et al., 1987). The greatest discrimination comes from applying multiple typing tools (Barbaree, 1993; Mamolen

et al., 1993). It has been suggested that certain monoclonal antibody types of *L. pneumophila* are associated with increased virulence (Dournon et al., 1988; Helbig et al., 1995).

Media Legionellae grow on BCYE agars. They do not grow on standard blood agar, nutrient agar, or other commonly employed laboratory media that do not contain both L-cysteine and iron salts. To document the nutritional pattern of a fresh isolate, parallel subcultures should be made to BCYE- α and either sheep blood agar or BCYE- α agar without L-cysteine. Growth only on BCYE- α agar with L-cysteine suggests the possibility of *Legionella* spp. Legionellae must be differentiated phenotypically from two groups of organisms that grow on some modified charcoal media that contain L-cysteine, but not on common laboratory media. The first, and most significant, for clinical diagnosis and laboratory safety is *F. tularensis* (Westerman and McDonald, 1983; K. McLeod, C.P. Patton, and J.C. Feeley, unpublished data), and possibly *F. novicida*. *Francisella* spp. are easily distinguished from *Legionella* spp. because they produce acid from carbohydrates and do not contain branched chain fatty acids (Jantzen et al., 1979). *F. tularensis* constitutes a major biohazard for unsuspecting laboratories, so this possibility must be kept in mind. A number of thermophilic, spore-forming bacilli that stain Gram-negative mimic legionellae on BCYE agars and do not grow on common laboratory media. They also have branched chain fatty acid profiles that superficially resemble those of one or more *Legionella* species (Thacker et al., 1981). These strains are distinguished from legionellae by their ability to grow at 50°C or higher, to form spores, and to grow on certain media that lack cysteine, such as tryptone broth (Thacker et al., 1981; Weaver, unpublished data). These environmental organisms rarely, if ever, produce human disease, but may cause confusion if they contaminate reagents or specimens.

Pigments Pigments are of minimal value in identification of legionellae. Brown pigment is formed on supportive clear agars that contain tyrosine (Baine et al., 1978). Yellow, blue-white, or red autofluorescence may be seen with some species if the colonies are examined in the dark with a long-wavelength ultraviolet lamp (Wood's light) (Tables BXII.γ.56 and BXII.γ.57).

Serologic identification Serologic characterization of isolates, most commonly by the direct immunofluorescent antibody (DFA) test, is the most common method of identifying isolated strains. Polyclonal antisera, probably directed primarily against a lipopolysaccharide antigen, provide a serotype-specific diagnosis. Reagents are available commercially for selected serogroups of *L. pneumophila*, *L. micdadei*, and many other species (Hookey et al., 1995). A commercially available monoclonal antibody to a species-specific antigen of *L. pneumophila* provides more consistent and easily interpreted results (Edelstein et al., 1985; Tenover et al., 1985, 1986). Although serologic identification is sufficiently accurate to use in clinical and environmental laboratories, cross-reactions among serotypes and species do occur occasionally. The only certain way to establish the nature of an isolate is by nucleic acid analysis.

A slide agglutination test can also be used for the rapid identification of legionellae (Wilkinson and Fikes, 1980), especially in laboratories that are unable to perform immunofluorescence exams. Unfortunately, the agglutination test is not commercially available in the United States.

Gas-liquid chromatography (GLC) GLC analysis of cellular fatty acids is a method used, along with DFA, to test suspect

isolates and to confirm the identification of known legionellae, new serogroups, and new species (Moss and Dees, 1979; Lambert and Moss, 1989) (Table BXII.γ.58). The procedure is, thus far, totally specific in identifying species of *Legionella* at the genus level and can identify most, but not all, legionellae to the species level. The procedure involves saponification of whole cells, methylation of the fatty acids, extraction of the resultant methyl esters with diethyl ether and then hexane, followed by GLC analysis on a nonpolar stationary-phase column. The retention time of the resultant elution peaks is recorded, and tentative identification is made by comparison to reference standards. Confirmation and quantitation of the fatty acids are obtained by subsequent procedures, including mass spectrometry, use of polar columns, hydrogenation, and acetylation.

Characterization of cellular ubiquinones is also useful in characterizing *Legionella* species (Table BXII.γ.59). The genus contains varying amounts of ubiquinones with 9-14 isoprene units in the side chains (Lambert and Moss, 1989).

DNA relatedness This is the only method whereby one can unequivocally identify legionellae to the level of species, confirm the existence of new serogroups in existing species, and identify new species. DNA relatedness, however, is not adaptable to routine laboratory use and should be used only for isolates that cannot be characterized with certainty by biochemical, antigenic, and fatty acid analysis.

DIFFERENTIATION OF THE GENUS *LEGIONELLA* FROM OTHER GENERA

Legionella pneumophila has no close relatives in other genera based on DNA relatedness (Brenner et al., 1979). The original four strains that we now know to represent three separate *Legionella* species were isolated between 1943 and 1959 (Tatlock, 1944; Jackson et al., 1952; Bozeman et al., 1968). At the time of their discovery, however, none of these strains was cultivatable on bacteriologic media. They were treated as rickettsiae and not further studied until 1978 (McDade et al., 1979; Hébert et al., 1980a,d). The "modern" study of what was to become the genus *Legionella* began in 1976 with the investigation of a large outbreak of respiratory disease at an American Legion Convention in Philadelphia, Pennsylvania (Fraser et al., 1977), resulting in the isolation of the causative agent, *L. pneumophila*, by Joseph McDade (McDade et al., 1977). Shortly thereafter, the Flint 1 strain was isolated as an unknown on bacteriologic media by Dumoff (Dumoff, 1979) and the Philadelphia strains were isolated on bacteriologic media by Weaver (Feeley et al., 1978). The first species, *L. pneumophila*, was named in 1979 (Brenner et al., 1979).

The named species of *Legionella* have similar phenotypic characteristics (Tables BXII.γ.56 and BXII.γ.57). These include growth on media designed for legionellae, optimum pH and temperature for growth, common flagellar antigens, pathogenicity for guinea pigs and embryonated hen's eggs, nonfermentative metabolism, presence of catalase, absence of urease, and *in vitro* antibiotic susceptibility patterns. The cellular fatty acid compositions of all species are characterized by relatively large amounts of branched chain acids. Approximately half of the recognized species have been directly or indirectly implicated as causative agents of human pneumonia. Available microbiological and biochemical tests are insufficient to separate all species with certainty. Once a species has been defined, it is quite simple to identify species and serogroups by DFA. As the number of recognized species has expanded, so has the number of immunological cross-reactions (Table BXII.γ.60). It is still appropriate to

identify clinical isolates by DFA, particularly if the species is *L. pneumophila*, but the possibility of cross-reactions should be recognized.

TAXONOMIC COMMENTS

A frequently used definition of a "genetic" species is "a group of strains whose DNAs are 70% or more related at optimal reassociation conditions and have 5% or less divergence in related sequences". Stackebrandt and Goebel (1994) consider that strains that have at least 70% relatedness will also have at least 96% sequence identity when conserved portions of 16S rRNA are analyzed. The technology of DNA hybridization is costly and time-consuming, whereas genetic sequences can now be determined quickly and relatively cheaply. Analysis of 16S rRNA, however, does not have the discriminatory ability of DNA-DNA hybridization at the species level. Stackebrandt and Goebel suggest that analysis of 16S rRNA genes be used as a screening tool to select likely candidates for hybridization studies.

Legionella species fulfill the current definition of bacterial species. DNA relatedness cannot be used to generate good genetic definitions for a genus or family. The most common method for establishing relationships at the genus level is now sequencing of 16S rDNA (Wayne et al., 1987). Some of the amoeba-associated pathogens, which are not available as isolated bacteria, are closely related to other *Legionella* species by analysis of 16S rDNA; others, representing as yet unnamed species, are more distantly related (Fry et al., 1991a, b).

New species and serogroups have been confirmed by DNA relatedness. Relatedness between strains of the same species is 75% or higher for all species except *L. bozemanii*, for which relatedness is 56-77%. The related sequences of all species, including *L. bozemanii*, exhibit 3% or less divergence (unpaired bases within heteroduplex DNA), and there is only a small decrease in relatedness when reactions are done at a supraoptimal incubation temperature. Using another approach to molecular taxonomy, Ludwig and Stackebrandt (1983) studied the phylogeny of four *Legionella* species that were closely related. Analysis of 16S rDNA from seven *Legionella* species demonstrated that they were closely related (>95%) to each other and that they fell into the *Gammaproteobacteria* (Fry et al., 1991b). Using several technical variations for detection of rDNA, several investigators have confirmed and extended these observations (Hookey et al., 1995, 1996; Birtles et al., 1996; Riffard et al., 1998).

A proposal was made to divide the family *Legionellaceae* into three genera: *Legionella*, *Tatlockia*, and *Fluoribacter* (Garrity et al., 1980; Brown et al., 1981). Other investigators have not supported the proposal and have recommended that a single genus, *Legionella*, be maintained (Brenner et al., 1985; Brenner, 1986). Analysis of 16S ribosomal RNA supported existence of a single genus (Fry et al., 1991b). Most importantly, the single-genus concept has been uniformly accepted within the community of microbiologists.

FURTHER READING

- Dowling, J.N., Saha, A.K. and Glew, R.H.. 1992. Virulence factors of the family *Legionellaceae*. *Microbiol. Rev.* 56: 32-60.
- Edelstein, P.H., Meyer, R.D. and Finegold, S.M.. 1980. Laboratory diagnosis of Legionnaires' disease. *Am. Rev. Respir. Dis.* 121: 317-327.
- Edelstein, P.H. 1993. Legionnaires' disease. *Clin. Infect. Dis.* 16: 741-747.
- Eisenstein, B.I. and Engleberg, N.C.. 1986. Applied molecular genetics: new tools for microbiologists and clinicians. *J. Infect. Dis.* 153: 416-430.